Generalised Linear Models

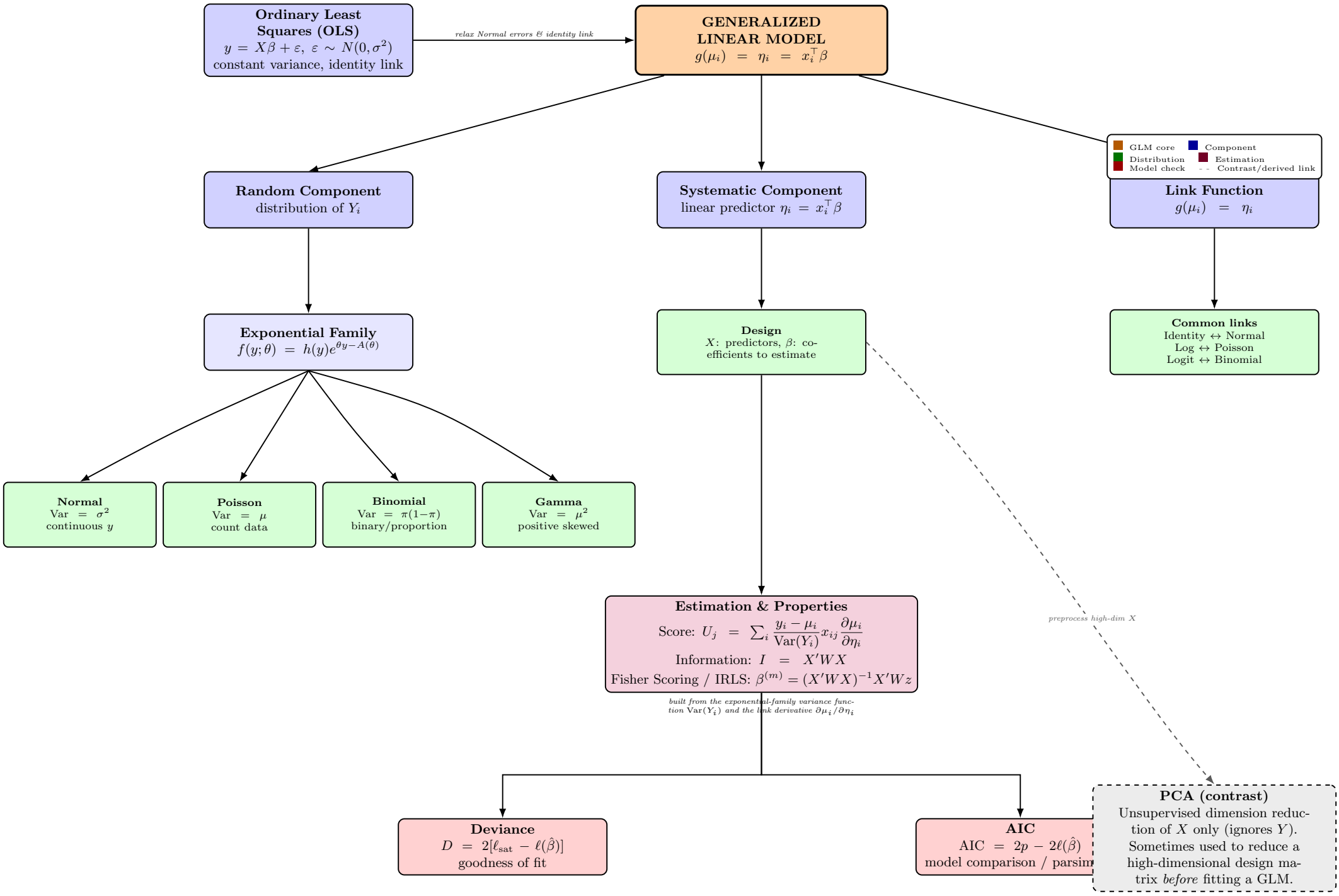
A Portfolio

An Exploration–Cognition–Reflection–Application journey through GLM theory

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The Generalized Linear Model Landscape

From Ordinary Least Squares to the Exponential Family, Link Functions, Estimation, and Model Evaluation



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Foreword

This portfolio began as an attempt to capture my journey from absolute uncertainty to a solid grasp of Generalised Linear Models. My aim is to document not just the concepts, but the problems I encountered, the questions I asked, and how I went about answering them.

Traditional note-taking usually amounts to writing down vague, cognitively disconnected concepts that teach the how without ever establishing the why. This portfolio takes the opposite approach: it moves slowly through the why before arriving at the how.

To make that work, I settled on a study technique I call **ECRA**, which breaks studying into four phases: **E**xploration, **C**ognition, **R**eflection and **A**pplication. Exploration is the informal, beginner's first contact with a concept (a video, a search) to build intuition. Cognition is where things get formal, with definitions and derivations. Reflection is where I step back and interrogate what actually landed and what didn't. Application is where the concept gets used, tested or simulated. Chapters follow this structure wherever it applies.

I have not approached this portfolio like a textbook, even though it covers essentially all the mathematics needed for a first course in GLMs. Instead, the structure follows the shape of the problem itself: try something, watch it fail, ask why it failed, then build the solution that addresses that specific failure. Each chapter is a pathway from problem to solution rather than a definition waiting to be memorised.

By the end of this portfolio, a reader should understand why we speak about generalised linear models at all, the mathematics underlying them, how to fit a GLM with code in R, and how to judge variable importance.

The class tests written throughout the semester (Class Tests 1 to 4) are attached in the appendix.

Without further ado, let's dive into the world of GLMs.

PART I.

Introduction to GLMs

CHAPTER 1

OLS

Exploration

At this point in the chapter I approach OLS as a total beginner trying to explore the concept of Ordinary Least Squares. To start, the best way to approach any exploration of new concepts is to research using YouTube University, so I searched "OLS" on YouTube and got this.

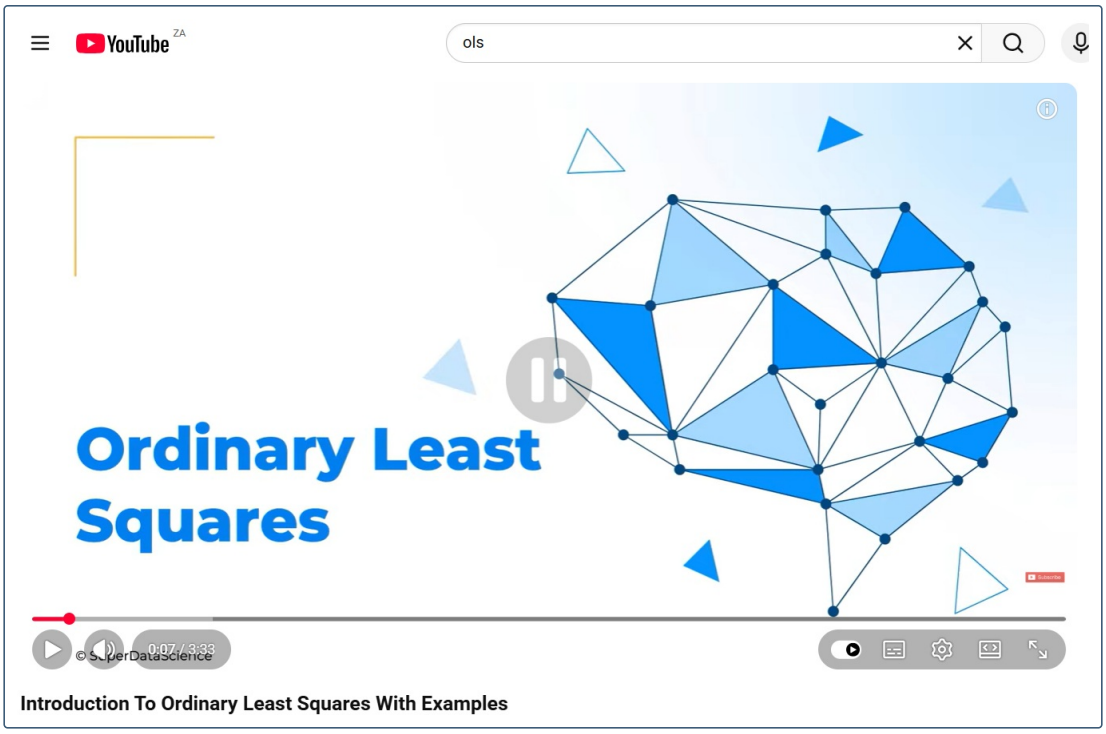


Figure 1.1.: OLS video from YouTube (Super Data Science, 2023).

Source: Super Data Science (2023), *Introduction to Ordinary Least Squares with Examples*, YouTube.

So what is "Ordinary Least Squares"?
This image is a practical example of OLS (linear regression) showing the line of best fit, model accuracy, and sum of squares (minimised).

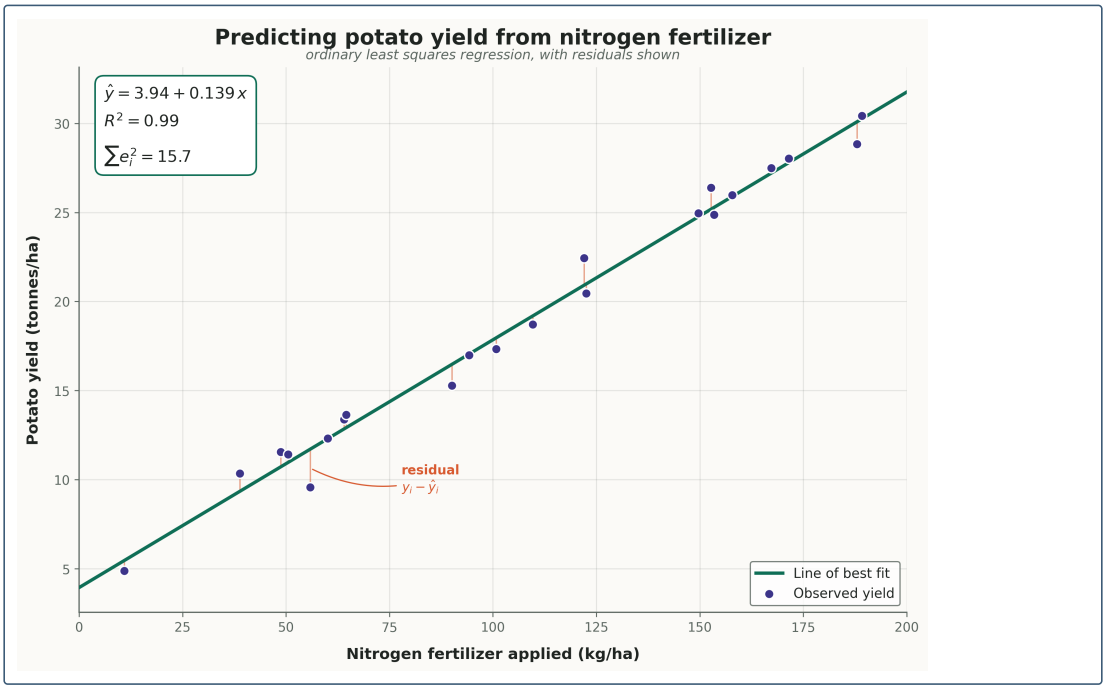


Figure 1.2.: A practical example of OLS (linear regression): the line of best fit, model accuracy, and the (minimised) sum of squares.

We move further to the Cognition phase, where things get more formal and definitions and some fun maths appear.

Cognition

Linear Regression

A model used in statistics to assess the connection between a scalar response (dependent variable) and one or more explanatory factors (independent variables or regressors) is called linear regression.

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Simple Linear Regression

A linear regression with just one explanatory variable. It is a clear system with no confounding, where cause is most likely linked directly to effect.

Multiple Linear Regression

A linear regression with two or more explanatory variables. It is more interesting and more prone to confounding, since cause doesn't always mean effect; it could be a combination of causes, or a cause could appear prominent only because of another variable.

To see why the exponential family occupies such a central place in GLM theory, it helps to start with the simple linear model.

The Linear Model as a Starting Point

The simple linear model is written

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where ϵ_i is an error term assumed to be normally distributed with mean 0 and constant variance σ^2, that is $\epsilon_i \sim N(0, \sigma^2)$. Because Y_i is a constant $(\beta_0 + \beta_1 x_i)$ plus a normally distributed random variable, Y_i itself is normally distributed:

$$Y_i \sim N(\mu_i, \sigma^2), \qquad \mu_i = \beta_0 + \beta_1 x_i$$

In other words, the linear combination of the explanatory variable and its coefficient does not directly define Y_i, it defines where the mean of Y_i sits. The error term provides the randomness and its distributional form, while the linear predictor $\beta_0 + \beta_1 x_i$ simply drags that mean to where the covariate x_i indicates it should be. This separation, a distributional assumption on one side and a linear predictor governing the mean on the other, is precisely what GLMs later generalise.

Extending to Multiple Explanatory Variables and Vector Responses

The same logic extends naturally once there is more than one explanatory variable. Instead of a single x_i, we have a row of covariates for each observation, and the mean becomes

$$\mu_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} = x_i^\top \beta$$

with $Y_i \sim N(\mu_i, \sigma^2)$ still holding, only now μ_i is a linear combination of several predictors rather than one. Written for the whole sample at once, this is $Y = X\beta + \epsilon$, with X the design matrix and β the vector of coefficients.

Where the Linear Model Falls Short

The assumptions underpinning the linear model, constant variance, a normally distributed response, and a mean that varies linearly with the predictors, do not hold for a great deal of real data. Counts of events, binary outcomes (success/failure), and proportions are common, none of which are well described by a normal distribution with constant variance. A count cannot be negative, and its variance tends to grow with its mean; a binary outcome only takes two values; a proportion is bounded between 0 and 1.

There is also the question of the relationship between the response mean and the linear predictor. In the classical linear model that relationship is direct: μ_i literally equals the linear combination of covariates. But for many response types above, a straight linear relationship on the original scale does not make sense; for a binary outcome, what varies linearly with the covariates is more naturally the log-odds of success (the logit) rather than the probability itself. Historically, fitting anything outside the normal linear model was difficult because closed-form estimators generally do not exist once the response distribution departs from normality or the mean is connected to the predictors through a nonlinear transformation. Iterative fitting procedures, developed to extend maximum likelihood estimation into settings without closed-form solutions, changed this: estimation still proceeds via an iteratively reweighted least-squares computation, echoing the linear model even as the distributional assumptions underlying it change entirely.

Deriving the Normal Equations by Hand

For simple linear regression, we are fitting $\hat{y}_i = \beta_0 + \beta_1 x_i$, with residual $e_i = y_i - \hat{y}_i$. Residuals are not summed directly, since positive and negative errors would cancel; instead the sum of squared residuals is minimised,

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

Theorem 1.1 (Normal Equations). *Differentiating $S(\beta_0, \beta_1)$ with respect to each parameter and setting both to zero gives the normal equations (Dobson and Barnett, 2018; Casella and Berger, 2002):*

$$\frac{\partial S}{\partial \beta_0} = -2 \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i) = 0, \qquad \frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^{n} x_i (y_i - \beta_0 - \beta_1 x_i) = 0$$

Expanding the first and solving for β_0 gives $\beta_0 = \bar{y} - \beta_1 \bar{x}$, which also shows the fitted line always passes through $(\bar{x}, \bar{y})$. Substituting this into the second equation and simplifying gives the slope,

$$\beta_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

the covariance of x and y divided by the variance of x. Because $S(\beta_0, \beta_1)$ is a sum of squares, it is convex, so the second-order conditions are automatically satisfied and this stationary point is guaranteed to be a minimum.

Matrix Form of OLS

The scalar derivation generalises directly to any number of predictors. Stacking the data gives $Y = X\beta + \epsilon$, with fitted values $\hat{Y} = X\beta$ and residual vector $e = Y - X\beta$. The sum of squared residuals is

$$S(\beta) = e^\top e = (Y - X\beta)^\top (Y - X\beta) = Y^\top Y - 2\beta^\top X^\top Y + \beta^\top X^\top X \beta$$

Differentiating with respect to β using the standard identities $\partial(\beta^\top X^\top Y)/\partial\beta = X^\top Y$ and $\partial(\beta^\top X^\top X\beta)/\partial\beta = 2X^\top X\beta$ (Casella and Berger, 2002), and setting the result to zero, gives the normal equations in matrix form, $X^\top X\beta = X^\top Y$. Provided $X^\top X$ is invertible,

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y$$

which is exactly the closed-form estimator introduced as a term in Chapter 3, now derived rather than quoted. The Hessian $2X^\top X$ is positive semi-definite by construction, confirming this stationary point is a minimum (Dobson and Barnett, 2018). Rearranging $X^\top X\hat{\beta} = X^\top Y$ also gives $X^\top(Y - X\hat{\beta}) = 0$, i.e. $X^\top e = 0$: the residual vector is orthogonal to every column of X, the orthogonality condition referenced under the Transposed Matrix term.

Reflection **Application**

Looking back on the process of learning Ordinary Least Squares, I can say with confidence that I now understand both the intuition behind it and the mathematics underpinning it. Two questions arose along the way and were already answered, in passing, during the Cognition phase. Naming a question clearly is often what turns a passively absorbed derivation into a properly internalised one.

Question: Why do we need models at all?

A model is a representation of a real-world process, constructed so that inferences can be made about the future behaviour of that process without repeating the underlying experiment every time. An experiment cannot be run on a future outcome, since that outcome does not yet exist empirically; data collection is also costly, so it would be far from optimal to repeat an experiment in full each time a new inference is required. The most intuitive model of all is a straight line, the line of best fit, from which it becomes possible both to predict future outcomes and to quantify how wrong those predictions are likely to be.

Question: Why are the residuals squared before being summed?

Each residual measures how wrong a single prediction is, and each residual carries a sign. Summing the residuals directly would allow positive and negative errors to cancel, leaving only the net direction in which predictions err rather than their overall magnitude. Squaring removes the sign while preserving, and emphasising, the size of the deviation. The resulting sum of squared errors is the quantity Ordinary Least Squares minimises.

To put the theory into practice, a dataset was simulated in R to echo the exploration-phase example: nitrogen fertilizer applied (kg/ha) as the predictor, and potato yield (tonnes/ha) as the response, generated from the true relationship $y = 3.94 + 0.139x + \epsilon$ with $\epsilon \sim N(0, 0.55^2)$, so that the simulated data would recover a fit close to the original figure while remaining a fresh, reproducible dataset in its own right.

Listing 1.1: Simulating the dataset and fitting it with lm()

```r
set.seed(2026)
n <- 25

# Nitrogen fertilizer applied, kg/ha
nitrogen <- round(runif(n, min = 10, max = 190), 1)

# True relationship (matches the exploration-phase figure)
beta0_true <- 3.94
beta1_true <- 0.139
sigma <- 0.55

yield <- beta0_true + beta1_true * nitrogen + rnorm(n, mean = 0, sd = sigma)
yield <- round(pmax(yield, 0), 2)

potato_yield_data <- data.frame(
  nitrogen_kg_ha = nitrogen,
  yield_tonnes_ha = yield
)

fit <- lm(yield_tonnes_ha ~ nitrogen_kg_ha, data = potato_yield_data)
summary(fit)
```

Listing 1.2: Output of summary(fit)

```
Coefficients:
```

```
                Estimate Std. Error t value Pr(>|t|)
(Intercept)    3.864663   0.203246   19.02  1.45e-15 ***
nitrogen_kg_ha 0.137992   0.002323   59.40  < 2e-16 ***

Residual standard error: 0.5296 on 23 degrees of freedom
Multiple R-squared: 0.9935, Adjusted R-squared: 0.9932
F-statistic: 3529 on 1 and 23 DF, p-value: < 2.2e-16
```

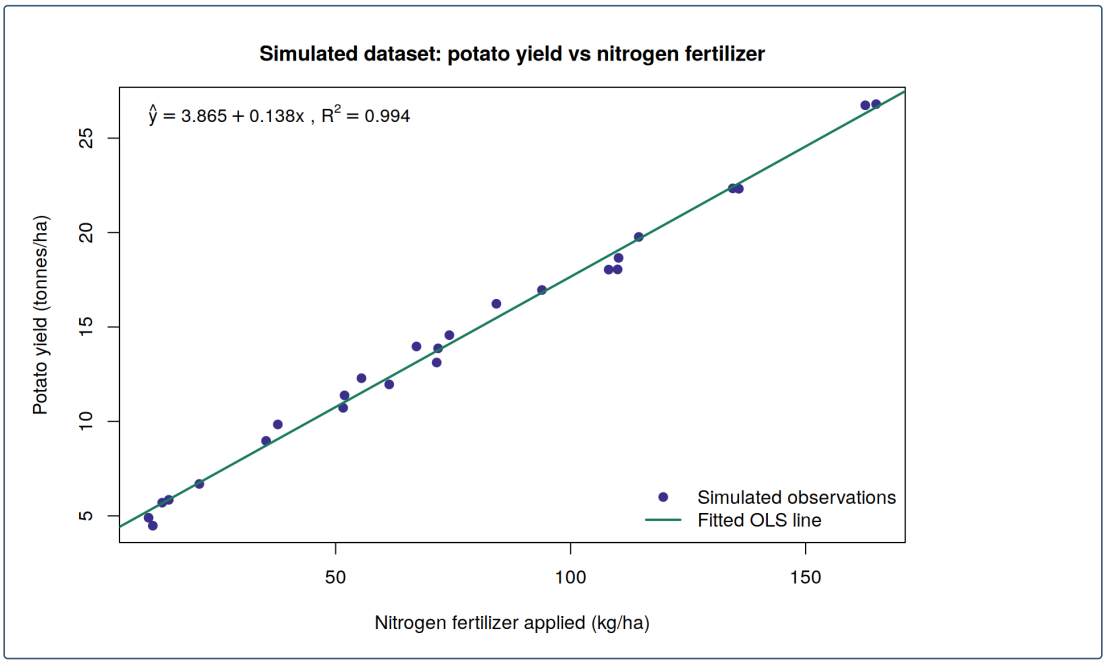


Figure 1.3.: Simulated nitrogen-yield dataset with the fitted OLS line.

The fitted line, $\hat{y} = 3.865 + 0.138x$ with $R^2 = 0.994$, lands almost exactly on the $\hat{y} = 3.94 + 0.139x$, $R^2 = 0.99$ seen in the exploration-phase figure, confirming that the by-hand normal equations, the matrix form $\hat{\beta} = (X^\top X)^{-1} X^\top Y$, and R's own `lm()` solver all converge on the same answer.

CHAPTER 2

Different Types of Response Variables

Exploration

To get a feel for this topic before formalising anything, I watched a video on different types of response variables on YouTube.

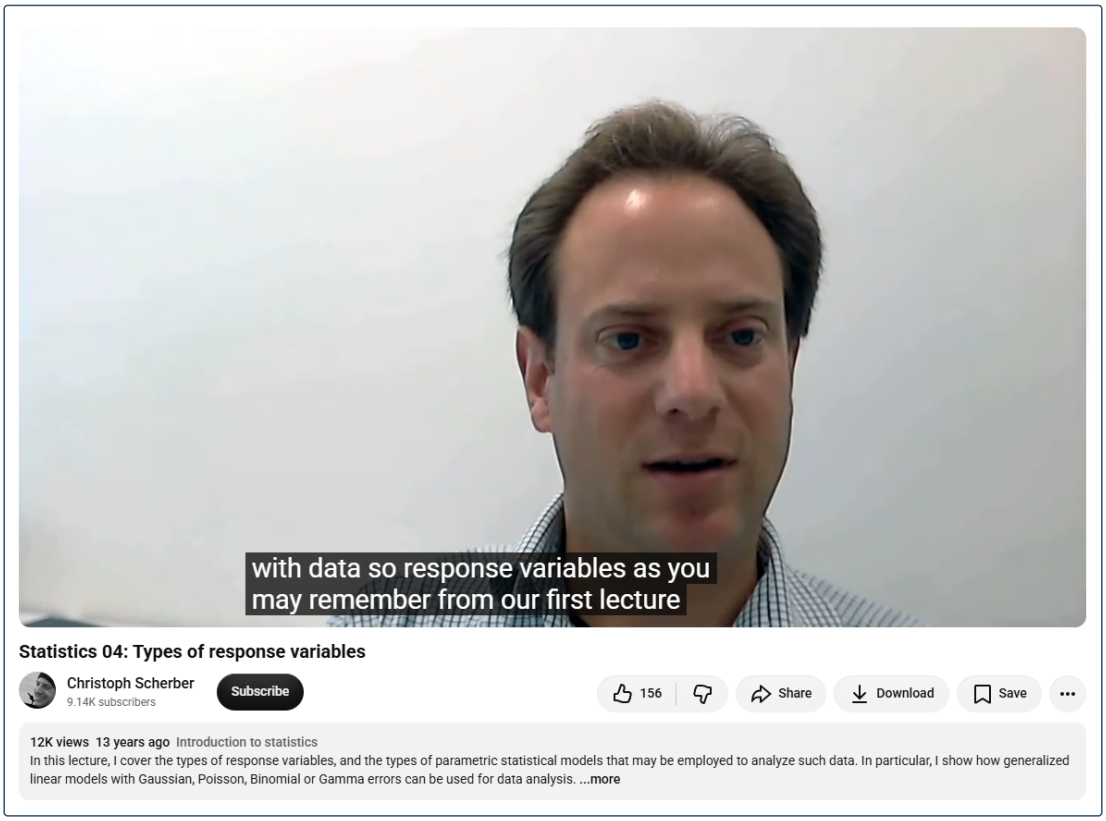


Figure 2.1.: Types of response variables, from a YouTube lecture (Scherber, nd).
Source: Scherber, C., *Statistics 04: Types of Response Variables*, YouTube.

From this video I picked up a few things about response variables:

1. A response variable is what we usually put on the y-axis, and it's dependent on the explanatory variables on the x-axis.
2. A response variable is the main output we're interested in, the main reason the model was built in the first place.
3. Since it's our aim, we can aim to do many different things with it: measure, count, partition. Because of this, we end up with different types of response variables, each carrying its own distribution.

We move on to the Cognition phase, where these intuitions get sorted into a proper table.

Cognition

Now, to build solid cognitive pillars on top of that intuitive background, the table below sets out the different response variables and the distributions that typically go with them.

Outcome	Distribution	Typical Link	Form of η
Continuous, symmetric, can be positive or negative	Normal	Identity	μ
Binary (yes/no), proportion out of n trials	Binomial (Bernoulli)	Logit	$\log\left(\frac{\pi}{1-\pi}\right)$
Count, no fixed upper limit	Poisson	Log	$\log(\mu)$
Count, overdispersed (Var $\gg$ Mean)	Negative Binomial	Log	$\log(\mu)$
Continuous, positive, right-skewed	Gamma	Log or Inverse	$\log(\mu)$

Table 2.1.: Common response variable types, their typical distributions, and typical link functions.

η is the same idea no matter which distribution is in play,

$$\eta = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k$$

It's just the linear combination of explanatory variables. The link function is what connects η to the mean of the outcome's distribution.

Reflection

So, to reflect on this, let's work through a couple of scenarios and decide, each time, what distribution the response follows, then write out the linear component and link function to get some firsthand decision-making experience.

Scenario 1: Loan Default

A bank wants to predict whether a customer defaults on a loan (yes/no), using annual income, credit score, and number of existing loans.

- **Outcome (Y):** Default (1 = yes, 0 = no).
- **Explanatory (X's):** Income, credit score, number of existing loans.
- **Distribution:** Binomial (Bernoulli, one trial per customer).
- **Link:** Logit.

$$\eta = \beta_0 + \beta_1(\text{income}) + \beta_2(\text{credit score}) + \beta_3(\text{number of loans})$$

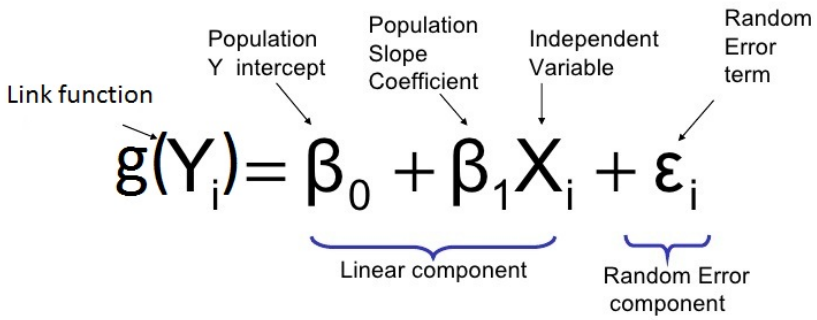


Figure 2.2.: How a link function sits inside the linear predictor (Chandan, nd).

Scenario 2: Traffic Accidents

- **Outcome (Y):** Number of accidents per month at an intersection (count, no natural cap).
- **Explanatory (X's):** Traffic volume, traffic light (yes/no), speed limit.
- **Distribution:** Poisson.
- **Link:** Log.

$$\eta = \beta_0 + \beta_1(\text{traffic volume}) + \beta_2(\text{traffic light}) + \beta_3(\text{speed limit})$$

Question: Why do we bother with different types of response variables?

As usual, to get at the real reason we generalise linear models, it helps to go back to why we make models at all: to get some outcome. In the simple linear model that outcome is usually just a continuous real number, typically unbounded except in special cases. But suppose we want to make inference on the proportion of rabbits with a certain deformity. Our outcome is now a binomial response, since each rabbit either has the deformity or doesn't, but what we actually want is the percentage of rabbits that get it. We still want to use a linear model to describe this, but for that to work we need to generalise the linear model using a link function, one that connects our binary outcomes to a linear predictor that gives back a continuous real number.

CHAPTER 3

The Language of GLMs

Exploration

If GLMs were a demographic of people, they would have their own language. But to understand the language of these people, we first need to understand why we want to speak to them in the first place: what they eat, what they stand for, and their purpose in the grand scheme of things. Okay, enough of the analogy, before we dive into the boring definitions, let's take a moment to examine the actual aim of GLMs.

Aim: As simply as possible, the aim of GLMs is to know how a bunch of X's affect a Y. That's the simplest definition.

Let's step into a time machine. This time machine takes us to 1972, when John Nelder and Robert Wedderburn formulated a framework for unifying various other statistical models, e.g. linear regression, logistic regression, and Poisson regression (Nelder and Wedderburn, 1972), along with a neat solution for maximum likelihood estimation known as the iteratively reweighted least squares method.

As we have already established intuitively, OLS predicts the expected value of a random variable Y as a linear combination of a set of observed values (a "bunch of X's"), also known as predictors. We can make a few assumptions based on this:

1. A constant change in a predictor leads to a constant change in the response.
2. In the case of OLS, these assumptions hold when the response can vary indefinitely in both directions, i.e. it follows the pattern of a normal distribution.
3. However, when the response doesn't follow this pattern, we can use transformations to make it follow that pattern.
4. When these transformations are not sufficient, we employ the use of GLMs.



Figure 3.1.: Components of GLMs, from a YouTube lecture (Wrobel, 2021).

Source: Wrobel, J. (2021, January 25), *L1.3 Components of GLMs*, YouTube.
Quick summary of this framework from a YouTube video I watched, before we dive into more definitions: assume that a known function of μ_i is related linearly to covariates,

$$g(\mu_i) = \eta_i = x_i^\top \beta$$

Then:

1. g = link function: links the linear predictor with the mean of the outcome.
2. We assume independence of Y_i.
3. The linearity assumption now applies to η_i, which need not equal μ_i.
4. X is treated as fixed.

Cognition

This chapter collects, in one place, the vocabulary used throughout the rest of the portfolio. Later chapters build on these terms without redefining them.

Generalised Linear Model

A generalised linear model is a generalisation of linear regression that is more versatile. The GLM generalises linear regression by connecting the linear model to the response variable via a link function, and by permitting the variance of each measurement to be a function of its predicted value (Nelder and Wedderburn, 1972; McCullagh and Nelder, 1989).

Log-linear Model

A log-linear model is a mathematical model that allows for the application of potential multivariate linear regression by taking the form of a function whose logarithm equals a linear combination of the model's parameters:

$$\exp\left\{c + \sum_i w_i f_i(X)\right\}$$

where $f_i(X)$ are functions of the vector of values X, and c and w_i are model parameters. Taking the natural log of this gives a linear combination that can be worked with directly (Agresti, 2015).

Definition: Exponential Family

An exponential family is a parametric set of probability distributions with the form

$$f_X(x \mid \theta) = h(x) \exp\left\{\eta(\theta) \cdot T(x) - A(\theta)\right\}$$

where $h(x)$ is non-negative and known, and $\eta(\theta)$, $T(x)$, $A(\theta)$ are known as well (McCullagh and Nelder, 1989; Wikipedia, 2026).

Definition: One-Parameter Exponential Family

A one-parameter exponential family is a special case of the exponential family in which the distribution depends on a single scalar parameter θ, and its probability density (or mass) function can be written

$$f(y;\theta) = \exp\{a(y)\,b(\theta) + c(\theta) + d(y)\}$$

where $a(y)$ is the sufficient statistic, $b(\theta)$ is the natural (canonical) parameter, $c(\theta)$ is the log-partition function ensuring the density normalises to one, and $d(y)$ is the base measure, a term depending on y alone. This form underlies several of the standard distributions used in GLMs, such as the Binomial and Poisson, for which the dispersion parameter is fixed at 1 (Taylor, nd).

Link Function

A link function links the predicted mean of the response variable μ to the linear predictor η in a GLM:

$$g(\mu) = \eta = X\beta = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n, \qquad \mu = g^{-1}(X\beta)$$

When g is chosen such that the linear predictor equals the natural parameter of the distribution, it is called a canonical link function for that distribution, e.g. the logit link for the Binomial, or the log link for the Poisson (Dobson and Barnett, 2018).

Transposed Matrix

For any matrix A with dimensions $m \times n$, the transpose $A^\top$ is the $n \times m$ matrix obtained by swapping its rows and columns: $(A^\top)_{ij} = A_{ji}$. Transposition is central to GLM estimation: for a design matrix X and coefficient vector β, the normal equations used to estimate β take the form $\hat{\beta} = (X^\top X)^{-1} X^\top y$, reflecting the orthogonality condition $X^\top(y - \mu(\beta)) = 0$ that arises from likelihood theory in GLMs (Casella and Berger, 2002).

Transposed Vector

A transposed vector is a vector whose rows and columns have been flipped, so a column vector becomes a row vector or vice versa. For a column vector of covariates x_i and column vector of coefficients β, $x_i^\top \beta = x_{i1}\beta_1 + x_{i2}\beta_2 + \cdots + x_{ip}\beta_p$, which is precisely the linear predictor η_i, a single scalar value (Casella and Berger, 2002).

Linear Predictor

Denoted $X\beta$, this is the linear combination of covariates and regression coefficients that links the explanatory variables to the response distribution's mean through the link function (Dobson and Barnett, 2018).

Parameters

Quantities that characterise a probability distribution or a statistical model, whose values determine its shape, location, or scale. In a GLM, parameters include the regression coefficients

(β), the dispersion parameter (ϕ), and any other parameter implicit in the chosen distribution (Casella and Berger, 2002).

Sufficiency

A property of a statistic whereby it captures all the information in a sample relevant to estimating a parameter, such that no other statistic calculated from the same sample provides additional information about that parameter (Casella and Berger, 2002).

Sufficient Statistic $a(y)$

The statistic $a(y)$ that satisfies the property of sufficiency for a parameter. In exponential family notation, it is the function of the data multiplied by the natural parameter in the exponent. In many cases $a(y) = y$, and the distribution is then said to be in canonical form (Nowak, nd, p. 2).

Natural Parameter $b(\theta)$

Also called the canonical parameter, this is the parameter that appears in the exponent of a distribution written in exponential family form. It is the parameter through which the linear predictor connects to the distribution when a canonical link function is used (McCullagh and Nelder, 1989).

Log-partition Function $c(\theta)$

Also called the cumulant function or normalising function, this ensures the exponential family density integrates (or sums) to one. Its derivatives generate the moments of the sufficient statistic: the first derivative gives the mean, the second gives the variance (McCullagh and Nelder, 1989).

Base Measure $d(y)$

The component of the exponent that depends on y alone, independent of θ. It functions as a normalising term ensuring the density integrates to 1 (McCullagh and Nelder, 1989).

Dispersion Parameter

A parameter, denoted ϕ, that scales the variance of the response distribution independently of its mean. It reflects variability beyond what is explained by the mean-variance relationship of the chosen distribution. In the one-parameter exponential family form the dispersion parameter is fixed at 1 (Dobson and Barnett, 2018).

Score Function

The first derivative of the log-likelihood function with respect to the parameter(s) of interest. Setting the score function equal to zero and solving yields the maximum likelihood estimate (Casella and Berger, 2002).

Maximum Likelihood Estimates

The parameter values that maximise the likelihood function, i.e. the values that make the observed data most probable under the assumed model. In GLMs, MLEs for β are typically obtained via iterative methods such as Iteratively Reweighted Least Squares (IRLS), since closed-form solutions rarely exist outside the Gaussian case (Casella and Berger, 2002).

Canonical Form

A distribution is said to be in canonical form when its sufficient statistic $a(y)$ equals y itself, in which case the natural parameter is also called the canonical parameter (McCullagh and Nelder, 1989).

Logit

The canonical link function for the binomial distribution, defined as the log-odds:

$$g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$$

It maps probabilities bounded between 0 and 1 onto the entire real line, making it suitable for linear modelling (Dobson and Barnett, 2018).

Probit

A link function defined as the inverse of the standard normal cumulative distribution function, $g(\mu) = \Phi^{-1}(\mu)$. Commonly used for binary response models as an alternative to the logit link (Dobson and Barnett, 2018).

Nuisance Parameter

A parameter in a statistical model that is not of primary interest but must be accounted for in order to correctly estimate the parameters that are of interest. In many GLMs, the dispersion parameter is treated as a nuisance parameter (Casella and Berger, 2002).

PART II.

The Mathematics Behind GLMs

CHAPTER 4

The Exponential Family

Exploration

This video touches on two insights: first, the definition of what it means to belong to the exponential family; second, a demonstration that if a random variable has an exponential distribution, it belongs to the exponential family.

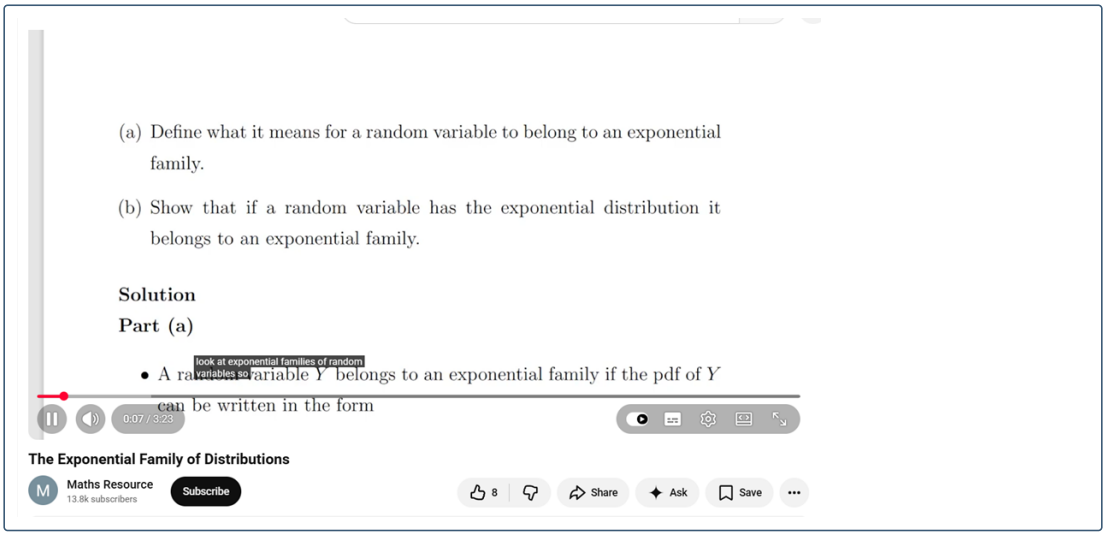


Figure 4.1.: Exponential Family of distributions, from a YouTube lecture (Maths Resource, 2022).

Source: Maths Resource. (2022, November 7), *The Exponential Family of Distributions*, YouTube.

Part A. A random variable Y belongs to an exponential family if the PDF of Y can be written in the form

$$f(y;\theta,\phi) = \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} - c(y;\phi)\right]$$

where a, b and c are functions.

Part B of the video shows that if a PDF of Y follows the exponential distribution, it can be written in exponential family form, nb: this doesn't mean it is the only distribution that can be written in this form, as we will investigate in the Cognition part of this chapter. So suppose the parameter of the exponential distribution Y is λ. Then,

$$f(y) = \lambda\exp(-\lambda y) = \exp[\log\lambda] \times \exp[-\lambda y] = \exp[\log\lambda - \lambda y] = \exp\left[\frac{\lambda y - \log\lambda}{-1}\right]$$

so that

$$\theta = \lambda, \qquad a(\phi) = -1, \qquad b(\theta) = \log\theta, \qquad c(y;\phi) = 0$$

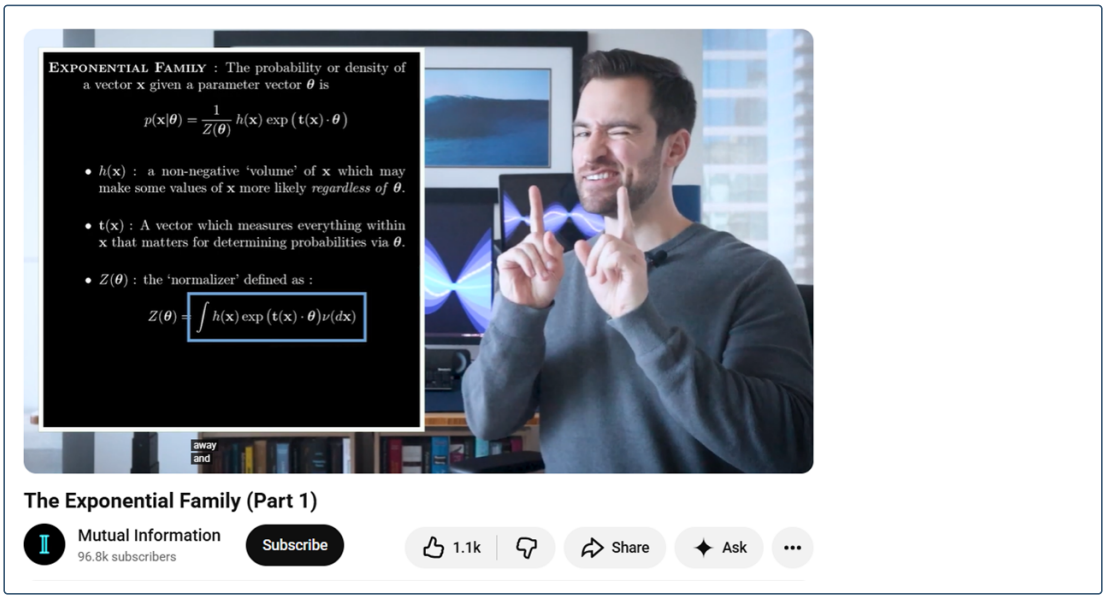


Figure 4.2.: The Exponential Family (Part 1), from a YouTube lecture (Mutual Information, 2021).

Source: Mutual Information. (2021, July 28), *The Exponential Family (Part 1)*, YouTube.

Notes from this video: a generalisation of most distributions, but it's more than just a generalisation, it also gives us properties which help MLE and other things, including GLMs in general.

Focusing on the generalisation of the exponential family: let's look at a random scenario featuring the Normal distribution. We get data points that look like a bell curve, but to get the exact distribution of that data we need parameters that maximize the likelihood of our data,

$$\mu^*, \sigma^* = \arg\max_{\mu,\sigma} \prod_{i=1}^{N} N(x_i \mid \mu, \sigma), \qquad N(x_i \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Next we rewrite this so that we separate expressions involving only parameters from expressions involving only observed values of x:

$$\frac{1}{\sigma\sqrt{2\pi}\exp\left(\frac{\mu^2}{2\sigma^2}\right)} \exp\left(\begin{bmatrix} x_i & x_i^2 \end{bmatrix} \begin{bmatrix} \mu/\sigma^2 \\ -1/2\sigma^2 \end{bmatrix}\right)$$

where the leading fraction is all params that do not multiply with the observed x, and the bracketed expression is a dot product of two vectors: ① depends only on x values, ② depends only on parameters. In summary, we are separating parameters from observed data.

We can rewrite the parameter vector in the form θ_1 and θ_2:

$$\frac{1}{Z(\theta_1, \theta_2)} \exp\left(\begin{bmatrix} x_i & x_i^2 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}\right)$$

So then we find the MLE of θ_1 and θ_2:

$$\theta_1^*, \theta_2^* = \arg\max_{\theta_1,\theta_2} \prod_{i=1}^{N} N(x_i \mid \theta_1, \theta_2)$$

So now we can generalize and say: if the probability density of a vector x given a parameter vector θ is

$$p(x \mid \theta) = \frac{1}{Z(\theta)} h(x) \exp\left(t(x) \cdot \theta\right)$$

then this distribution is of the exponential family, where:

- $h(x)$: non-negative "volume" of x which may make some values of x more likely regardless of θ; it does not depend on parameters.
- $t(x)$: a vector which measures everything within x that matters for determining probabilities via θ (sufficient statistic). When $h(x) = 1$, $t(x)$ is fully sufficient; if not, it is still sufficient with respect to the parameters.
- $Z(\theta)$: normalising function, defined as $Z(\theta) = \int h(x) \exp(t(x) \cdot \theta)\, \nu(dx)$; it ensures the distribution integrates to 1.

NB: if we have parameters where $Z(\theta) = \infty$, we discard those, we only pay attention to finite results for $Z(\theta)$.

Cognition

4.1 Definition and Properties

A probability distribution belongs to the exponential family if its density (or mass) function can be written in the general form

$$f_X(x \mid \theta) = h(x) \exp\left\{\eta(\theta) \cdot T(x) - A(\theta)\right\}$$

Here $h(x)$ is a known, non-negative function of the data alone, while $\eta(\theta)$, $T(x)$ and $A(\theta)$ are known functions governing the natural parameter, the sufficient statistic, and the normalising term respectively (McCullagh and Nelder, 1989). In this general version θ may be a vector and $T(x)$ a vector of sufficient statistics, so this is sometimes called the multi-parameter or full exponential family.
For GLMs, though, what matters is the one-parameter exponential family, since this is the restricted form underlying the standard GLM distributions: Binomial, Poisson, Normal, and Gamma. Restricting θ to a scalar and writing the exponent as a sum rather than a dot product gives the working form used throughout this portfolio:

$$f(y; \theta) = \exp\left\{a(y)\, b(\theta) + c(\theta) + d(y)\right\}$$

Matching the terms defined in Part I:

- $a(y)$ is the sufficient statistic: the function of the data that gets multiplied by the natural parameter.

- $b(\theta)$ is the natural (canonical) parameter, the function of θ through which the linear predictor will eventually connect to the distribution.
- $c(\theta)$ is the log-partition function, which guarantees $f(y;\theta)$ integrates (or sums) to one over its support.
- $d(y)$ is the base measure, the leftover piece depending on y alone, with no θ in it.

What makes this form useful is the separation it forces between data and parameter. y and θ never appear multiplied together except through the product $a(y)\,b(\theta)$; everything else involving y sits in $d(y)$, and everything else involving θ sits in $c(\theta)$. This separation is also what makes $a(y)$ sufficient in the first place: once $a(y)$ is known, nothing more about y is needed to work out how the likelihood depends on θ.
A distribution is in canonical form when $a(y) = y$. In that case $b(\theta)$ is called the canonical parameter, and, as covered later when link functions are discussed, the link function that sets the linear predictor equal to $b(\theta)$ directly is the canonical link for that distribution.
$c(\theta)$ is worth dwelling on a little longer, because it isn't really a free choice, it's forced by the requirement that $f(y;\theta)$ be a valid density. Given $a(y)$, $b(\theta)$ and $d(y)$, normalisation requires

$$\int \exp\{a(y)\,b(\theta) + d(y)\}\,dy = \exp\{-c(\theta)\}$$

(or the equivalent sum in the discrete case), so $c(\theta)$ ends up completely determined by the rest of the exponent. This is also why $c(\theta)$ is called the cumulant function: differentiating it with respect to θ generates the moments of $a(y)$. Under the canonical parameterisation, $\mathbb{E}[a(Y)] = -c'(\theta)$ and $\mathrm{Var}[a(Y)] = -c''(\theta)$, a fact that becomes useful once the mean-variance relationships for individual GLM distributions are worked out further on.

Proof: $c(\theta)$ as the Cumulant (Normalising) Function

The claim above, that $c(\theta)$ is forced by normalisation rather than being a free choice, is not asserted without justification; it follows directly from requiring $f(y;\theta)$ to be a valid density. The derivation below is a proof of that result.
Since $f(y;\theta)$ is a probability density (or mass) function, it must integrate, or sum in the discrete case, to 1:

$$\int f(y;\theta)\,dy = 1$$

Substituting the one-parameter exponential family form gives

$$\int \exp\{a(y)b(\theta) + c(\theta) + d(y)\}\,dy = 1$$

Since $c(\theta)$ does not depend on y, it can be factored out of the integral using $\exp\{A+B\} = \exp\{A\}\cdot\exp\{B\}$:

$$\exp\{c(\theta)\}\int \exp\{a(y)b(\theta) + d(y)\}\,dy = 1$$

Dividing both sides by the remaining integral isolates $\exp\{c(\theta)\}$:

$$\exp\{c(\theta)\} = \left(\int \exp\{a(y)b(\theta) + d(y)\}\, dy \right)^{-1}$$

Taking the logarithm of both sides,

$$\log\left[\exp\{c(\theta)\}\right] = \log\left[\left(\int \exp\{a(y)b(\theta) + d(y)\}\, dy \right)^{-1} \right]$$

which simplifies, using $\log(x^{-1}) = -\log(x)$, to

$$\therefore\ c(\theta) = -\log \int \exp\{a(y)b(\theta) + d(y)\}\, dy$$

Equivalently, moving the negative sign across gives

$$-c(\theta) = \log \int \exp\{a(y)b(\theta) + d(y)\}\, dy$$

and exponentiating both sides recovers the normalisation identity stated above:

$$\exp\{-c(\theta)\} = \int \exp\{a(y)b(\theta) + d(y)\}\, dy$$

This confirms that, given $a(y)$, $b(\theta)$, and $d(y)$, the function $c(\theta)$ is not an independent modelling choice but is fully determined by the requirement that the density integrate to one. ■

4.2 Why the Exponential Family Matters

Toward a Unified Framework

Nelder and Wedderburn's insight, in 1972, was that a wide range of these non-normal response distributions, Binomial, Poisson, Gamma, and Normal among them, could all be written in the same one-parameter exponential family form introduced above. Once a distribution is expressed that way, the same general machinery (score equations, Fisher information, iteratively reweighted least squares) can be used to estimate its parameters, regardless of which member of the family is in play. The generalised linear model is precisely this: a linear predictor for the mean, exactly as in the classical linear model, combined with a chosen distribution from the exponential family and a link function that connects the mean of that distribution to the linear predictor, allowing the connection to be nonlinear (such as the log-odds link mentioned above) where the data demand it.

This is why the exponential family matters so much to GLM theory. It is not simply a convenient algebraic form, it is the structural bridge that lets the well-understood machinery of the linear model be reused, almost unchanged in spirit, across an entire class of response distributions that would otherwise each require their own separate theory (Nelder and Wedderburn, 1972; Dobson and Barnett, 2018).

4.3 Normal Distribution

The Normal distribution, also known as the Gaussian distribution, is a continuous probability distribution for a real-valued random variable. It occupies a central place in statistics, largely because of the Central Limit Theorem: the average of many statistically independent observations of a random variable, provided that variable has a finite mean and variance, converges to a normal distribution as the number of observations grows large. Consequently, physical quantities that arise as the sum of many independent processes, such as height or measurement error, tend to be approximately normally distributed (Casella and Berger, 2002).

Probability Density Function

$$f(y;\mu) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left\{-\frac{(y-\mu)^2}{2\sigma^2}\right\}$$

Rewriting in One-Parameter Exponential Form

Taking logs of both sides gives

$$\log f(y;\mu) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(y-\mu)^2}{2\sigma^2}$$

Expanding the square in the exponent separates the terms depending on y alone, μ alone, and both together:

$$\log f(y;\mu) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{y^2}{2\sigma^2} + \frac{y\mu}{\sigma^2} - \frac{\mu^2}{2\sigma^2}$$

Exponentiating both sides and grouping terms according to the one-parameter exponential family form $f(y;\theta) = \exp\{a(y)b(\theta) + c(\theta) + d(y)\}$ then gives

$$f(y;\mu) = \exp\left\{y \cdot \frac{\mu}{\sigma^2} - \frac{\mu^2}{2\sigma^2} - \left[\frac{1}{2}\log(2\pi\sigma^2) + \frac{y^2}{2\sigma^2}\right]\right\}$$

so that, with $\theta = \mu$,

$$a(y) = y, \quad b(\mu) = \frac{\mu}{\sigma^2}, \quad c(\mu) = -\frac{\mu^2}{2\sigma^2}, \quad d(y) = -\left[\frac{1}{2}\log(2\pi\sigma^2) + \frac{y^2}{2\sigma^2}\right]$$

Since $a(y) = y$, the Normal distribution is in canonical form. When the dispersion is treated as fixed, as in the one-parameter form used throughout this portfolio, so that $\sigma^2 = 1$, $b(\mu)$ reduces to μ itself, meaning the natural parameter coincides with the mean. The canonical link for the Normal distribution is therefore the identity link $g(\mu) = \mu$, which is consistent with how the mean is modelled directly in the classical linear model discussed in Part III.

Proof: Normalising Term $c(\mu)$

The value of $c(\mu)$ stated above is not an arbitrary choice; it follows from the general normalisation identity proved above, applied here with $a(y) = y$, $b(\mu) = \mu/\sigma^2$, and $d(y) = -\left[\frac{1}{2}\log(2\pi\sigma^2) + y^2/(2\sigma^2)\right]$.

Starting from the identity $\exp\{-c(\mu)\} = \int \exp\{a(y)b(\mu)+d(y)\}\,dy$ and substituting these terms gives

$$\exp\{-c(\mu)\} = \int \exp\left\{\frac{y\mu}{\sigma^2} - \frac{y^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)\right\} dy$$

Since the last term does not depend on y, it can be pulled outside the integral:

$$\exp\{-c(\mu)\} = \frac{1}{\sqrt{2\pi\sigma^2}} \int \exp\left\{\frac{y\mu}{\sigma^2} - \frac{y^2}{2\sigma^2}\right\} dy$$

Completing the square in the remaining exponent, since $\frac{y\mu}{\sigma^2} - \frac{y^2}{2\sigma^2} = -\frac{(y-\mu)^2}{2\sigma^2} + \frac{\mu^2}{2\sigma^2}$, gives

$$\exp\{-c(\mu)\} = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{\frac{\mu^2}{2\sigma^2}\right\} \int \exp\left\{-\frac{(y-\mu)^2}{2\sigma^2}\right\} dy$$

The remaining integral is the standard Gaussian integral, which evaluates to $\sqrt{2\pi\sigma^2}$, so this reduces to

$$\exp\{-c(\mu)\} = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left\{\frac{\mu^2}{2\sigma^2}\right\} \cdot \sqrt{2\pi\sigma^2} = \exp\left\{\frac{\mu^2}{2\sigma^2}\right\}$$

Equating exponents and solving for $c(\mu)$ therefore gives

$$-c(\mu) = \frac{\mu^2}{2\sigma^2} \implies \therefore\ c(\mu) = -\frac{\mu^2}{2\sigma^2} \quad \blacksquare$$

This confirms that $c(\mu)$ is forced by the same normalisation constraint established generally above, worked through here with the specific $a(y)$ and $d(y)$ that arise from the Normal density.

4.4 Binomial Distribution

The Binomial distribution is a discrete probability distribution for the number of successes in a sequence of n independent trials, each with a boolean outcome, success or failure, where success occurs with probability p and failure with probability $1-p$, since the two outcomes must sum to 1. The special case $n = 1$ is the Bernoulli distribution, so the Binomial can be thought of as a generalisation of a sequence of Bernoulli trials (Casella and Berger, 2002).

Probability Mass Function

$$f(y; n, p) = \binom{n}{y} p^y (1-p)^{n-y}, \qquad y = 0, 1, \ldots, n$$

Rewriting in One-Parameter Exponential Form

Taking logs of both sides gives

$$\log f(y;n,p) = \log \binom{n}{y} + y\log(p) + (n-y)\log(1-p)$$

Expanding the last term and collecting everything that multiplies y separates the exponent into a piece linear in y and a piece that does not depend on y at all:

$$\begin{aligned}\log f(y;n,p) &= y\log(p) - y\log(1-p) + n\log(1-p) + \log\binom{n}{y} \\ &= y\log\left(\frac{p}{1-p}\right) + n\log(1-p) + \log\binom{n}{y}\end{aligned}$$

Exponentiating both sides and grouping terms according to $f(y;\theta) = \exp\{a(y)b(\theta)+c(\theta)+d(y)\}$ then gives

$$f(y;n,p) = \exp\left\{y\log\left(\frac{p}{1-p}\right) + n\log(1-p) + \log\binom{n}{y}\right\}$$

so that, with $\theta = p$,

$$a(y) = y, \quad b(p) = \log\left(\frac{p}{1-p}\right), \quad c(p) = n\log(1-p), \quad d(y) = \log\binom{n}{y}$$

Since $a(y) = y$, the Binomial distribution is in canonical form, and $b(p)$ is its canonical parameter. Because $b(p) = \log\left(\frac{p}{1-p}\right)$ is exactly the logit function defined in Part II, the canonical link for the Binomial distribution is the logit link, which is why logistic regression models the log-odds of success as a linear predictor rather than the probability itself.

Proof: Normalising Term $c(p)$

The value of $c(p)$ stated above is not an arbitrary choice; it follows from the general normalisation identity proved above, applied here with $a(y) = y$, $b(p) = \log\left(\frac{p}{1-p}\right)$, and $d(y) = \log\binom{n}{y}$. Since the Binomial distribution is discrete, the identity holds with a sum in place of an integral. Starting from $\exp\{-c(p)\} = \sum_{y=0}^{n}\exp\{a(y)b(p) + d(y)\}$ and substituting these terms gives

$$\exp\{-n\log(1-p)\} = \sum_{y=0}^{n}\exp\left\{y\log\left(\frac{p}{1-p}\right) + \log\binom{n}{y}\right\}$$

Evaluating the exponent on the right, using $\exp\{A+B\} = \exp\{A\}\exp\{B\}$,

$$\exp\left\{y\log\left(\frac{p}{1-p}\right) + \log\binom{n}{y}\right\} = \exp\left\{y\log\left(\frac{p}{1-p}\right)\right\}\cdot\exp\left\{\log\binom{n}{y}\right\} = \left(\frac{p}{1-p}\right)^{y}\binom{n}{y}$$

so the right-hand side becomes

$$\sum_{y=0}^{n} \left(\frac{p}{1-p}\right)^{y} \binom{n}{y} = \left(1 + \frac{p}{1-p}\right)^{n}$$

by the Binomial theorem. Simplifying the base,

$$\left(1 + \frac{p}{1-p}\right)^{n} = \left(\frac{1-p+p}{1-p}\right)^{n} = \left(\frac{1}{1-p}\right)^{n} = (1-p)^{-n}$$

This matches the left-hand side exactly, since

$$\exp\{-n\log(1-p)\} = (1-p)^{-n}$$

Equating both sides and taking logs confirms the identity holds:

$$\log\left[\exp\{-n\log(1-p)\}\right] = \log\left[(1-p)^{-n}\right] \implies -n\log(1-p) = -n\log(1-p) \quad \blacksquare$$

This proves that $c(p) = n\log(1-p)$ is the function that makes $\sum_{y=0}^{n} f(y;n,p) = 1$.

4.5 Poisson Distribution

The Poisson distribution is a discrete probability distribution that expresses the probability of a number of events occurring in a fixed interval of time (or space), given that these events occur with a known constant mean rate and independently of the time since the last event (Casella and Berger, 2002).

Probability Mass Function

$$f(y;\lambda) = \frac{\lambda^{y} e^{-\lambda}}{y!}, \qquad \lambda = \text{rate parameter},\ y = 0, 1, 2, \ldots$$

Rewriting in One-Parameter Exponential Form

Taking logs of both sides gives

$$\log f(y;\lambda) = y\log(\lambda) - \lambda - \log(y!)$$

This is already separated into a piece linear in y, a piece depending on λ alone, and a piece depending on y alone, so exponentiating and grouping terms according to $f(y;\theta) = \exp\{a(y)b(\theta) + c(\theta) + d(y)\}$ gives

$$f(y;\lambda) = \exp\left\{y\log(\lambda) - \lambda - \log(y!)\right\}$$

so that, with $\theta = \lambda$,

$$a(y) = y, \quad b(\lambda) = \log(\lambda), \quad c(\lambda) = -\lambda, \quad d(y) = -\log(y!)$$

Since $a(y) = y$, the Poisson distribution is in canonical form, and $b(\lambda) = \log(\lambda)$ is its canonical parameter. The canonical link for the Poisson distribution is therefore the log link, $g(\mu) = \log(\mu)$, which is why Poisson regression models the log of the mean count as a linear predictor.

Proof: Normalising Term $c(\lambda)$

The value of $c(\lambda)$ stated above is not an arbitrary choice; it follows from the general normalisation identity proved above, applied here with $a(y) = y$, $b(\lambda) = \log(\lambda)$, and $d(y) = -\log(y!)$. Since the Poisson distribution is discrete with unbounded support $y = 0, 1, 2, \ldots$, the identity holds with an infinite sum in place of an integral.
Starting from $\exp\{-c(\lambda)\} = \sum_{y=0}^{\infty} \exp\{a(y)b(\lambda) + d(y)\}$ and substituting these terms gives

$$\exp\{-(-\lambda)\} = \sum_{y=0}^{\infty} \exp\{y \log(\lambda) - \log(y!)\}$$

Decomposing the exponent on the right using exponential laws,

$$\exp\{y \log(\lambda) - \log(y!)\} = \frac{\exp\{y \log(\lambda)\}}{\exp\{\log(y!)\}} = \frac{\lambda^y}{y!}$$

so the identity becomes

$$\exp\{\lambda\} = \sum_{y=0}^{\infty} \frac{\lambda^y}{y!}$$

The right-hand side is exactly the Maclaurin series expansion of the exponential function, which is known to equal e^{λ}:

$$\sum_{y=0}^{\infty} \frac{\lambda^y}{y!} = e^{\lambda} = \exp\{\lambda\}$$

Equating both sides confirms the identity holds,

$$\exp\{\lambda\} = \exp\{\lambda\} \quad \blacksquare$$

This proves that $c(\lambda) = -\lambda$ is the function that makes $\sum_{y=0}^{\infty} f(y; \lambda) = 1$.

4.6 Gamma Distribution

The Gamma distribution is a two-parameter family of continuous probability distributions with many applications across fields such as econometrics and Bayesian statistics (Casella and Berger, 2002). In econometrics, the (α, θ) shape-scale parameterisation is commonly used to model waiting times, such as time until failure, where it often takes the form of a special case of the Gamma distribution, the Erlang distribution, for integer α. In Bayesian statistics, the parameterisation takes the form (α, β), with $\beta = 1/\theta$ the rate parameter, and the Gamma distribution is widely used as a conjugate prior for several inverse scale parameters, which facilitates analytical tractability in posterior computations. In GLMs, it is more convenient to work

with a mean-shape parameterisation, writing the Gamma distribution in terms of its mean μ and shape ν, with ν treated as a nuisance parameter (Dobson and Barnett, 2018).

Probability Density Function

$$f(y;\mu) = \frac{1}{\Gamma(\nu)}\, y^{-1} \left(\frac{\nu y}{\mu}\right)^{\nu} \exp\left\{-\frac{\nu y}{\mu}\right\}$$

Rewriting in One-Parameter Exponential Form

Taking logs of both sides gives

$$\log f(y;\mu) = -\log\Gamma(\nu) - \log(y) + \nu\left[\log(\nu y) - \log(\mu)\right] - \frac{\nu y}{\mu}$$

Expanding $\log(\nu y) = \log(\nu) + \log(y)$ and collecting terms separates the exponent into a piece linear in y, a piece depending on μ alone, and a piece depending on y alone:

$$\log f(y;\mu) = -\frac{\nu y}{\mu} - \nu\log(\mu) + \left[(\nu-1)\log(y) + \nu\log(\nu) - \log\Gamma(\nu)\right]$$

Exponentiating both sides and grouping terms according to $f(y;\theta) = \exp\{a(y)b(\theta)+c(\theta)+d(y)\}$ then gives

$$f(y;\mu) = \exp\left\{y\cdot\left(-\frac{\nu}{\mu}\right) - \nu\log(\mu) + \left[(\nu-1)\log(y) + \nu\log(\nu) - \log\Gamma(\nu)\right]\right\}$$

so that, with $\theta = \mu$,

$$a(y) = y, \quad b(\mu) = -\frac{\nu}{\mu}, \quad c(\mu) = -\nu\log(\mu), \quad d(y) = (\nu-1)\log(y) + \nu\log(\nu) - \log\Gamma(\nu)$$

Since $a(y) = y$, the Gamma distribution is in canonical form, and $b(\mu) = -\nu/\mu$ is its canonical parameter. Because $b(\mu)$ is proportional to $-1/\mu$, the canonical link for the Gamma distribution is the negative reciprocal link, $g(\mu) = -1/\mu$, in contrast to the identity and log links seen for the Normal and Poisson distributions above.

Proof: Normalising Term $c(\mu)$

The value of $c(\mu)$ stated above is not an arbitrary choice; it follows from the general normalisation identity proved above, applied here with $a(y) = y$, $b(\mu) = -\nu/\mu$, and $d(y) = (\nu-1)\log(y) + \nu\log(\nu) - \log\Gamma(\nu)$.
Starting from $\exp\{-c(\mu)\} = \int_0^\infty \exp\{a(y)b(\mu) + d(y)\}\, dy$ and substituting these terms gives

$$\exp\{\nu\log(\mu)\} = \int_0^\infty \exp\left\{-\frac{\nu y}{\mu} + (\nu-1)\log(y) + \nu\log(\nu) - \log\Gamma(\nu)\right\} dy$$

Decomposing the exponent on the right using exponential laws separates out the part of the integrand that depends on y from the part that does not:

$$\exp\left\{-\frac{\nu y}{\mu}\right\} \times \exp\left\{(\nu-1)\log(y) + \nu\log(\nu) - \log\Gamma(\nu)\right\} = \exp\left\{-\frac{\nu y}{\mu}\right\} \times \frac{y^{\nu-1}\nu^{\nu}}{\Gamma(\nu)}$$

Plugging this back into the integral and pulling the constants $\nu^{\nu}/\Gamma(\nu)$, which do not depend on y, outside it,

$$\int_0^{\infty} \frac{\nu^{\nu}}{\Gamma(\nu)} y^{\nu-1} \exp\left\{-\frac{\nu y}{\mu}\right\} dy = \frac{\nu^{\nu}}{\Gamma(\nu)} \int_0^{\infty} y^{\nu-1} \exp\left\{-\frac{\nu y}{\mu}\right\} dy$$

The remaining integral is Euler's integral of the second kind, which states that for $a, b > 0$,

$$\int_0^{\infty} y^{a-1} e^{-by}\, dy = \frac{\Gamma(a)}{b^a}$$

Applying this with $a = \nu$ and $b = \nu/\mu$ gives

$$\frac{\nu^{\nu}}{\Gamma(\nu)} \int_0^{\infty} y^{\nu-1} \exp\left\{-\frac{\nu y}{\mu}\right\} dy = \frac{\nu^{\nu}}{\Gamma(\nu)} \cdot \frac{\Gamma(\nu)}{(\nu/\mu)^{\nu}} = \frac{1}{\mu^{-\nu}} = \mu^{\nu}$$

This matches the left-hand side exactly, since $\exp\{\nu\log(\mu)\} = \mu^{\nu}$. Equating both sides confirms the identity holds:

$$\exp\{\nu\log(\mu)\} = \mu^{\nu} \quad \blacksquare$$

This proves that $c(\mu) = -\nu\log(\mu)$ is the function that makes $\int_0^{\infty} f(y;\mu)\, dy = 1$.
Having built up the Normal, Binomial, Poisson, and Gamma distributions in one-parameter exponential family form, it is worth stepping back and asking what general properties fall out of that form itself, independent of which specific distribution is being used. Three results turn out to matter for everything that follows: the mean and variance of the sufficient statistic $a(Y)$ can be read off directly from $b(\theta)$ and $c(\theta)$ without ever touching an integral, and the score function, the quantity that drives maximum likelihood estimation, has an expected value of exactly zero at the true parameter.

4.7 Mean of $a(Y)$

Recall the one-parameter exponential family density from earlier in this chapter,

$$f(y;\theta) = \exp\{a(y)b(\theta) + c(\theta) + d(y)\}.$$

To find $\mathbb{E}(Y)$ directly would normally require evaluating $\sum_y yf(y)$ or $\int yf(y)\, dy$, which can be difficult or intractable depending on the distribution. The exponential family form lets this be sidestepped entirely by working with $a(Y)$ instead: since $a(Y)$ is a sufficient statistic for θ, it is the natural quantity to study, and when the distribution is in canonical form ($a(y) = y$), a general result for $\mathbb{E}[a(Y)]$ is at the same time a general result for $\mathbb{E}(Y)$.
Why not just differentiate $c(\theta)$? It is tempting to think that, since differentiating the normalising term $c(\theta)$ will turn out to produce the mean of $a(Y)$, this could be extracted and differentiated directly. The reason this shortcut does not work on its own is that the derivative of $c(\theta)$ alone

is only a function of θ; it says nothing by itself. It is only once it is placed back inside the full density, and the identity that every density must integrate to 1 is used, that a relationship between $c(\theta)$ and the moments of $a(Y)$ emerges. That identity is the actual engine of the derivation below.

Theorem 4.1 (Mean of the Sufficient Statistic)**.**

$$\mathbb{E}[a(Y)] = -\frac{c'(\theta)}{b'(\theta)}$$

Proof. Every valid density satisfies

$$\int f(y;\theta)\,dy = 1 \quad \text{for all } \theta.$$

Differentiating both sides with respect to θ, and, under standard regularity conditions, interchanging the order of differentiation and integration,

$$\int \frac{\partial f(y;\theta)}{\partial \theta}\,dy = 0.$$

Differentiating the exponential family density itself with respect to θ, using the chain rule on the exponent,

$$\frac{\partial f(y;\theta)}{\partial \theta} = f(y;\theta)\left[a(y)b'(\theta) + c'(\theta)\right].$$

Substituting this back in and splitting the integral, pulling the θ-only terms $b'(\theta)$ and $c'(\theta)$ outside,

$$b'(\theta)\int a(y)f(y;\theta)\,dy + c'(\theta)\int f(y;\theta)\,dy = 0.$$

The first integral is $\mathbb{E}[a(Y)]$ by definition, and the second is 1, so $b'(\theta)\,\mathbb{E}[a(Y)] + c'(\theta) = 0$, giving

$$\mathbb{E}[a(Y)] = -\frac{c'(\theta)}{b'(\theta)}$$

This is the relationship referred to above: $c'(\theta)$ alone is meaningless without $b'(\theta)$ to divide it by, but together they give the exact mean, with no integration required once $b(\theta)$ and $c(\theta)$ are known. ■

4.8 Variance of $a(Y)$

To find the variance, the same identity is differentiated a second time with respect to θ:

$$\int \frac{\partial^2 f(y;\theta)}{\partial \theta^2}\,dy = 0.$$

From the mean derivation above, $\dfrac{\partial f}{\partial \theta} = f(y;\theta)\cdot v(\theta,y)$, where $v(\theta,y) = a(y)b'(\theta)+c'(\theta)$. Writing $u = f(y;\theta)$ so that $u' = uv$, the product rule gives $\dfrac{\partial^2 f}{\partial \theta^2} = u'v + uv' = u(v^2+v')$, that is,

$$\frac{\partial^2 f}{\partial \theta^2} = f(y;\theta)\left[(a(y)b'(\theta)+c'(\theta))^2 + a(y)b''(\theta)+c''(\theta)\right].$$

Substituting this into the zero-integral identity above and splitting into two pieces,

$$\underbrace{\int f(y;\theta)\left[a(y)b'(\theta)+c'(\theta)\right]^2 dy}_{(I)} + \underbrace{\int f(y;\theta)\left[a(y)b''(\theta)+c''(\theta)\right] dy}_{(II)} = 0.$$

Integral (I)

Using the mean result above, $c'(\theta) = -b'(\theta)\mathbb{E}[a(Y)]$, so the bracket simplifies to $a(y)b'(\theta) + c'(\theta) = b'(\theta)\,(a(y) - \mathbb{E}[a(Y)])$. Therefore

$$(I) = [b'(\theta)]^2 \int f(y;\theta)\,(a(y)-\mathbb{E}[a(Y)])^2\,dy = [b'(\theta)]^2\,\mathrm{Var}[a(Y)],$$

since $\mathbb{E}\left[(a(Y)-\mathbb{E}[a(Y)])^2\right]$ is precisely the definition of $\mathrm{Var}[a(Y)]$.

Integral (II)

Splitting and pulling out the θ-only terms exactly as before,

$$(II) = b''(\theta)\,\mathbb{E}[a(Y)] + c''(\theta).$$

Combining the Two Integrals

Putting (I) and (II) together,

$$[b'(\theta)]^2\,\mathrm{Var}[a(Y)] + b''(\theta)\,\mathbb{E}[a(Y)] + c''(\theta) = 0.$$

Substituting the mean result $\mathbb{E}[a(Y)] = -c'(\theta)/b'(\theta)$ and isolating $\mathrm{Var}[a(Y)]$ gives

$$\mathrm{Var}[a(Y)] = \frac{b''(\theta)\,c'(\theta) - b'(\theta)\,c''(\theta)}{[b'(\theta)]^3}$$

4.9 Score Function

The score is the first derivative of the log-likelihood with respect to the parameter of interest,

$$U(\theta) = \frac{\partial}{\partial\theta}\ell(\theta) = \frac{\partial}{\partial\theta}\ln L(\theta),$$

where $L(\theta)$ is the likelihood and $\ell(\theta) = \ln L(\theta)$ is the log-likelihood.

- If $U(\theta) > 0$, the log-likelihood is increasing at θ.
- If $U(\theta) < 0$, the log-likelihood is decreasing at θ.
- If $U(\theta) = 0$, the log-likelihood is flat (stationary) at θ; this is how the maximum likelihood estimate is characterised.

An important property: $\mathbb{E}[U(\theta)] = 0$**.** A zero expected score reflects the fact that the true parameter sits at the centre of the information provided by the data: on average, the log-likelihood has no tendency to push away from the true θ. This holds for any regular density, not only the exponential family: since

$$U(\theta) = \frac{1}{f(y;\theta)} \frac{\partial f(y;\theta)}{\partial \theta},$$

$$\mathbb{E}[U(\theta)] = \int f(y;\theta)\, U(\theta)\, dy = \int \frac{\partial f(y;\theta)}{\partial \theta}\, dy = 0,$$

directly from the zero-integral identity used throughout this chapter. The exponential family proof below is the same identity, made concrete using the explicit form of $f(y;\theta)$.

Theorem 4.2 (Zero Expected Score)**.** $\mathbb{E}[U(\theta)] = 0$ *for any regular density, exponential family or otherwise.*

Proof. For $f(y;\theta) = \exp[a(y)b(\theta) + c(\theta) + d(y)]$, the log-density is linear in $b(\theta)$ and $c(\theta)$, so the score is

$$U(\theta) = \frac{\partial}{\partial \theta} \log f(Y;\theta) = a(Y)b'(\theta) + c'(\theta).$$

Taking the expectation, using linearity, and substituting the mean result $\mathbb{E}[a(Y)] = -c'(\theta)/b'(\theta)$,

$$\mathbb{E}[U(\theta)] = b'(\theta)\,\mathbb{E}[a(Y)] + c'(\theta) = b'(\theta)\left(-\frac{c'(\theta)}{b'(\theta)}\right) + c'(\theta) = -c'(\theta) + c'(\theta) = 0. \blacksquare$$

$\blacksquare$

4.10 Fisher's Information

Fisher Information answers a single question: how much does one observation tell us about the parameter θ? It admits two equivalent definitions, both of which follow directly from the exponential family results already derived above.

As the Variance of the Score

Fisher Information is defined as the variance of the score,

$$\mathcal{I}(\theta) = \mathrm{Var}[U(\theta)].$$

Because $U(\theta) = a(Y)b'(\theta) + c'(\theta)$ and $c'(\theta)$ does not depend on Y, $\mathrm{Var}[U(\theta)] = [b'(\theta)]^2 \,\mathrm{Var}[a(Y)]$. Substituting the variance result derived above,

$$\mathcal{I}(\theta) = [b'(\theta)]^2 \cdot \frac{b''(\theta)\,c'(\theta) - b'(\theta)\,c''(\theta)}{[b'(\theta)]^3} = \frac{b''(\theta)\,c'(\theta) - b'(\theta)\,c''(\theta)}{b'(\theta)}.$$

As the Expected Curvature of the Log-Likelihood

Differentiating the score once more with respect to θ,

$$\frac{\partial^2 \ell}{\partial \theta^2} = a(Y)b''(\theta) + c''(\theta).$$

Taking the expectation and substituting $\mathbb{E}[a(Y)] = -c'(\theta)/b'(\theta)$ from the mean result above,

$$\mathbb{E}\left[\frac{\partial^2 \ell}{\partial \theta^2}\right] = b''(\theta)\,\mathbb{E}[a(Y)] + c''(\theta) = -\frac{b''(\theta)\,c'(\theta)}{b'(\theta)} + c''(\theta) = -\mathcal{I}(\theta).$$

So the two definitions coincide:

$$\mathcal{I}(\theta) = \mathrm{Var}[U(\theta)] = -\mathbb{E}\left[\frac{\partial^2 \ell}{\partial \theta^2}\right]$$

A sharply peaked log-likelihood (high curvature, a large negative second derivative) corresponds to high information; a flat log-likelihood leaves the parameter poorly determined. For n independent observations, information is additive, $\mathcal{I}_n(\theta) = n \cdot \mathcal{I}_1(\theta)$, and the Cramér–Rao bound states that no unbiased estimator can beat

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{\mathcal{I}_n(\theta)},$$

with the maximum likelihood estimator achieving this bound asymptotically.

Reflection | **Application**

Question: If differentiating the normalising function $c(\theta)$ automatically produces the mean and variance of $a(Y)$, why don't we just extract $c(\theta)$, differentiate it, and call it a day?

Because just the derivative of $c(\theta)$ alone doesn't tell us the full story, it's just a function of θ. It's only once it's placed back inside the whole distribution that we're able to see the relationships and connections between the data and the parameters.

Question: What is the difference between the log-linear model and the log-partition function?

The log-partition function is a unique function of θ, the parameter of interest, with zero dependency on y, that forces the density to integrate to 1. The log-linear structure, on the other hand, describes what's inside the exponent of the one-parameter exponential form: it's the linear combination exposed when taking the log of $f(y;\theta)$, since the term $a(y)b(\theta)$ is linear in $b(\theta)$. A model is called log-linear specifically when the natural parameter itself is written as a linear combination of covariates.

Forms of the Exponential Family: a mathematical guide for an interactive portfolio application.

Project idea

The application is designed to show how different mathematical representations of the exponential family are connected. Its central workflow is form $\rightarrow$ transformation $\rightarrow$ form, with the algebra displayed at each step.
Important distinction: Jordan canonical form is a linear-algebra canonical form, not itself an exponential-family form. It is included as a related linear-algebra branch when a matrix transformation is relevant.

1. General Exponential-Family Form

A useful starting representation is

$$f(x;\theta) = h(x)c(\theta)\exp\left\{\sum_{j=1}^{k} Q_j(\theta)T_j(x)\right\}$$

Here $h(x)$ contains terms depending only on the observation, $c(\theta)$ is parameter-dependent, $T_j(x)$ are statistics of the observation, and $Q_j(\theta)$ are functions of the parameters. The important structural feature is that the parameter dependence enters through the statistics $T_j(x)$.

2. One-Parameter Exponential Family

When $k = 1$, the generalized expression reduces to

$$f(x;\theta) = h(x)c(\theta)\exp\{Q(\theta)T(x)\}$$

Define the natural parameter $\eta = Q(\theta)$. Then

$$f(x;\theta) = h(x)c(\theta)\exp\{\eta T(x)\}$$

Write $c(\theta) = \exp\{-A(\eta)\}$. This gives the canonical form

$$f(x;\eta) = h(x)\exp\{\eta T(x) - A(\eta)\}$$

Thus the key transition is $\theta \rightarrow \eta = Q(\theta)$.

3. Canonical / Natural Parameterization

The canonical one-parameter representation is

$$f(x;\eta) = h(x)\exp\{\eta T(x) - A(\eta)\}$$

The components are $T(x)$, the sufficient statistic; η, the natural or canonical parameter; and $A(\eta)$, the log-partition function. Normalization requires

$$1 = \int h(x)\exp\{\eta T(x) - A(\eta)\}\,dx$$

Therefore

$$A(\eta) = \log \int h(x)\exp\{\eta T(x)\}\,dx$$

The application can reveal this derivation when the user selects "Why does $A(\eta)$ appear?"

4. Generalized / Multiparameter Exponential Family

For k parameters,

$$f(x;\eta) = h(x)\exp\left\{\sum_{j=1}^{k}\eta_j T_j(x) - A(\eta)\right\}$$

Introduce the vectors

$$\eta = (\eta_1, \eta_2, \ldots, \eta_k)^\top, \qquad T(x) = (T_1(x), T_2(x), \ldots, T_k(x))^\top$$

Then the summation becomes an inner product:

$$\sum_j \eta_j T_j(x) = \eta^\top T(x)$$

Hence

$$f(x;\eta) = h(x)\exp\{\eta^\top T(x) - A(\eta)\}$$

This is a useful visual transition: scalar notation becomes vector notation without changing the statistical model.

5. Original Parameters to Natural Parameters

A multiparameter model can be reparameterized by defining

$$\eta = (Q_1(\theta), Q_2(\theta), \ldots, Q_k(\theta))^\top$$

This creates the transformation $\theta \to \eta$. Example: for $X \sim N(\mu, \sigma^2)$, one natural-parameter representation uses

$$T(x) = (x, x^2)^\top, \qquad \eta = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2}\right)^\top$$

Thus the original parameters (μ, σ^2) are replaced by the natural parameters.

6. Changing the Sufficient-Statistic Representation

The representation is not unique. Suppose

$$S(x) = UT(x) + v$$

where U is invertible. The corresponding natural parameter can be transformed as

$$\zeta = (U^{-1})^{\top}\eta$$

The application can display both transformations at once:

$$T(x) \to UT(x) + v, \qquad \eta \to (U^{-1})^{\top}\eta$$

This demonstrates that different-looking exponential-family representations can describe the same statistical model.

7. Matrix Representation

The transformation $S(x) = UT(x) + v$ is controlled by the matrix U. The application can show

$$T(x) \to UT(x) \to UT(x) + v$$

while simultaneously showing

$$\eta \to (U^{-1})^{\top}\eta$$

The key idea is that a change in the sufficient-statistic basis requires a corresponding change in the natural parameter so that the statistical representation remains equivalent.

8. Jordan Canonical Form: Linear-Algebra Branch

Jordan canonical form should be presented as a related linear-algebra concept, not as another definition of an exponential family. For a matrix M, Jordan form is obtained through a similarity transformation:

$$J = P^{-1}MP, \qquad \text{equivalently} \qquad M = PJP^{-1}$$

A Jordan block has the form

$$J_{\lambda} = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$$

Conceptually, this gives another example of a transformation from an original representation to a canonical representation through a change of basis. The portfolio application can therefore place this beside the exponential-family transformations while clearly distinguishing the two mathematical ideas.

9. Proposed Application Flow

Exponential Family
↓
General parameterization
↓
One-parameter form
↓
Canonical / natural form
↓
Multiparameter form
↓
Vector / matrix representation
↓
Equivalent sufficient-statistic representation
↓
Linear-algebra branch: matrix transformation / Jordan form

Interface principle: each arrow should display only the mathematics needed to explain the transition: Before → operation → After → short explanation. This keeps the project focused and feasible to implement quickly.

10. Suggested Interactive Examples

Transformation	What the user sees
General → one-parameter	Set $k = 1$ and collapse the summation.
Original → canonical	Define $\eta = Q(\theta)$ and absorb $c(\theta)$ into $A(\eta)$.
Scalar → vector	Replace $\sum_j \eta_j T_j(x)$ with $\eta^\top T(x)$.
Statistic → equivalent statistic	Apply $S(x) = UT(x) + v$ and $\zeta = (U^{-1})^\top \eta$.
Matrix → Jordan form	Apply $J = P^{-1}MP$ and show the change of basis.

Table 4.1.: Suggested interactive examples for the application.

Sources: (Fithian, nd; UC Berkeley Department of Statistics, nd; MIT Department of Mathematics, nd).

Case study: the Normal distribution. Rather than differentiating $\log p(x;\mu)$ directly, the Normal case is worked out here from its one-parameter exponential family form, to stay consistent with how every other distribution in this portfolio is handled. From earlier in this chapter, with $\theta = \mu$ and σ^2 treated as known,

$$a(y) = y, \quad b(\mu) = \frac{\mu}{\sigma^2}, \quad c(\mu) = -\frac{\mu^2}{2\sigma^2},$$

so that

$$b'(\mu) = \frac{1}{\sigma^2}, \quad b''(\mu) = 0, \quad c'(\mu) = -\frac{\mu}{\sigma^2}, \quad c''(\mu) = -\frac{1}{\sigma^2}.$$

Substituting these into the general information formula derived above,

$$\mathcal{I}(\mu) = \frac{b''(\mu)\,c'(\mu) - b'(\mu)\,c''(\mu)}{b'(\mu)} = \frac{0 - \left(\frac{1}{\sigma^2}\right)\left(-\frac{1}{\sigma^2}\right)}{1/\sigma^2} = \frac{1/\sigma^4}{1/\sigma^2} = \frac{1}{\sigma^2}.$$

For n independent observations, $\mathcal{I}_n(\mu) = n/\sigma^2$, and the Cramér–Rao bound gives $\mathrm{Var}(\hat{\mu}) \geq \sigma^2/n$.

A useful feature of this case falls straight out of the exponential family form: because $b''(\mu) = 0$, the curvature $a(Y)b''(\mu) + c''(\mu)$ reduces to $c''(\mu)$ alone, with no Y in it at all. The observed information therefore equals the expected information exactly, for every sample, not just on average, a structural property of the Normal distribution that does not hold for a distribution such as the Poisson.

A simulation with $\mu = 5$ under two variances, $\sigma^2 = 25$ and $\sigma^2 = 1$, illustrates these results. Panel 1 shows the log-likelihood sharpening as variance decreases, holding n fixed. Panel 2 shows the score fluctuating more widely across repeated single-observation draws when σ^2 is small, exactly the larger variance that defines higher Fisher Information. Panel 3 shows the information as flat in μ for both scenarios, confirming $\mathcal{I}(\mu)$ does not depend on where the true mean sits. Panel 4 tracks the Cramér–Rao bound: empirical variance of 4000 simulated MLEs against the theoretical bound σ^2/n, on a log-log scale, for both variances. Panel 5 plots $\mathcal{I}(\mu) = 1/\sigma^2$ against σ^2 directly, marking the two scenarios and making the inverse relationship visible.

[Figure: normal_fisher_panels_5.png – not embedded in source]

Numerically, the theoretical information values $\mathcal{I}(\mu) = 0.04$ and $\mathcal{I}(\mu) = 1$ match a Monte Carlo estimate of $\mathrm{Var}(\hat{\mu})$ at $n = 100$ closely (0.2435 against a bound of 0.25 for $\sigma^2 = 25$; 0.0102 against a bound of 0.01 for $\sigma^2 = 1$): a 25-fold difference in variance produces exactly a 25-fold difference in information.

PART III.

Building the GLM

CHAPTER 5

Parameter Estimation

Exploration

Fitting a model means writing a log-likelihood and finding the θ that maximises it, i.e. solving $U(\theta) = 0$. For most GLMs this has no closed form, so we need a numerical method: Newton–Raphson, which repeatedly replaces the hard nonlinear problem, locally, by an easy linear one (the tangent line at the current guess) and iterates toward the root.

Why the score specifically? In MLE the root-finding target is the score $U(\theta) = \partial\ell/\partial\theta$ itself, so we substitute U for f and θ for x in the Newton–Raphson update. $U'(\theta)$ gives the curvature of the log-likelihood, telling us how big a step to take.

Why replace $U'(\theta)$ by its expectation? The observed $U'(\theta)$ can be noisy and unstable. For exponential-family models, $\mathbb{E}[U'(\theta)] = -\mathcal{I}(\theta)$, and since $\mathcal{I}(\theta) = \mathrm{Var}(U(\theta))$ is a variance it is guaranteed non-negative, making the algorithm far more stable. This substitution turns Newton–Raphson into the *Method of Scoring* (Fisher scoring). For GLMs the information matrix has the closed form $\mathcal{I} = X'WX$, which is what eventually lets the algorithm be re-expressed as weighted least squares.

Why the exponential family and link function? Writing $\ell_i = y_i b(\theta_i) + c(\theta_i) + d(y_i)$ lets the mean, variance, score and information be derived once in general notation, then specialised per distribution. The mean μ_i is connected to the linear predictor $\eta_i = x_i^\top \beta$ through a link $g(\mu_i) = \eta_i$, since μ_i often lives on a restricted scale while η_i can be any real number.

Why does the score look like a weighted residual? Differentiating ℓ with respect to β_j passes through $\theta_i \to \mu_i \to \eta_i$, and almost everything cancels except

$$U_j = \sum_{i=1}^{n} \frac{(y_i - \mu_i)}{\mathrm{Var}(Y_i)} x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i}\right)$$

a sum of residuals, each weighted inversely by variance and by how sensitive the mean is to the linear predictor: noisier or less-sensitive observations count for less.

Why does this reduce to IRLS? Once $U(\beta)$ and $\mathcal{I}(\beta) = X'WX$ are known, the scoring update $\beta^{(m)} = \beta^{(m-1)} + \mathcal{I}^{-1}U$ rearranges, via a pseudo-response z, into the ordinary weighted-least-squares formula $\beta^{(m)} = (X'WX)^{-1}X'Wz$. Fitting a GLM by maximum likelihood is therefore equivalent to repeatedly solving a weighted regression, where W and z are recomputed each iteration from the current β, hence *Iteratively Reweighted Least Squares*.

Cognition

The general MLE recipe

Given data and a log-likelihood $\ell(\theta)$: take the log-likelihood; differentiate to get $U(\theta) = \partial\ell/\partial\theta$; set $U(\theta) = 0$ and solve, checking $\partial^2\ell/\partial\theta^2 < 0$ confirms a maximum. For GLMs this generally has no closed form, so we turn to Newton's method.

Deriving the Newton–Raphson formula

To solve $f(x) = 0$, with $x^{(m)} = x^{(m-1)} + h$, a first-order Taylor expansion gives

$$f(x^{(m)}) \approx f(x^{(m-1)}) + \left(x^{(m)} - x^{(m-1)}\right) f'(x^{(m-1)})$$

Setting this to zero and solving for $x^{(m)}$,

$$x^{(m)} = x^{(m-1)} - \frac{f(x^{(m-1)})}{f'(x^{(m-1)})}$$

Newton–Raphson applied to the MLE

Substituting U for f and θ for x,

$$\theta^{(m)} = \theta^{(m-1)} - \frac{U(\theta^{(m-1)})}{U'(\theta^{(m-1)})}$$

Theorem 5.1 (Exponential Family Identity)**.** *For distributions in the exponential family,* $\mathbb{E}(U') = -\mathcal{I}$*, where* $\mathcal{I} = \mathrm{Var}(U)$.

Substituting $\mathbb{E}[U'(\theta)] = -\mathcal{I}(\theta)$ gives the **Method of Scoring**:

$$\theta^{(m)} = \theta^{(m-1)} + \frac{U(\theta^{(m-1)})}{\mathcal{I}(\theta^{(m-1)})}$$

Estimation in GLMs

For independent Y_i in the exponential family, $\ell_i = y_i b(\theta_i) + c(\theta_i) + d(y_i)$, so that

$$\mathbb{E}(Y_i) = \mu_i = -\frac{c'(\theta_i)}{b'(\theta_i)}, \qquad \mathrm{Var}(Y_i) = \frac{b''(\theta_i)c'(\theta_i) - c''(\theta_i)b'(\theta_i)}{[b'(\theta_i)]^3}$$

with link function $g(\mu_i) = x_i^\top \beta = \eta_i$, and full log-likelihood $\ell = \sum_i \ell_i$.

Deriving the GLM score U_j

By the chain rule through $\theta_i \to \mu_i \to \eta_i \to \beta_j$,

$$U_j = \frac{\partial \ell}{\partial \beta_j} = \sum_{i=1}^{n} \frac{\partial \ell_i}{\partial \theta_i} \cdot \frac{\partial \theta_i}{\partial \mu_i} \cdot \frac{\partial \mu_i}{\partial \beta_j}$$

The three factors: $\partial \ell_i / \partial \theta_i = b'(\theta_i)(y_i - \mu_i)$ (using $c'(\theta_i) = -b'(\theta_i)\mu_i$); $\partial \theta_i / \partial \mu_i = [b'(\theta_i)\,\mathrm{Var}(Y_i)]^{-1}$ (inverting $\partial \mu_i / \partial \theta_i$); and $\partial \mu_i / \partial \beta_j = (\partial \mu_i / \partial \eta_i)\, x_{ij}$ since $\eta_i = \sum_j x_{ij}\beta_j$. Multiplying the three together, $b'(\theta_i)$ cancels, leaving

$$U_j = \sum_{i=1}^{n} \frac{y_i - \mu_i}{\mathrm{Var}(Y_i)}\, x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i}\right)$$

The Fisher Information matrix for a GLM

Definition: Information Matrix

$I_{jk} = \mathbb{E}[U_j U_k]$.

Expanding the double sum, independence of the Y_i kills every cross term ($i \neq \ell$), leaving only the diagonal, $\mathbb{E}[(y_i - \mu_i)^2] = \mathrm{Var}(Y_i)$, so

$$I_{jk} = \sum_{i=1}^{n} \frac{x_{ij} x_{ik}}{\mathrm{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2$$

giving the Fisher scoring update $\beta^{(m)} = \beta^{(m-1)} + \left[I^{(m-1)} \right]^{-1} U^{(m-1)}$.

Rewriting in matrix form

The information matrix is $I = X'WX$, with W diagonal, $w_{ii} = \mathrm{Var}(Y_i)^{-1} (\partial \mu_i / \partial \eta_i)^2$: exactly the information matrix of a weighted linear regression. Multiplying the update through by $I^{(m-1)}$,

$$I^{(m-1)} \beta^{(m)} = I^{(m-1)} \beta^{(m-1)} + U^{(m-1)}$$

The right-hand side can be written compactly as $X'Wz$, where the pseudo-response is

$$z = X\beta^{(m-1)} + (y - \mu) \left(\frac{\partial \eta}{\partial \mu} \right) = \eta^{(m-1)} + (y - \mu) \left(\frac{\partial \eta}{\partial \mu} \right)$$

a linearised approximation of the link-scale response: the current linear predictor, corrected toward the observed data. Hence the update becomes $X'WX\,\beta^{(m)} = X'Wz$, giving

$$\beta^{(m)} = (X'WX)^{-1} X'Wz$$

Since W and z both depend on the current $\beta^{(m-1)}$, they are recomputed every iteration: this is **Iteratively Reweighted Least Squares (IRLS)**, the standard way GLMs are fitted in practice.

Quantity	Formula
Newton–Raphson update	$x^{(m)} = x^{(m-1)} - f(x^{(m-1)})/f'(x^{(m-1)})$
MLE via Newton's method	$\theta^{(m)} = \theta^{(m-1)} - U(\theta^{(m-1)})/U'(\theta^{(m-1)})$
Fisher information identity	$\mathbb{E}(U') = -\mathcal{I}, \mathcal{I} = \mathrm{Var}(U)$
Method of Scoring	$\theta^{(m)} = \theta^{(m-1)} + U(\theta^{(m-1)})/\mathcal{I}(\theta^{(m-1)})$
GLM score	$U_j = \sum_i \frac{y_i - \mu_i}{\mathrm{Var}(Y_i)} x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i}\right)$
GLM information	$I_{jk} = \sum_i \frac{x_{ij}x_{ik}}{\mathrm{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i}\right)^2 = X'WX$
Pseudo-response	$z = \eta^{(m-1)} + (y - \mu)(\partial \eta/\partial \mu)$
IRLS update	$\beta^{(m)} = (X'WX)^{-1}X'Wz$
Sampling distribution of the score	$U \dot{\sim} N(0, \mathcal{I}), U^2/\mathcal{I} \dot{\sim} \chi^2_1$

Table 5.1.: Summary of the Newton–Raphson to IRLS derivation.

Sampling distribution of the score statistic

The score $U(\theta)$ is a function of the data, so it is itself a random variable with a sampling distribution. Three facts pin that distribution down: its mean (Theorem 4.2), its variance (the Bartlett identity, Section 4.10), and, because U is a sum of independent mean-zero terms, its shape (the central limit theorem).

Definition: Sampling distribution of the score

For independent observations, $U = \sum_i U_i$ with $U_i = \partial \ell_i / \partial \theta$, and

$$\mathbb{E}[U(\theta)] = 0, \qquad \mathrm{Var}[U(\theta)] = \mathcal{I}(\theta), \qquad \frac{U(\theta)}{\sqrt{\mathcal{I}(\theta)}} \dot{\sim} N(0,1).$$

Squaring a standard Normal gives a χ^2_1, so $U(\theta)^2/\mathcal{I}(\theta) \dot{\sim} \chi^2_1$. With p parameters, $\mathbf{U} \dot{\sim} N_p(\mathbf{0}, \mathcal{I})$ and $\mathbf{U}^\top \mathcal{I}^{-1} \mathbf{U} \dot{\sim} \chi^2_p$. For Normal responses the result is exact rather than approximate.

Example: Poisson. For $Y_1, \ldots, Y_n$ i.i.d. Poisson(θ),

$$U(\theta) = \frac{\sum_i y_i}{\theta} - n = \frac{n(\bar{y} - \theta)}{\theta}, \qquad \mathcal{I}(\theta) = \frac{n}{\theta}, \qquad \frac{U^2}{\mathcal{I}} = \frac{n(\bar{y} - \theta)^2}{\theta}.$$

Because $\sum_i Y_i \sim$ Poisson$(n\theta)$, U is a shifted, rescaled Poisson variable: skewed when $n\theta$ is small and approaching the Normal as $n\theta$ grows. With $\theta = 3.5$ and $n = 12$ its exact distribution is already close to $N(0,1)$ (Figure 5.1). The listing computes it exactly, with no simulation, so the printed values are reproducible.

Listing 5.1: Exact sampling distribution of the score (Poisson, $\theta = 3.5$, $n = 12$)

```
theta <- 3.5; n <- 12; I <- n / theta          # n*theta = 42
k <- 0:120                                     # S = sum(Y) ~ Poisson(n*theta)
U <- k / theta - n                             # score for each possible S
```

```
p <- dpois(k, n * theta)                          # its exact probabilities
c(mean = sum(p * U), var = sum(p * U^2), I = I)     # approx 0, 3.4286, 3.4286
sum(p[U^2 / I > qchisq(0.95, 1)])                 # exact size of the 5% score
    test: 0.0530
w <- 1 / (theta * sqrt(I))                        # spacing of U/sqrt(I) values
plot(U / sqrt(I), p / w, type = "h"); curve(dnorm(x), add = TRUE)
```

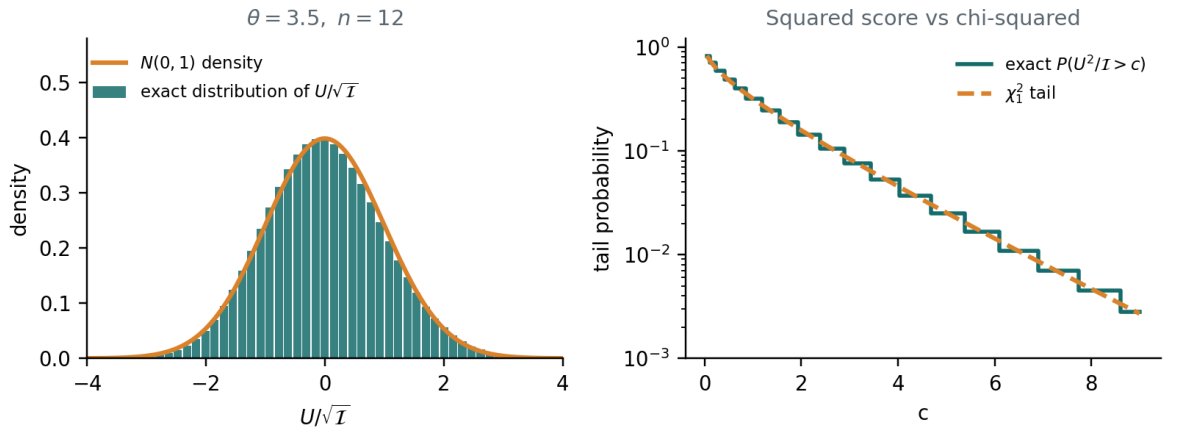


Figure 5.1.: Left: exact distribution of $U/\sqrt{\mathcal{I}}$ against the $N(0,1)$ density. Right: exact tail of $U^2/\mathcal{I}$ against the χ^2_1 tail.

Why it matters.

- **Fisher scoring.** The step $U/\mathcal{I}$ divides a mean-zero random quantity by its variance, so on average it moves θ toward the root of U.
- **The score test.** To test $H_0 : \theta = \theta_0$, compare $U(\theta_0)^2/\mathcal{I}(\theta_0)$ with χ^2_1; only the null model is fitted. Here the exact size of the nominal 5% test is 0.0530 (the lattice nature of counts makes it slightly off).
- **Deviance.** The same quadratic-form argument, applied to a Taylor expansion of the log-likelihood about $\hat{\theta}$, is what makes the deviance approximately χ^2 (Wilks' theorem, next section).

Reflection | **Application**

The two facts used to build Fisher scoring, $\mathbb{E}[U(\theta)] = 0$ and $\mathbb{E}[U'(\theta)] = -\mathcal{I}(\theta)$, are together known as the **Bartlett identities**.

Definition: Bartlett Identities

First. Since $\int f(x;\theta)\,dx = 1$ for every θ, differentiating under the integral gives $\mathbb{E}[U(\theta)] = 0$.
Second. Differentiating a second time,

$$\mathbb{E}[U'(\theta)] + \mathbb{E}[U(\theta)^2] = 0 \implies \mathbb{E}[U'(\theta)] = -\mathbb{E}[U(\theta)^2] = -\operatorname{Var}(U(\theta)) = -\mathcal{I}(\theta),$$

using $\mathbb{E}[U(\theta)] = 0$ so $\mathbb{E}[U(\theta)^2] = \operatorname{Var}(U(\theta))$. This is exactly the identity $\mathbb{E}(U') = -\mathcal{I}$ used to turn Newton–Raphson into the Method of Scoring.

Question: Why replace $U'(\theta)$ with its expectation $-\mathcal{I}(\theta)$ instead of the observed second derivative?

The observed $U'(\theta)$ is computed from a single sample and can be noisy, unstable, or even the wrong sign far from the maximum, sending Newton–Raphson in the wrong direction. The Bartlett identity shows $\mathbb{E}[U'(\theta)] = -\mathcal{I}(\theta) = -\operatorname{Var}(U(\theta))$, and a variance is always non-negative (positive semi-definite in the multivariate case). Using $\mathcal{I}(\theta)$ in place of the observed curvature guarantees a well-behaved, stable step direction at every iteration.

Question: Why is fitting a GLM "secretly" a sequence of weighted least-squares problems?

The Fisher scoring update $\beta^{(m)} = \beta^{(m-1)} + \mathcal{I}^{-1}U$ can be algebraically rearranged (multiplying through by $\mathcal{I}$) into the normal equations $X'WX\beta^{(m)} = X'Wz$ of a weighted linear regression of a pseudo-response z on X with weights W. Because W and z depend on the current β, this weighted regression must be solved again at every step, hence *iteratively* reweighted least squares.

Question: Why does Fisher scoring reach $\bar{y}$ in one step when Newton–Raphson takes several?

For the Poisson mean, $U/\mathcal{I} = (\sum_i y_i/\theta - n)\,\theta/n = \bar{y} - \theta$, so the scoring step $\theta + U/\mathcal{I}$ lands exactly on $\bar{y}$ from any start. Newton–Raphson divides by the observed $U'(\theta) = -\sum_i y_i/\theta^2$ instead, which gives $\theta \mapsto 2\theta - \theta^2/\bar{y}$: it creeps toward $\bar{y}$ and, if the start exceeds $2\bar{y}$, overshoots to a negative θ (from $\theta_0 = 8$ the first step is -2.29, outside the parameter space). Replacing U' by its expectation $-\mathcal{I}$ is exactly the stabilising move that the Bartlett identity justifies.

Applied exercise: three canonical GLMs. For each model we derive $\partial\mu_i/\partial\eta_i$, then plug into

$$U_j = \sum_i \frac{y_i - \mu_i}{\operatorname{Var}(Y_i)}\, x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i}\right), \qquad w_{ii} = \frac{1}{\operatorname{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i}\right)^2$$

(a) Normal, identity link

$Y_i \sim N(\mu_i, \sigma^2)$, $\operatorname{Var}(Y_i) = \sigma^2$, $g(\mu_i) = \mu_i = \eta_i \Rightarrow \partial\mu_i/\partial\eta_i = 1$.

$$U_j = \frac{1}{\sigma^2}\sum_i (y_i - \mu_i)x_{ij}, \qquad w_{ii} = \frac{1}{\sigma^2} \text{ (constant)}.$$

Takeaway: weights don't depend on β, so IRLS converges in a single step, this is just ordinary least squares.

(b) Poisson, log link

$Y_i \sim \text{Poisson}(\mu_i)$, $\operatorname{Var}(Y_i) = \mu_i$, $g(\mu_i) = \log\mu_i = \eta_i \Rightarrow \partial\mu_i/\partial\eta_i = \mu_i$.

$$U_j = \sum_i (y_i - \mu_i)x_{ij}, \qquad w_{ii} = \mu_i.$$

Takeaway: observations with larger predicted counts get more weight, matching the mean-variance link of the Poisson.

(c) Binomial, logit link

$Y_i \sim \text{Bernoulli}(\pi_i)$, $\text{Var}(Y_i) = \pi_i(1-\pi_i)$, $g(\pi_i) = \log \frac{\pi_i}{1-\pi_i} = \eta_i \Rightarrow \partial\pi_i/\partial\eta_i = \pi_i(1-\pi_i)$.

$$U_j = \sum_i (y_i - \pi_i) x_{ij}, \qquad w_{ii} = \pi_i(1-\pi_i).$$

Takeaway: weight is largest when $\pi_i \approx 0.5$ (most uncertain prediction) and shrinks toward 0 as $\pi_i \to 0$ or 1 (confident predictions contribute little to further updates).

Model	$\text{Var}(Y_i)$	U_j	w_{ii}
Normal	σ^2	$\sigma^{-2}\sum_i (y_i - \mu_i)x_{ij}$	σ^{-2}
Poisson	μ_i	$\sum_i (y_i - \mu_i)x_{ij}$	μ_i
Binomial	$\pi_i(1-\pi_i)$	$\sum_i (y_i - \pi_i)x_{ij}$	$\pi_i(1-\pi_i)$

Table 5.2.: Score and weight for three canonical-link GLMs.

Pattern to notice: in all three canonical-link cases, $\text{Var}(Y_i)$ and $\partial\mu_i/\partial\eta_i$ cancel partially or fully in U_j, leaving the simple raw residual $(y_i - \mu_i)$. This cancellation is a special feature of the *canonical* link, where $\theta_i = \eta_i$.

Worked example: deriving and computing an MLE (Poisson)

Twelve plots each record a count y_i, modelled as $Y_i \sim \text{Poisson}(\theta)$. The data are hard-coded so the output is reproducible.

By hand. The log-likelihood, score and second derivative are

$$\ell(\theta) = \sum_i y_i \log\theta - n\theta - \sum_i \log y_i!, \qquad U(\theta) = \frac{\sum_i y_i}{\theta} - n, \qquad U'(\theta) = -\frac{\sum_i y_i}{\theta^2}.$$

Setting $U(\hat\theta) = 0$ gives $\hat\theta = \bar y$. Since $U'(\theta) < 0$ for every $\theta > 0$ the stationary point is a maximum. The Fisher information is $\mathcal{I}(\theta) = -\mathbb{E}[U'] = n/\theta$, so $\text{SE}(\hat\theta) = \sqrt{\hat\theta/n}$.

Listing 5.2: Closed form, numerical check, Newton–Raphson versus Fisher scoring, and glm()

```
y <- c(2, 4, 3, 5, 1, 4, 6, 3, 2, 5, 4, 3)    # counts per plot (hard-coded)
n <- length(y); s <- sum(y)

loglik <- function(th) sum(dpois(y, th, log = TRUE))
U      <- function(th) s / th - n             # score
Uprime <- function(th) -s / th^2              # observed second derivative
Info   <- function(th) n / th                 # Fisher information, = -E[U']

mean(y)                                       # 1. closed form: 3.5
optimize(loglik, c(0.1, 20), maximum = TRUE)$maximum # 2. numerical check:
    about 3.5

nr <- sc <- 1                                 # 3. both methods start at
    theta0 = 1
for (m in 1:6) {
  nr <- nr - U(nr) / Uprime(nr)               #    Newton-Raphson
  sc <- sc + U(sc) / Info(sc)                 #    Fisher scoring
  cat(m, sprintf("%.6f %.6f\n", nr, sc))
}

```

```r
fit <- glm(y ~ 1, family = poisson)          # 4. same MLE through the GLM machinery
c(coef(fit), exp(coef(fit)))                 #    beta0 = log(3.5), then 3.5
sqrt(vcov(fit))                              #    SE(beta0) = 1/sqrt(sum(y))
c(deviance(fit), AIC(fit), as.numeric(logLik(fit)))
```

m	Newton–Raphson	Fisher scoring	Quantity	Value
1	1.714286	3.500000	$\hat\theta = \bar y$	3.5
2	2.588921	3.500000	$\ell(\hat\theta)$	-21.8339
3	3.262839	3.500000	$\mathcal{I}(\hat\theta) = n/\hat\theta$	3.4286
4	3.483930	3.500000	$\mathrm{SE}(\hat\theta)$	0.5401
5	3.499926	3.500000	$\hat\beta_0 = \log\hat\theta$ (SE 0.1543)	1.2528
6	3.500000	3.500000	Deviance (11 df); D/df	7.049; 0.64
			AIC	45.67

Table 5.3.: Output of the listing, starting from $\theta_0 = 1$.

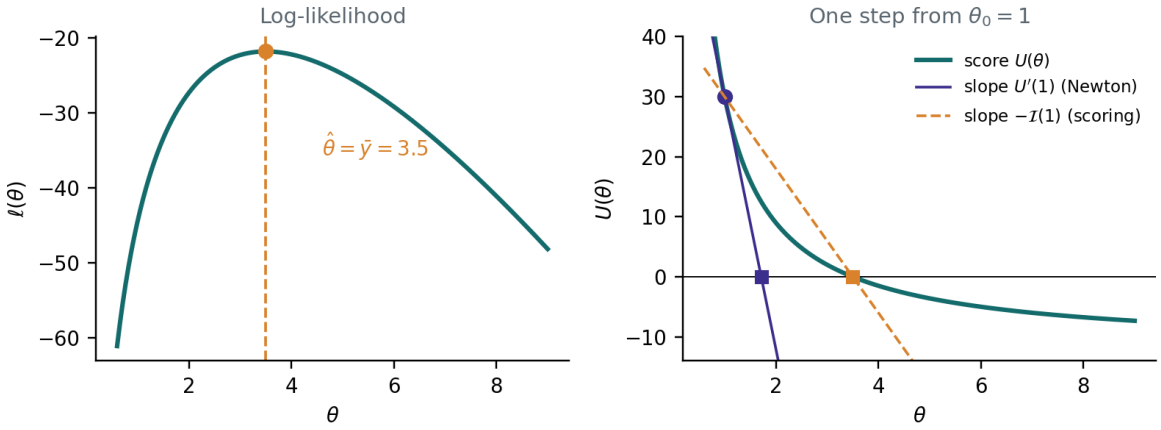


Figure 5.2.: Left: the log-likelihood peaks at $\hat\theta = \bar y$. Right: from $\theta_0 = 1$, Newton–Raphson follows the observed slope $U'(1)$ and stops short at 1.71; Fisher scoring follows the expected slope $-\mathcal{I}(1)$ and lands on 3.5.

The intercept-only Poisson GLM uses the log link, so $\beta_0 = \log\theta$ and, by invariance of the MLE, $\hat\beta_0 = \log\bar y = 1.2528$, with $\mathrm{SE}(\hat\beta_0) = 1/\sqrt{\sum_i y_i} = 0.1543$. A deviance of 7.05 on 11 degrees of freedom shows no sign of overdispersion for this toy sample.

5.1 Deviance

Exploration

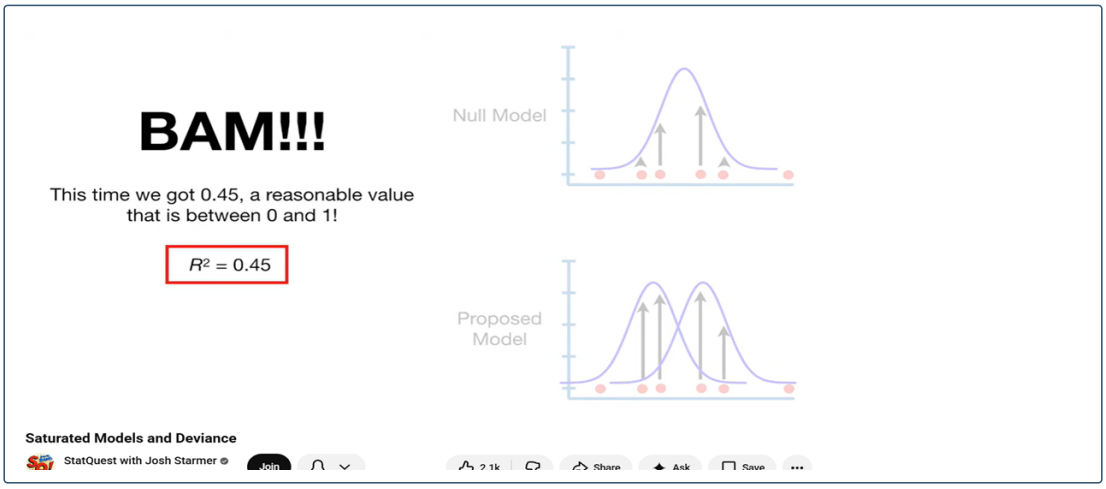


Figure 5.3.: Saturated models and deviance, from a YouTube lecture (StatQuest with Josh Starmer, 2018a).

Source: StatQuest with Josh Starmer. (2018, July 9), *Saturated Models and Deviance*, YouTube.

What is a saturated model? With n data points and n free parameters, one per point, we can set $\hat{\mu}_i = y_i$ for every i: a perfect fit, zero error. This is the *saturated model*. It explains nothing on its own, but it is the ceiling: the best log-likelihood any model from this distributional family could ever reach on this data. Every simpler model (a GLM with $p \ll n$ coefficients) is judged against that ceiling.

What is deviance, intuitively? If the saturated model has the best possible log-likelihood and our fitted model has a smaller one, the gap is what we gave up for simplicity. Deviance is twice that gap, $D = 2[\ell_{\text{sat}} - \ell(\hat{\mu})]$. Small deviance, good fit; large deviance, poor fit, exactly the role residual sum of squares plays in OLS. The factor of 2 isn't decoration: doubling the gap is precisely what makes deviance asymptotically chi-squared (Theorem 5.2), turning it into a quantity usable for formal testing.

Why a reference distribution? A raw deviance value means nothing alone, we need to know the range expected from noise even under a correct model. The chi-squared distribution supplies that range, the same way a residual sum of squares means little until compared to an F or χ^2 reference.

Cognition

The quadratic-form argument used for the sampling distribution of the score statistic earlier in this chapter, applied to the log-likelihood, is why D is approximately χ^2.

Definition: Saturated Model

For n independent observations from an exponential-family distribution with mean μ_i, the saturated model sets $\hat{\mu}_i^{\text{sat}} = y_i$ for all i. Its log-likelihood $\ell_{\text{saturated}} = \sum_i \ell_i(y_i; y_i)$ is the maximum attainable for this sample under this family.

Definition: Deviance

For a fitted model with estimated means $\hat{\mu}_i$,

$$D = 2\left[\ell_{\text{saturated}} - \ell(\hat{\mu})\right] = 2\sum_{i=1}^{n}\left[\ell_i(y_i; y_i) - \ell_i(\hat{\mu}_i; y_i)\right]$$

When the family has a dispersion parameter ϕ (e.g. σ^2 for the Normal), the scaled deviance is $D^* = D/\phi$. For the Normal with known variance, D reduces exactly to the residual sum of squares (see the Normal derivation in the Application below), so deviance generalises "sum of squared residuals" to any exponential-family response.

Definition: Chi-Squared Distribution

If $Z_1, \ldots, Z_k$ are independent $N(0,1)$, then $X = \sum_{j=1}^{k} Z_j^2 \sim \chi^2_k$, with $\mathbb{E}(X) = k$, $\text{Var}(X) = 2k$, supported on $[0, \infty)$, right-skewed, approaching Normal as k grows.

Why deviance is approximately chi-squared.

Theorem 5.2 (Wilks' Theorem, informal)**.** *Under standard regularity, for a correctly specified model with n observations and p parameters, $D^* \dot{\sim} \chi^2_{n-p}$. More generally, comparing nested models with $p_0 < p_1$ parameters, $D_0 - D_1 \dot{\sim} \chi^2_{p_1-p_0}$ under the null that the smaller model is adequate.*

For the Normal case this is exact rather than asymptotic: if $Y_i \sim N(\mu_i, \sigma^2)$ and the model is correct, each standardised residual $(y_i - \hat{\mu}_i)/\sigma$ behaves like a standard Normal draw, so

$$D/\sigma^2 = \sum_{i=1}^{n}\left(\frac{y_i - \hat{\mu}_i}{\sigma}\right)^2 \sim \chi^2_{n-p}$$

exactly the sum-of-squared-standard-Normals form that defines χ^2. This is why deviance buys us two things in practice: (i) an overall goodness-of-fit test, comparing D^* to χ^2_{n-p}; and (ii) a way to compare nested GLMs via the deviance difference, analogous to an F-test/ANOVA but valid for any exponential-family response.

Reflection | **Application**

For each distribution below, the per-observation log-likelihood is evaluated at the saturated fit ($\mu = y_i$) and the model fit ($\mu = \hat{\mu}_i$), then combined via $D = 2\sum_i[\ell_i(y_i; y_i) - \ell_i(\hat{\mu}_i; y_i)]$.
Poisson. $\ell_i = -\mu + y_i \log \mu - \log(y_i!)$; the $\log(y_i!)$ term cancels on subtraction:

$$D = 2\sum_{i=1}^{n}\left[y_i \log\left(\frac{y_i}{\hat{\mu}_i}\right) - (y_i - \hat{\mu}_i)\right]$$

Normal (known σ^2). $\ell_i = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(y_i-\mu)^2}{2\sigma^2}$; the saturated fit kills the quadratic term, the constant cancels on subtraction, and taking $\sigma^2 = 1$:

$$D = \sum_{i=1}^{n}(y_i - \hat{\mu}_i)^2 = \text{Residual Sum of Squares}$$

Exponential. $f(y; \mu) = \mu^{-1}e^{-y/\mu}$, so $\ell_i = -\log \mu - y_i/\mu$; subtracting and doubling:

$$D = 2\sum_{i=1}^{n}\left[\frac{y_i}{\hat{\mu}_i} - \log\left(\frac{y_i}{\hat{\mu}_i}\right) - 1\right]$$

Reflection. All three share one ingredient, $2[\ell_i(y_i) - \ell_i(\hat{\mu}_i)]$, specialised per distribution: squared error for the Normal, a scaled log-ratio "surprise" term for the Poisson, a relative-error-minus-log-ratio term for the Exponential (which is the Gamma deviance at shape 1, since Exponential is a special case of Gamma). Same unifying idea as the rest of this portfolio: one general formula, specialised by plugging in each distribution's log-likelihood.

Distribution	**Deviance** D
Poisson	$2\sum_i [y_i \log(y_i/\hat{\mu}_i) - (y_i - \hat{\mu}_i)]$
Normal ($\sigma^2 = 1$)	$\sum_i (y_i - \hat{\mu}_i)^2$
Exponential	$2\sum_i [y_i/\hat{\mu}_i - \log(y_i/\hat{\mu}_i) - 1]$

Table 5.4.: Deviance formulas for three exponential-family distributions.

CHAPTER 6

Generalized Linear Models for Binary Data

Exploration

To keep the structure consistent we start with the exploration of logistic regression and binary data, but before we even go into it, what is binary data? A binary outcome is any outcome that can potentially only take on two values. They take many forms: a logic gate, a yes or no question, 1 and 0 in computing, Boolean variables, True or False. So if we wanted to model this, how do we approach it? That is where logistic regression comes in, and as we dive deeper you'll start to see why we need logistic regression instead of linear regression and how we go about it. So without further ado, let's go.
To start the Exploratory phase we watch a video on logistic regression.

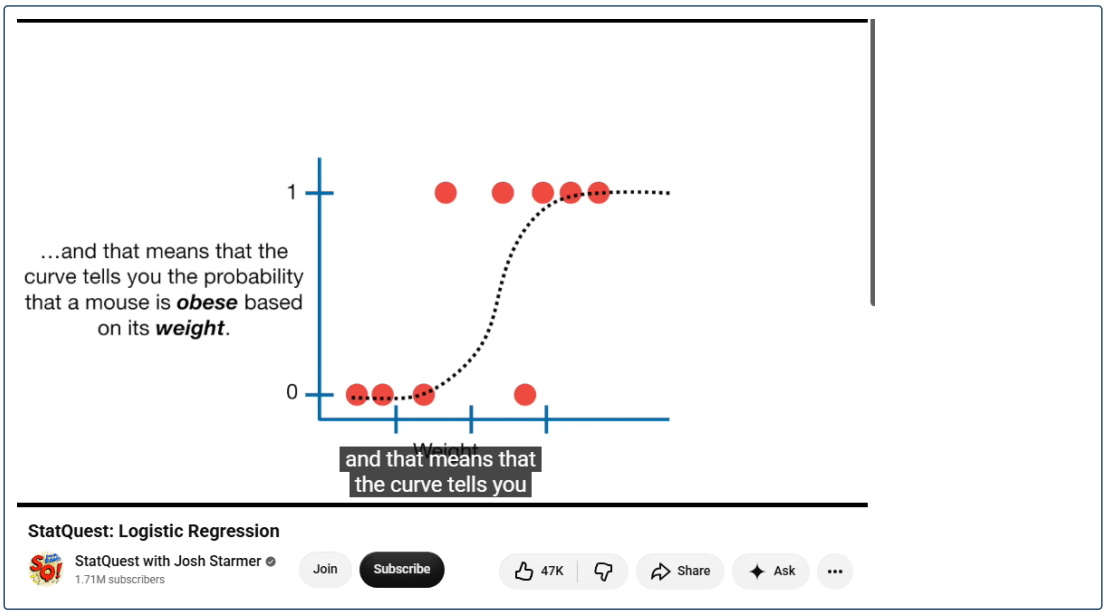


Figure 6.1.: Logistic Regression Explainer (StatQuest with Josh Starmer, 2018b).
Source: StatQuest with Josh Starmer. (2018, March 5), *StatQuest: Logistic Regression*, YouTube. https://www.youtube.com/watch?v=yIYKR4sgzI8

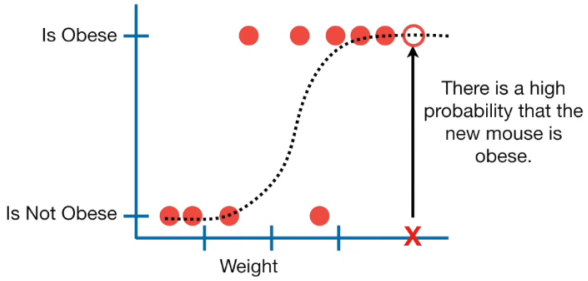
Video Summary

Suppose we have a scenario where we have some data on the weight and size of mice, with the independent variable being weight and our response being size, so we want to predict size using weight (active recall moment: this is an example of a simple linear regression).

- Firstly we calculate the R^2 and determine if weight and size are correlated. Large values imply a large effect, meaning weight has a big effect on size.

- Secondly we calculate the p-value to determine if the R^2 value is statistically significant.
- We could then use the line to predict future outcomes given a new size observation.

NB: using data to predict something is a form of machine learning, so simple linear regression is the most basic form of machine learning, just a small nugget.
Now logistic regression is similar to linear regression, except it predicts whether something is True or False, instead of continuous variables like weight or size. So how do we predict size using weight in a logistic regression?
By design, since binary data has two possible outcomes, we will have two categories and then have a threshold for which we decide what category a mouse falls in based on its weight. So our categories are "is obese" and "is not obese".



Interesting fact to note: logistic regression doesn't fit a straight line through the data points, it fits an S-like curve that ranges between 0 and 1.
So based on this graph a small weight will fall in the "is not obese" category, and a large weight will fall in the "is obese" category. Kind of intuitive if you think about it.
For the classification to be possible we need a threshold, for example: if $P(\text{is obese} \mid \text{weight}) \geq 0.5$ we classify those who satisfy this condition as obese and vice versa.
Just like linear regression we can have a simple model where obesity is predicted by weight, or a multivariate model where obesity is predicted by a number of other factors like genotype, and so on.
So we can conclude that, just like linear regression, logistic regression can work with continuous predictors. That is exactly why it is so perfect: it maps multiple continuous variables to a simple decision, yes or no. How interesting is that.

Fitting the Line

Another big difference between linear regression and logistic regression is not only that linear regression fits a straight line and logistic regression fits an S shaped curve, it is how the line is fitted.
For linear regression we use ordinary least squares (residuals) to fit the best line on the data points, and we calculate R^2 with the residuals, and use it to compare simple models to complicated models.
Logistic regression doesn't have the same concept of a residual, so it can't use OLS and we can't calculate R^2. Instead it uses MLE (Maximum Likelihood Estimation): we create a likelihood product and choose the curve with the maximum likelihood.

Log Odds

Next thing to dissect is the idea of log odds. The information for this section was taken from GeeksforGeeks (GeeksforGeeks, 2022), and it explains the idea of log odds and the odds ratio,

so let's dive in.

Source: GeeksforGeeks. (2022, September 26), *Log odds*. https://www.geeksforgeeks.org/machine-learning/log-odds/

What are Odds?

Usually depicting the odds of success, odds are defined as the chance of success divided by the chance of failure; more formally, the ratio between the probability of success and failure. If the probability of success is $P(A)$, then the odds ratio of event A is

$$\text{Odds Ratio} = \frac{P(A)}{1 - P(A)} \qquad \text{or} \qquad \frac{P(A)}{P(A')}$$

Example

1. If the chance of winning a rugby match is 90%, then the odds ratio is

$$\text{Odds Ratio} = \frac{90}{10} = 9$$

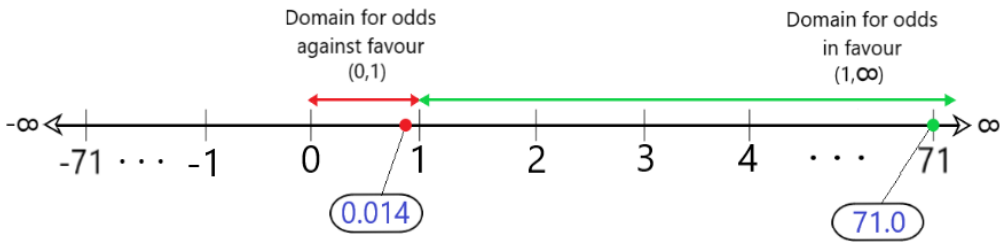
2. Assume the odds of my team winning the rugby tournament are 1 to 5. The odds will be 0.2, but the probability of my team winning is $\frac{1}{6}$. Mathematically we can derive the odds from the probability:

$$\text{Odds} = \frac{P(\text{win})}{P(\text{loss})} = \frac{\frac{1}{6}}{\frac{5}{6}} = \frac{1}{5}$$

So we just showed that we can derive the odds from the probability, and that the probability of success and the odds of success are linked, but they are not the same thing.

Log Odds

The problem with the odds ratio is the asymmetry created between the odds of winning and the odds of losing, like our 90:10 ratio earlier, so we need a better way to represent odds.
Example. 72 countries participate in the Commonwealth Games. Considering every country has an equal chance of winning, the odds of the Indian team winning gold are 1 to 71, or 0.014. Assuming a hypothetical scenario where the odds of the Indian team winning are 71 to 1, we get this:



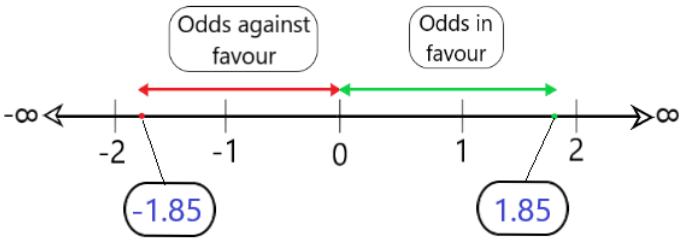
What we observe from this is that when the odds are against our favour the value is bounded between 0 and 1, which is a small value, and when the odds are in our favour the value is bounded between 1 and infinity, which could potentially be a very large number. To solve this problem of asymmetry, log odds came into the picture.

Definition: Log Odds

The log odds is the logarithm of the odds ratio (GeeksforGeeks, 2022),

$$\text{Log Odds} = \log\left(\frac{P(A)}{1-P(A)}\right)$$

$$\begin{cases} 1:71 = \log\left(\frac{1}{71}\right) = -1.85 \\ 71:1 = \log\left(\frac{71}{1}\right) = 1.85 \end{cases}$$



Because of this, the log odds of any event when plotted as a histogram gives a normal distribution, and that's why it's so useful.

Sigmoid Function

Let's explore further the idea of a sigmoid function. The information for this section was taken from GeeksforGeeks (GeeksforGeeks, 2025).

Source: GeeksforGeeks. (2025, July 23), *Sigmoid function*. https://www.geeksforgeeks.org/machine-learning/derivative-of-the-sigmoid-function/

Definition: Sigmoid Function

A mathematical function that maps any real valued number into a value between 0 and 1. It is characterised by an S shaped curve, which makes it particularly useful in scenarios where we need to convert outputs into probabilities. This function is often called the logistic function (GeeksforGeeks, 2025). Represented mathematically as

$$\sigma(x) = \frac{1}{1+e^{-x}}, \qquad -\infty < x < \infty$$

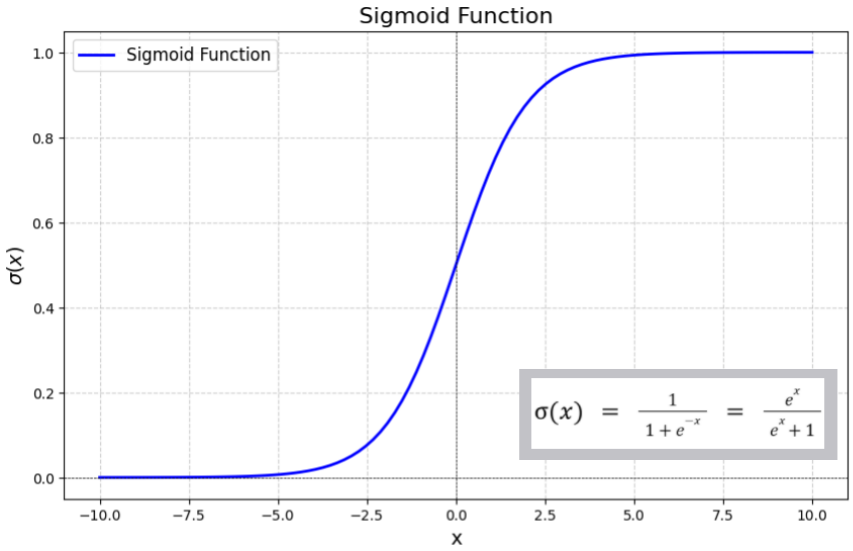
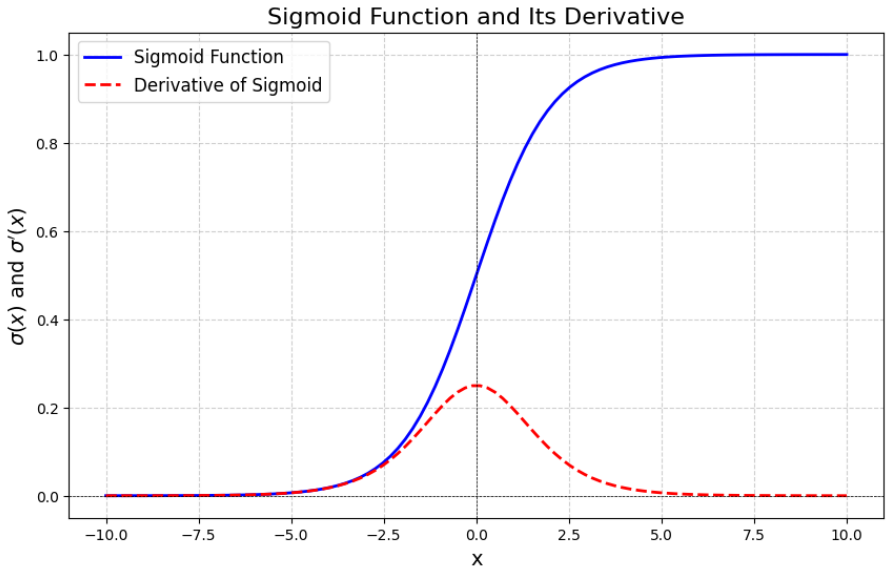


Figure 6.2.: Graph of the sigmoid function (GeeksforGeeks, 2025).

Properties of the Sigmoid Function

1. **Domain:** all real numbers $(-\infty, \infty)$.
2. **Range:** probability output, so it is strictly bounded between 0 and 1.
3. **Midpoint (intercept):** $\sigma(0) = 0.5$.
4. **Monotonicity:** it is strictly monotonically increasing everywhere.
5. **Differentiability:** the sigmoid is differentiable, yielding a bell shaped curve with maximum gradient of 0.25 at $x = 0$. The derivative in mathematical form is

$$\sigma'(x) = \sigma(x)\,(1 - \sigma(x))$$



Uses and Applications of the Sigmoid Function

Activation function: in machine learning and neural networks for binary classification problems, smoothing outputs, and introducing non-linearity into models. It is useful in backpropagation because it is the only function that appears in its own derivative and it can be differentiated at any point. During backpropagation the weightings and biases are recalculated using the gradient of the activation function, and the sigmoid is the best for this. It does present an issue called the vanishing gradient problem, where the gradient is so small it is insignificant and causes the learning process to hit a plateau.
Logistic regression: the sigmoid function is the inverse of the log odds, they are mathematical inverse operations of each other. Let's look at the difference in the table below.

Feature	Sigmoid Function	Logit Function
Formula	$\sigma(x) = \frac{1}{1+e^{-x}}$	$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$
Domain	All real numbers	Probabilities $[0, 1]$
Range	Probabilities $[0, 1]$	All real numbers
Core concept	Converts log odds into a probability	Converts a probability into log odds

Table 6.1.: Sigmoid function versus logit function.

Sigmoid Function to Log Odds

$$p = \frac{1}{1+e^{-x}}$$

Flip both sides:

$$\frac{1}{p} = 1 + e^{-x}$$

$$e^{-x} = \frac{1-p}{p}$$

Take the logarithm of both sides:

$$-x = \ln\left(\frac{1-p}{p}\right)$$

and when we multiply both sides by -1 we get

$$x = \ln\left(\frac{p}{1-p}\right)$$

a negative log flips its fraction: $-\ln(a) = \ln\left(\frac{1}{a}\right)$.

Cognition

This section explains the ideas and the maths behind logit and probit models. We begin with the basic set-up. Then we look at tolerance distributions, which is where the two models differ. After that we cover the parts that both models share: general logistic regression, estimation, goodness of fit, residuals and overdispersion.

6.1 Probability Distribution

Definition: Binary Random Variable

A binary random variable takes the value 1 for a "success" and 0 for a failure,

$$Z = \begin{cases} 1 & \text{if the outcome is a "success",} \\ 0 & \text{if the outcome is a failure,} \end{cases}$$

with $P(Z = 1) = \pi$ and $P(Z = 0) = 1 - \pi$. The random variable has the Binomial distribution $Z \sim \text{Bin}(\pi, 1)$ (Dobson and Barnett, 2018).

If there are n such random variables $Z_1, \ldots, Z_n$ which are independent, with $P(Z_j = 1) = \pi_j$, then their joint probability is

$$\prod_{j=1}^{n} \pi_j^{z_j}(1-\pi_j)^{1-z_j} = \exp\left\{\sum_{j=1}^{n} z_j \log\left(\frac{\pi_j}{1-\pi_j}\right) + \sum_{j=1}^{n} \log(1-\pi_j)\right\}$$

The product binomial distribution is a member of the exponential family (Dobson and Barnett, 2018).
When all the π_j's are equal,

$$Y = \sum_{j=1}^{n} Z_j,$$

so that Y represents the number of successes in n trials. The random variable Y has the $\text{Bin}(\pi, n)$ distribution:

$$P(Y = y) = \binom{n}{y} \pi^y (1-\pi)^{n-y}, \qquad y = 0, 1, \ldots, n.$$

Finally, we consider the general case of N independent random variables $Y_1, \ldots, Y_N$ corresponding to the number of successes in N different strata or groups of size n_i. If $Y_i \sim \text{Bin}(\pi_i, n_i)$, $i = 1, \ldots, N$, the log-likelihood function is

$$\ell(\pi; y) = \sum_{i=1}^{N} \left[y_i \log\left(\frac{\pi_i}{1-\pi_i}\right) + n_i \log(1-\pi_i) + \log\binom{n_i}{y_i} \right],$$

where $y = (y_1, \ldots, y_N)^\top$.
The term $\log[\pi/(1-\pi)]$ is called the log-odds. It shows up here as the natural parameter of the binomial distribution (Schall, nd, pp. 67–68).

6.2 GLM for Binary Data

We want to model the probability of success $\pi_1, \ldots, \pi_N$ in the N groups in terms of explanatory variables which characterise the subgroup. Since $\mathbb{E}(Y_i) = n_i\pi_i$, we model the probabilities π_i as

$$g(\pi_i) = x_i^\top \beta,$$

where x_i is a vector of explanatory variables, β is a vector of parameters and g is the link function.
The quantity $\eta_i = x_i^\top \beta$ is the *linear predictor*, and g is the *link function* (Dobson and Barnett, 2018; Schall, nd). Every GLM has three parts: a distribution (here, binomial), a linear predictor and a link. Logit and probit share the first two. They differ only in the link.
The names are short for what the models do. "Probit" means *probability unit*. Bliss introduced the word in 1934. "Logit" means *logistic unit*. Berkson introduced it in 1944, copying the idea behind probit (Wikipedia, ndb,n).
We could model this with the identity link, where $g(\pi_i) = \pi_i$, but it has some problems because π_i is strictly bounded between 0 and 1. The fitted values from the identity link function, $\hat{\pi}_i = x_i^\top \hat{\beta}$, might be smaller than 0 or larger than 1.

Definition: Tolerance Distribution

To ensure that probability estimates stay in the bounds $[0, 1]$, π is often modelled using a cumulative distribution function such that

$$\pi = \int_{-\infty}^{t} f(s)\, ds,$$

where $f(s)$ is a pdf which is called the tolerance distribution (Dobson and Barnett, 2018).

6.3 Tolerance distributions

The name *tolerance distribution* comes from experiments on doses of poison. Picture an animal with its own hidden tolerance T. If the dose x is bigger than the tolerance ($T < x$), the animal dies. If not, it lives. Tolerance differs from animal to animal. So the chance of dying at dose x is $\Pr(T < x)$ (Schall, nd, pp. 69–70). A different choice for the distribution of T gives a different model.
Probit. Suppose the tolerance follows a Normal(μ, σ^2) distribution. Then

$$\pi_i = \Pr(T < x_i) = \Phi\left(\frac{x_i - \mu}{\sigma}\right)$$
$$\iff \quad \Phi^{-1}(\pi_i) = \underbrace{-\frac{\mu}{\sigma}}_{\beta_1} + \underbrace{\frac{1}{\sigma}}_{\beta_2} x_i,$$

where Φ is the cdf of the standard normal distribution (Schall, nd, p. 70). The link is Φ^{-1}, the inverse of that cdf. The dose $x = \mu = -\beta_1/\beta_2$ is called the median lethal dose, or LD(50). Half of the animals are expected to die at this dose (Dobson and Barnett, 2018; Schall, nd).
Logit. Now suppose the tolerance follows a logistic distribution, with density

$$f(s) = \frac{\beta_2 e^{\beta_1+\beta_2 s}}{[1+e^{\beta_1+\beta_2 s}]^2}.$$

Integrating this gives

$$\pi = \frac{e^{\beta_1+\beta_2 x}}{1+e^{\beta_1+\beta_2 x}} \quad \Longleftrightarrow \quad \log\left(\frac{\pi}{1-\pi}\right) = \beta_1 + \beta_2 x.$$

The link is the logit, which is the log of the odds $\pi/(1-\pi)$ (Schall, nd, pp. 70–71). The logistic function turns log-odds back into a probability. A key feature of the model is that each one-unit rise in a predictor multiplies the odds by the same amount (Wikipedia, nda).

The hidden-variable view. There is another way to tell the same story. Imagine a hidden continuous variable $Y^* = x^\top\beta + \varepsilon$. We see $Y = 1$ when Y^* is above zero, and $Y = 0$ otherwise. If the error ε is standard normal, we get the probit model. If it is standard logistic, we get the logit model (Wikipedia, ndb,n).

Using a *standard* error loses nothing. A different mean can be moved into the intercept. A different spread can be absorbed by rescaling the coefficients (Wikipedia, ndb,n).

Both distributions are symmetric and bell-shaped. The logistic has heavier tails (Figure 6.3). This makes it a little less sensitive to unusual data points (Wikipedia, nda). The logistic distribution is also wider. With scale 1, its standard deviation is $\pi/\sqrt{3} \approx 1.81$. That is why logit coefficients usually come out about 1.8 times bigger than probit coefficients for the same data.

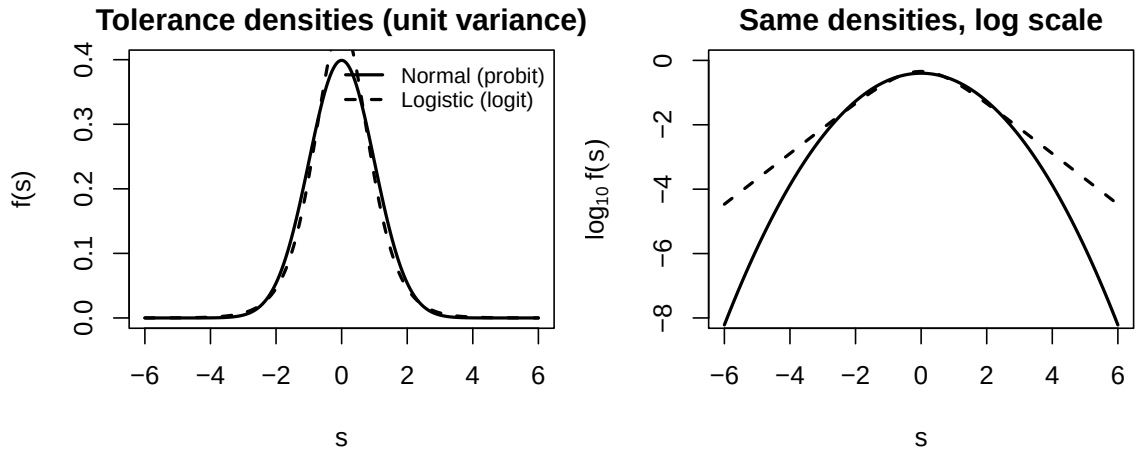


Figure 6.3.: Normal and logistic tolerance densities, both scaled to have variance 1. They look almost the same in the middle (left). They differ in the tails (right).

Other tolerance distributions exist. A uniform distribution gives a straight-line model that only works over a short range. An extreme-value distribution gives the complementary log-log link, $\log[-\log(1-\pi)] = \beta_1 + \beta_2 x$. It behaves like logit and probit when π is near 0.5. It differs when π is near 0 or 1 (Dobson and Barnett, 2018, pp. 151–153). If you are not sure which link to use, the advice is to try several (Dobson and Barnett, 2018, p. 167).

One more point. Suppose the true errors are logistic but we fit a probit model. The probit coefficients will be off. Even so, the estimated effects on the probability stay close to those from the true logit model (Wikipedia, ndb).

6.4 General logistic regression

A model with one predictor is a special case of the general logistic regression model:

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = x_i^\top \beta.$$

Here x_i is the i-th row of the design matrix. Its entries can be continuous variables or dummy variables for the levels of a category. This is like the step from simple linear regression to the general linear model (Schall, nd, p. 74). Probit works the same way, with Φ^{-1} in place of the logit.

Reading the coefficients. In the logit model, β_j is the change in the log-odds when x_j goes up by one unit. So e^{β_j} is the number that the odds are multiplied by (Schall, nd, p. 85). For a 0/1 variable that separates Group 1 from Group 2, the coefficient is the log odds ratio:

$$\log \omega = \log \frac{\pi_1/(1-\pi_1)}{\pi_2/(1-\pi_2)} = \beta_p.$$

Taking $e^{\hat{\beta}_p}$ gives the odds ratio itself. We can do the same to the ends of its confidence interval (Schall, nd, pp. 84–85).

We could also compare two groups with the risk difference $\pi_1 - \pi_2$ or the risk ratio π_1/π_2. But the logistic model is the easiest one to use for odds ratios (Schall, nd, p. 84). The lecture notes show an example with two tick treatments on sheep. The odds ratio was 0.968, with a 95% confidence interval from 0.837 to 1.187. The interval includes 1, so the two treatments are hard to tell apart (Schall, nd, pp. 86–87).

Probit coefficients do not have an odds-ratio meaning. They are changes on the normal-quantile (z-score) scale of the hidden variable.

6.5 Parameter estimation

Both models are usually fitted by maximum likelihood. Neither has a simple formula for the answer, unlike least squares (Wikipedia, ndb,n). The general GLM method gives us a step-by-step recipe (Schall, nd, pp. 39–42).

Take independent observations with $E(Y_i) = \mu_i$ and $g(\mu_i) = \eta_i = x_i^\top \beta$. The slope of the log-likelihood, called the score, is

$$U_j = \frac{\partial \ell}{\partial \beta_j} = \sum_{i=1}^{n} \frac{y_i - \mu_i}{\mathrm{Var}(Y_i)} \frac{\partial \mu_i}{\partial \eta_i} x_{ij}.$$

The information matrix is $I = X^\top W X$, where W is a diagonal matrix of weights:

$$w_{ii} = \frac{1}{\mathrm{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i}\right)^2.$$

The scoring method updates the estimate with $\beta^{(m)} = \beta^{(m-1)} + [I^{(m-1)}]^{-1} U^{(m-1)}$. This can be rewritten as a weighted least-squares problem:

$$X^\top W^{(m-1)} X\,\beta^{(m)} = X^\top W^{(m-1)} z^{(m-1)}, \qquad z_i = \eta_i + (y_i - \mu_i)\frac{\partial \eta_i}{\partial \mu_i}.$$

The weights W and the variable z depend on β. So we cannot solve this once. We start with a guess, solve, update the weights, and repeat. We stop when the estimates hardly change. This is called iteratively reweighted least squares (IRLS), and it is what R's `glm()` does (Schall, nd, p. 42). Other methods also work, such as Newton's method and L-BFGS (Wikipedia, nda).

Only two things change from model to model: $\mathrm{Var}(Y_i)$ and $\partial\mu_i/\partial\eta_i$. For a Bernoulli response, $\mathrm{Var}(Y_i) = \mu_i(1-\mu_i)$. With the logit link, $\partial\mu_i/\partial\eta_i = e^{\eta_i}/(1+e^{\eta_i})^2$. The weights then become $w_{ii} = e^{\eta_i}/(1+e^{\eta_i})^2$ (Schall, nd, pp. 45–46). With the probit link, $\partial\mu_i/\partial\eta_i = \varphi(\eta_i)$, the standard normal density. The weights are then $w_{ii} = \varphi(\eta_i)^2/\{\Phi(\eta_i)[1-\Phi(\eta_i)]\}$. It is the same algorithm with different weights.

For probit, the log-likelihood has a single peak. Standard optimizers therefore find it quickly. Under the usual conditions the estimator is consistent, approximately normal and efficient (Wikipedia, ndb).

Sometimes the fit fails to converge. Common causes are too many predictors for the number of cases, predictors that are highly correlated, empty cells with categorical predictors, and complete separation. Complete separation means the predictors sort every case perfectly. Then the likelihood only reaches its maximum when the coefficients are infinite (Wikipedia, nda).

Inference. In large samples, $\hat\beta \sim N(\beta, (X^\top W X)^{-1})$. This gives us Wald tests and Wald confidence intervals (Schall, nd, Sec. 4.5). The Wald test can behave badly when coefficients are large or data are sparse. The likelihood-ratio test is the recommended alternative. It compares the deviances of two nested models (Wikipedia, nda). Likelihood-ratio confidence intervals work in the same spirit (Schall, nd, Sec. 4.7).

6.6 Goodness of fit

Deviance. To check a model, we compare it with the *saturated* model. The saturated model uses the same distribution and link, but it has one parameter for each covariate pattern. So it fits the data as well as any model can. The deviance is $D = 2[\ell(b_{\max}; y) - \ell(b; y)]$ (Schall, nd, Sec. 4.6). For binomial data it becomes

$$D = 2\sum_{i=1}^{N}\left[y_i \log\frac{y_i}{n_i\hat\pi_i} + (n_i - y_i)\log\frac{n_i - y_i}{n_i - n_i\hat\pi_i}\right].$$

Pearson chi-square. A second measure is the Pearson statistic:

$$X^2 = \sum_{i=1}^{N}\frac{(y_i - n_i\hat\pi_i)^2}{n_i\hat\pi_i(1-\hat\pi_i)}.$$

It comes from fitting by weighted least squares instead of maximum likelihood. For large samples, D and X^2 are about the same. If the model is correct, both follow roughly a $\chi^2(N-p)$ distribution (Schall, nd, pp. 78–79). This gives a handy rule of thumb. Both statistics should be about as large as their degrees of freedom (Schall, nd, p. 80). Many people prefer X^2. The deviance is thrown off by very small counts. Both measures work poorly when expected counts are tiny, below about 1 (Dobson and Barnett, 2018, p. 163).

When the data are 0/1. Sometimes every observation has its own covariate pattern. This often happens with continuous predictors. Then each y_i is 0 or 1, and neither D nor X^2 tells us much (Dobson and Barnett, 2018, p. 163). The usual fix is the Hosmer–Lemeshow test. We sort

the observations by predicted probability and split them into about ten groups of similar size. Then we compute a Pearson statistic on the $g \times 2$ table of groups and compare it with $\chi^2(g-2)$ (Dobson and Barnett, 2018, pp. 163–164). Some statisticians now call this test obsolete. The groups are chosen arbitrarily, and the test has low power (Wikipedia, nda).

Comparing with a model that has no predictors. Let $\tilde{\pi} = \sum y_i / \sum n_i$ be the single probability from the intercept-only model. The likelihood-ratio chi-square is $C = 2[\ell(\hat{\pi}; y) - \ell(\tilde{\pi}; y)]$. It tests whether any explanatory variable is needed. If none is, C is roughly $\chi^2(p-1)$ (Dobson and Barnett, 2018, p. 164). A pseudo-R^2 is also common:

$$\text{pseudo-}R^2 = \frac{\ell(\tilde{\pi}; y) - \ell(\hat{\pi}; y)}{\ell(\tilde{\pi}; y)}.$$

It shows how much the log-likelihood improves over the intercept-only model. Be careful with it. Its distribution is hard to find, and it goes up whenever we add parameters. It can also look surprisingly small for a model that fits well, because it measures how well we predict single outcomes (Dobson and Barnett, 2018, p. 164). There are several versions (Cox–Snell, Nagelkerke, McFadden and Tjur), and none is the accepted standard (Wikipedia, nda).

Information criteria. AIC and BIC reward a good fit but charge a price for each extra parameter p:

$$\text{AIC} = -2\ell(\hat{\pi}; y) + 2p, \qquad \text{BIC} = -2\ell(\hat{\pi}; y) + p\ln(\text{number of observations}).$$

R uses these exact formulas. Stata divides AIC by the number of subgroups, so values from the two programs cannot be compared (Dobson and Barnett, 2018, p. 165). These criteria are not meant for testing nested models. They are very useful for comparing models that are *not* nested, such as logit against probit (Dobson and Barnett, 2018, p. 165). Lower values are better.

6.7 Residuals

There are two main kinds of residuals for logistic regression. They match the two fit statistics above. Suppose there are m covariate patterns. For pattern k, let y_k be the successes, n_k the trials and $\hat{\pi}_k$ the fitted probability. The *Pearson* and *deviance* residuals are

$$X_k = \frac{y_k - n_k\hat{\pi}_k}{\sqrt{n_k\hat{\pi}_k(1-\hat{\pi}_k)}},$$

$$d_k = \operatorname{sign}(y_k - n_k\hat{\pi}_k)\left\{2\left[y_k \log\frac{y_k}{n_k\hat{\pi}_k} + (n_k - y_k)\log\frac{n_k - y_k}{n_k - n_k\hat{\pi}_k}\right]\right\}^{1/2}.$$

Their squares add up to the fit statistics: $\sum X_k^2 = X^2$ and $\sum d_k^2 = D$ (Schall, nd, p. 79).

Each residual can be standardised by dividing by $\sqrt{1-h_k}$. Here h_k is the leverage, the k-th diagonal element of the hat matrix $H = W^{1/2}X(X^\top WX)^{-1}X^\top W^{1/2}$. This gives the standardised residuals r_{Pk} and r_{Dk}.

We use them as we would in linear regression. We plot them against each continuous predictor to check that the relationship is linear on the link scale. We plot them against variables left out of the model. We plot them in time order to look for patterns over time. Standardised residuals should look roughly like a standard normal variable if the n_k are not too small. So a normal probability plot is also useful (Dobson and Barnett, 2018, p. 166).

With 0/1 data, or very small n_k, the residuals take only a few values, so the plots say little.

In that case we rely on the overall statistics instead (Dobson and Barnett, 2018, pp. 166–167). Influence measures such as delta-beta, delta-chi-squared and delta-deviance can also help (Dobson and Barnett, 2018, p. 167).

6.8 Overdispersion

Overdispersion means the data vary more than the binomial model allows. If $Y \sim \text{Bin}(\pi, M)$, the variance should be $M\pi(1-\pi)$. In real data it is often larger. The lecture notes suggest we assume some overdispersion is there until we show it is not. To check, we compare D or X^2 with their degrees of freedom. A χ^2 variable with $N-p$ degrees of freedom has an average value of $N-p$. So a statistic far above that is a warning sign (Schall, nd, p. 80). Dobson and Barnett give the same advice. They add that the cause can be a missing variable, the wrong link, or responses that are not independent (Dobson and Barnett, 2018, p. 167).

A common cause: clusters. Families, households, litters and colonies are all clusters. Members of a cluster tend to be alike. Suppose M people come from M/k clusters of size k. The success probability changes from cluster to cluster, with $E(\pi_i) = \pi$ and $\text{Var}(\pi_i) = \tau^2\pi(1-\pi)$. Then

$$E(Y) = M\pi, \qquad \text{Var}(Y) = M\pi(1-\pi)\left[1 + (k-1)\tau^2\right] = \sigma^2 M\pi(1-\pi).$$

So the variance is multiplied by a dispersion parameter $\sigma^2 = 1 + (k-1)\tau^2$. It depends on the cluster size and on how much the clusters differ. It does not depend on the total sample size (Schall, nd, pp. 80–81). The beta-binomial distribution is another option. In it the variance ratio grows with M. McCullagh and Nelder found the constant factor worked better in their examples (Schall, nd, p. 81).

What to do about it. We do not need to pick a special distribution. We assume the mean is unchanged and the variance is multiplied by an unknown σ^2. We fit the model as usual. Only the sampling distribution changes: $\hat{\beta} \sim N(\beta, \sigma^2(X^\top W X)^{-1})$. Standard errors become larger by a factor of σ, and Wald statistics are divided by σ^2 (Schall, nd, p. 81). We estimate the dispersion with

$$\hat{\sigma}^2 = \frac{X^2}{N-p} \qquad \text{or} \qquad \hat{\sigma}^2 = \frac{D}{N-p}.$$

The Pearson version is preferred. The deviance version is unreliable when the n_i are close to one (Schall, nd, p. 82). In R, this is the `quasibinomial` family. It adds a parameter ϕ so that $\text{Var}(Y_i) = n_i\pi_i(1-\pi_i)\phi$ (Dobson and Barnett, 2018, p. 167).

The notes give an example with ticks on sheep, where each sheep is a cluster. Both X^2 and D were almost six times their degrees of freedom. That is a clear case of overdispersion (Schall, nd, p. 84). Note that you cannot check for overdispersion in 0/1 data that has not been grouped.

Reflection | **Application**

The two models side by side

Everything in the last section applies to both models. The one real difference is the tolerance distribution. It decides the link and how we read the coefficients. Table 6.2 sums this up.

	Probit	**Logit**
Tolerance distribution	Normal(μ, σ^2)	Logistic
Tolerance density $f(s)$	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{s-\mu}{\sigma}\right)^2\right]$	$\frac{\beta_2\, e^{\beta_1+\beta_2 s}}{[1+e^{\beta_1+\beta_2 s}]^2}$
Probability π **(the cdf)**	$\Phi(\beta_1 + \beta_2 x)$	$\frac{e^{\beta_1+\beta_2 x}}{1+e^{\beta_1+\beta_2 x}}$
Link function $g(\pi)$	$\Phi^{-1}(\pi)$	$\log\left[\frac{\pi}{1-\pi}\right]$
Linear predictor	$\Phi^{-1}(\pi) = \beta_1 + \beta_2 x$	$\log\left[\frac{\pi}{1-\pi}\right] = \beta_1 + \beta_2 x$
Parameters and tolerance	$\beta_1 = -\mu/\sigma,\ \beta_2 = 1/\sigma$	location $-\beta_1/\beta_2$, scale $1/\beta_2$
Hidden error ε	Standard normal	Standard logistic (heavier tails)
Median effective dose	$-\beta_1/\beta_2$	$-\beta_1/\beta_2$
Meaning of a coefficient	Change in the normal quantile (z-score) per unit of x	Change in log-odds per unit of x; e^β is an odds ratio
Estimation	Maximum likelihood by IRLS	Maximum likelihood by IRLS (weights $\mu_i(1-\mu_i)$ for 0/1 data)
Origin	Bliss (1934), dose-response studies	Berkson (1944), now the default in many fields

Table 6.2.: Probit and logit compared (after Dobson and Barnett, 2018, Schall, nd, Wikipedia, ndb and Wikipedia, nda).

Reflection

In most cases, the choice between the two hardly changes the fitted probabilities. The two curves look almost the same, except in the tails. The textbook says logit gives results very much like probit, and it is a little easier to compute (Dobson and Barnett, 2018, p. 152).
The history is interesting. Probit came first and was popular for dose-response work. At first, logit was seen as the weaker choice. By 1970 it was as common as probit in statistics journals, and later it became more common. This was mainly because logit spread into other fields and is easy to compute and explain. It did not replace probit in dose-response work (Wikipedia, nda).
So how should we choose? Here are some simple guidelines:

- If we want to report odds ratios, as is common in medicine, we use logit. The number e^β has a clear meaning.
- If we believe a hidden normal trait crosses a threshold, probit fits that idea.
- If both seem fine, we can let AIC break the tie.

One warning: the coefficients are on different scales. We should not compare a logit coefficient directly with a probit coefficient. The logit one should be about 1.8 times bigger. What we *can* compare are the fitted probabilities, the effects on the probability, and the value of x where $\pi = 0.5$.

Example in R: seed germination

The scenario. A grower plants 50 seeds at each of eight soil-moisture levels, from 10% to 45%. This is done in two soil types, sandy and loam. The grower then counts how many seeds germinate. We want to know three things. Do moisture and soil type affect germination? Does a logit or a probit link describe the data better? Do the model checks look fine?
The data are *simulated* in the script, and `set.seed(2026)` makes the results repeatable. They were generated from a logistic model, so a logit win would not be a surprise. We should keep that in mind when we read the AIC result. To use your own data, replace the `dat` data frame. It needs a column of successes (`germ`), a column of trials (`n`) and the predictors.
The data are grouped: 16 covariate patterns with 50 seeds each. This means the deviance, the Pearson statistic, the residuals and the overdispersion check all give useful answers. They would not for ungrouped 0/1 data.
The code.

```
# ------------------------------------------------------------------
# Logit vs probit on a simulated seed-germination experiment
# ------------------------------------------------------------------
# Scenario: seeds are planted at 8 soil-moisture levels in two soil
# types (sandy, loam). 50 seeds per combination; we count how many
# germinate. Data are simulated so the example is fully reproducible.

set.seed(2026)

dat <- expand.grid(moisture = seq(10, 45, by = 5),      # soil moisture (%)
                   soil     = c("sandy", "loam"))       # soil type
dat$n    <- 50                                          # seeds planted
eta_true <- -5 + 0.20 * dat$moisture + 0.8 * (dat$soil == "loam")
dat$germ <- rbinom(nrow(dat), size = dat$n, prob = plogis(eta_true))
dat$prop <- dat$germ / dat$n
head(dat)

# --- 1. Fit both models: same linear predictor, different link ----
fit_logit  <- glm(cbind(germ, n - germ) ~ moisture + soil,
                  family = binomial(link = "logit"),  data = dat)
fit_probit <- glm(cbind(germ, n - germ) ~ moisture + soil,
                  family = binomial(link = "probit"), data = dat)

summary(fit_logit)
summary(fit_probit)

# --- 2. Compare with AIC (same number of parameters, p = 3) -------
comp <- data.frame(
  Model     = c("Logit", "Probit"),
  LogLik    = c(logLik(fit_logit), logLik(fit_probit)),
  Deviance  = c(deviance(fit_logit), deviance(fit_probit)),
  df        = c(df.residual(fit_logit), df.residual(fit_probit)),
  AIC       = c(AIC(fit_logit), AIC(fit_probit)),
  BIC       = c(BIC(fit_logit), BIC(fit_probit))
)
comp$dAIC <- comp$AIC - min(comp$AIC)
print(comp, digits = 4, row.names = FALSE)

# --- 3. Interpretation on the odds scale (logit only) -------------
round(exp(cbind(OR = coef(fit_logit), confint(fit_logit))), 3)


# --- 3b. Moisture level giving a 50% germination rate (-b0/b1) ------
b_l <- coef(fit_logit); b_p <- coef(fit_probit)
m50 <- rbind(Logit  = c(sandy = unname(-b_l[1] / b_l[2]),
                        loam  = unname(-(b_l[1] + b_l[3]) / b_l[2])),
```

```
             Probit = c(sandy = unname(-b_p[1] / b_p[2]),
                        loam  = unname(-(b_p[1] + b_p[3]) / b_p[2])))
round(m50, 2)
round(b_l / b_p, 2)         # logit / probit coefficient ratio (about 1.8)

# --- 4. Goodness of fit: deviance and Pearson X^2 vs chi-square ---
gof <- function(fit) {
  X2  <- sum(residuals(fit, type = "pearson")^2)
  dev <- deviance(fit)
  df  <- df.residual(fit)
  c(Deviance = dev, Pearson_X2 = X2, df = df,
    p_deviance = pchisq(dev, df, lower.tail = FALSE),
    p_pearson  = pchisq(X2,  df, lower.tail = FALSE))
}
round(rbind(Logit = gof(fit_logit), Probit = gof(fit_probit)), 3)

# McFadden-type pseudo R^2 (log-lik of fitted vs intercept-only model)
null_logit <- update(fit_logit, . ~ 1)
1 - as.numeric(logLik(fit_logit)) / as.numeric(logLik(null_logit))

# --- 5. Over-dispersion check ---------------------------------------
phi_hat <- sum(residuals(fit_logit, type = "pearson")^2) / df.residual(fit_
    logit)
phi_hat                                      # should be close to 1
fit_quasi <- update(fit_logit, family = quasibinomial(link = "logit"))
summary(fit_quasi)$dispersion                # same idea, estimated by R

# --- 6. Residuals ---------------------------------------------------
dat$res_dev   <- residuals(fit_logit, type = "deviance")
dat$res_pear  <- residuals(fit_logit, type = "pearson")
dat$res_stdev <- rstandard(fit_logit, type = "deviance")   # divides by sqrt
    (1 - h)
cbind(dat[, c("moisture", "soil")],
      round(dat[, c("prop", "res_pear", "res_dev", "res_stdev")], 3))

# --- 7. Figure: fitted curves and residuals -------------------------
grid <- expand.grid(moisture = seq(10, 45, length.out = 200),
                    soil = c("sandy", "loam"))
grid$logit  <- predict(fit_logit,  grid, type = "response")
grid$probit <- predict(fit_probit, grid, type = "response")

pdf("fig_fit.pdf", width = 7.5, height = 3.3)
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
cols <- c(sandy = "grey30", loam = "firebrick")
plot(dat$moisture, dat$prop, pch = ifelse(dat$soil == "sandy", 16, 17),
     col = cols[as.character(dat$soil)],
     xlab = "Soil moisture (%)", ylab = "Proportion germinated",
     main = "Fitted curves")
for (s in c("sandy", "loam")) {
  g <- grid[grid$soil == s, ]
  lines(g$moisture, g$logit,  col = cols[s], lwd = 2, lty = 1)
  lines(g$moisture, g$probit, col = cols[s], lwd = 2, lty = 2)
}
legend("topleft", bty = "n", cex = 0.8,
       legend = c("sandy", "loam", "logit", "probit"),
       col = c(cols, "black", "black"), pch = c(16, 17, NA, NA),
       lty = c(NA, NA, 1, 2), lwd = 2)
plot(fitted(fit_logit), dat$res_stdev, pch = 16,
     xlab = "Fitted probability", ylab = "Standardised deviance residual",
     main = "Residuals (logit model)")
abline(h = c(-2, 0, 2), lty = c(3, 1, 3))
dev.off()
```

The estimates. Both models use the same linear predictor, $\beta_0 + \beta_1$ moisture $+ \beta_2$ loam. So we can put their coefficients next to each other.

	Logit (SE)	Probit (SE)	Ratio
Intercept	-5.395 (0.393)	-3.025 (0.200)	1.78
Moisture (per 1%)	0.2138 (0.0143)	0.1198 (0.0071)	1.78
Soil: loam vs. sandy	0.687 (0.214)	0.388 (0.119)	1.77

Table 6.3.: Coefficient estimates from the two fits. The ratio is the logit estimate divided by the probit estimate.

The ratio is about 1.8 for every coefficient. This is the scale difference we expected from the spread of the logistic distribution.
In the logit model, each extra percentage point of moisture multiplies the odds of germination by $e^{0.2138} = 1.24$ (95% interval 1.21 to 1.28). Loam soil has about twice the odds of sandy soil (odds ratio 1.99, 95% interval 1.31 to 3.04). Half of the seeds are expected to germinate at the moisture level $-\hat{\beta}_0/\hat{\beta}_1$. The two links give almost the same answer. For sandy soil it is 25.2% (logit 25.24, probit 25.25). For loam it is 22.0% (logit 22.02, probit 22.01).
Comparing the models.

Model	Log-lik.	Deviance	Pearson X^2	AIC	BIC	ΔAIC
Logit	-32.40	15.74	14.52	70.81	73.13	0.00
Probit	-33.62	18.17	18.74	73.24	75.56	2.43

Table 6.4.: Fit statistics. Both models have 13 residual degrees of freedom and $p = 3$ parameters.

Both models have three parameters. So the AIC comparison only depends on the log-likelihoods. The logit model has the lower AIC, by about 2.4 points. That is a small edge for logit, and it is not surprising given how we made the data. It is not a final verdict, because the probit model also fits acceptably. Figure 6.4 (left) shows how close the two curves are.
Fit, overdispersion and residuals. The logit deviance is 15.74 on 13 degrees of freedom ($p = 0.26$). The Pearson statistic is 14.52 ($p = 0.34$). Both are close to the degrees of freedom, so there is no sign of a poor fit. The probit values are slightly worse but still fine (deviance $p = 0.15$, Pearson $p = 0.13$).
The estimated dispersion is $\hat{\sigma}^2 = X^2/(N - p) = 1.12$. R's `quasibinomial` fit gives the same number. This is close to 1, so overdispersion is not a worry here, and the usual standard errors can be trusted. That makes sense, because our simulated data have no clusters. With clustered data, like the sheep in the lecture notes, we would expect a value well above 1. We would then report the quasibinomial standard errors. The McFadden-type pseudo-R^2 is high (0.89). As noted earlier, we should treat it as a rough guide only.
The right-hand panel of Figure 6.4 shows the standardised deviance residuals. They show no trend against the fitted probability. Two are right at ± 2: 2.01 for sandy soil at 35% moisture, and -1.99 for loam at 20%. With 16 residuals, one or two this far out is normal.

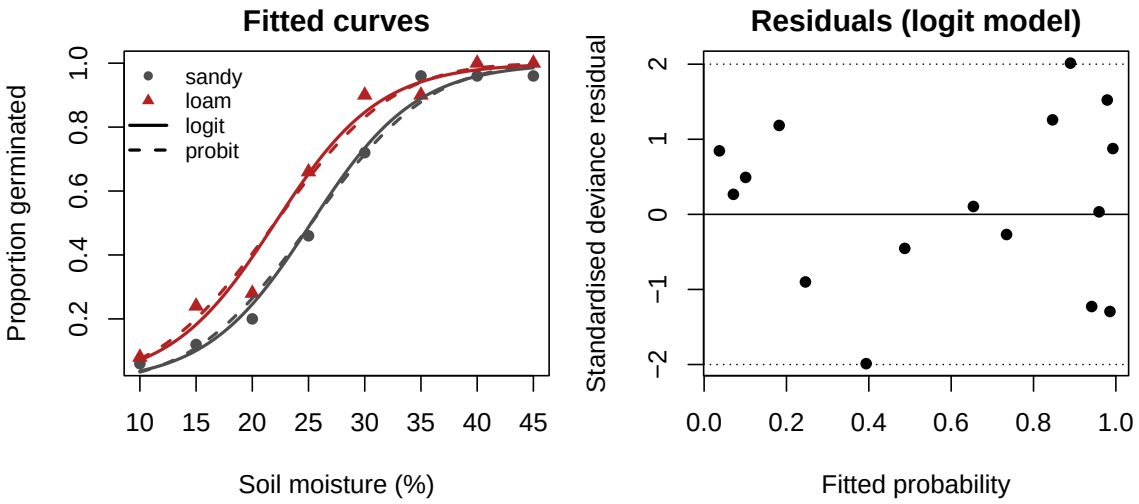


Figure 6.4.: Left: observed germination proportions with the logit (solid) and probit (dashed) fitted curves. Right: standardised deviance residuals from the logit model, plotted against the fitted probability.

Summary. The two links gave almost the same curves and the same 50% points. The coefficients differed only by the expected factor of about 1.8. AIC slightly favoured logit. The model checks found no problems with either model. In a case like this, we would probably pick logit and report the odds ratios.

CHAPTER 7

Generalized Linear Models for Count Data

This chapter moves through four phases: Exploration, Cognition, Reflection and Application. The first phase is the **Exploration phase**, where we get a feel for the topic through short explainer videos before any formal theory.

Exploration

We begin with a video that explains Poisson regression.

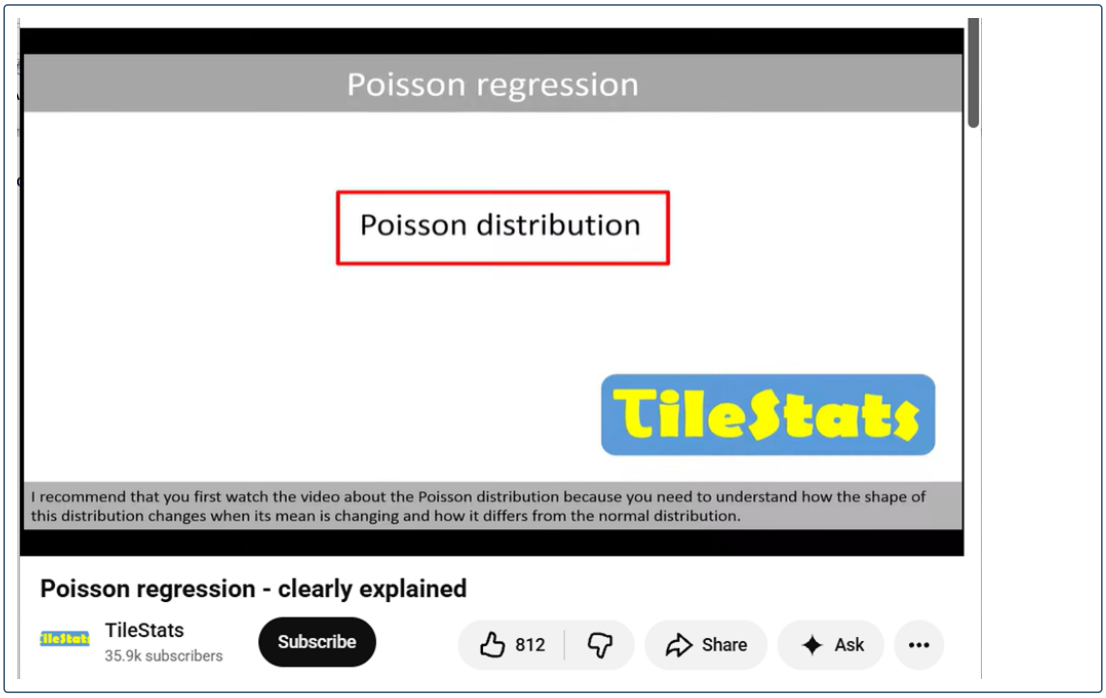


Figure 7.1.: Poisson regression explainer (TileStats, 2021).

Source: TileStats. (2021, March 18), *Poisson Regression – Clearly Explained*, YouTube. https://www.youtube.com/watch?v=0bpz_Uvo2rQ

Video Summary

Why do we bother learning about Poisson regression? First, let us revise what the Poisson distribution does: it gives the probability of a number of events happening within a certain time period or a defined space (area), for example the number of car accidents per year, or the number of needles falling between two lines.
For this explainer we look at an example of rabbit births over the period of spring, and as usual we try to fit a linear regression, but we soon encounter some problems:

- The line does not fit the data points well, since count data usually has high peaks and low tails.
- The line also predicts negative counts of births, and you cannot have negative births.

So how do we intend to solve this? Let us try fitting an exponential model to the data,

$$y = e^{\beta_0 + \beta_1 \text{ week}}$$

where e^{β_0} is the intercept and e^{β_1} is the multiplicative factor. This solves both problems, but now we are left with the error term, which is normally distributed and therefore assumes that the spread is constant along the curve,

$$y = e^{\beta_0 + \beta_1 \text{ week}} + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2)$$

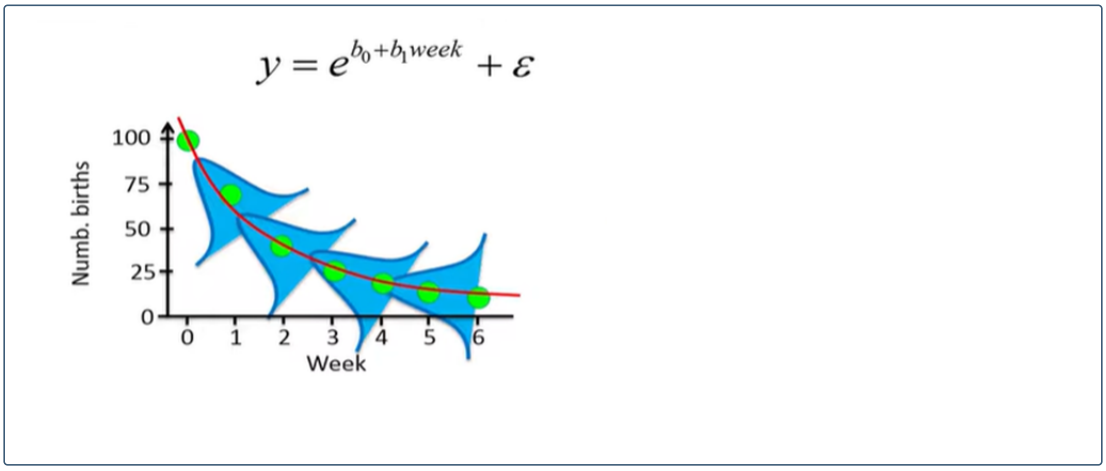


Figure 7.2.: Exponential model fitted to rabbit births, with normally distributed error around the curve (TileStats, 2021).

Source: TileStats. (2021, March 18), *Poisson Regression – Clearly Explained*, YouTube.

This leaves two additional problems:

- The variance of count data is not constant: higher counts vary more, so the variance around the curve is not constant.
- A normally distributed error term still allows negative values.

To solve this we need a distribution that does not take negative values and whose variance increases as the expected count increases. The standard choice is the Poisson distribution, because its variance equals its mean. In the case where we cannot assume the two are equal (overdispersion), we use the Negative Binomial instead.
Because Poisson regression is a GLM, it has three components:

- **Probability distribution:** Poisson.
- **Linear predictor:** $\eta = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n$, so that the mean is $\mu = e^{\eta}$.
- **Link function:** the log link, $\ln(\mu) = \eta$.

Before we dive into the Cognition phase, let us explore the idea of a multinomial distribution.



Figure 7.3.: Multinomial distribution explainer (Whats Up Dude, 2020).

Source: Whats Up Dude. (2020), *What Is The Multinomial Probability Distribution Formula Example Explained*, YouTube. https://youtu.be/BZxCIkSkMgo

Video Summary

We have heard of the binomial time and time again. The binomial has one distinct characteristic: it has two outcomes for a fixed number of trials. Now we compare the multinomial model to it.

Multinomial

- More than two outcomes with a fixed number of trials.
- It must have a specific number of outcomes, the trials must be independent, and the probabilities of the outcomes must stay fixed.

Multinomial Formula

$$P(X) = \frac{n!}{\prod_{j=1}^{k} x_j!} \prod_{j=1}^{k} p_j^{x_j}, \qquad \text{where } \sum_{j=1}^{k} x_j = n \text{ and } \sum_{j=1}^{k} p_j = 1.$$

Example

For better intuition, let us do a worked example. Assume we gather data on people who go to the gym: 20% do strength training, 40% cardio, 15% yoga and 25% sport. If 8 people walk into the gym, what is the probability that 1 person does strength training, 4 do cardio, 1 does yoga, and 2 do sport?

Event ID	Activity	Count	Probability
E_1	Strength training	1	0.20
E_2	Cardio	4	0.40
E_3	Yoga	1	0.15
E_4	Sport	2	0.25

Table 7.1.: Gym activity outcomes and their probabilities.

Therefore the probability is

$$\frac{8!}{1!\,4!\,1!\,2!}\,(0.2)^1(0.4)^4(0.15)^1(0.25)^2 \;=\; 840 \times 0.000048 \;\approx\; 0.04\;(4\%).$$

So we can say there is a 4 percent chance that, of 8 people walking into the gym, 1 does strength training, 4 do cardio, 1 does yoga, and 2 do sport.
This is the basic application of the multinomial model; we will see more of it in the Cognition phase.

Cognition

The first phase, exploration, gave us the idea. This second phase gives us the maths behind it. We start with the Poisson distribution, then connect it to the multinomial from the second video. After that we build the regression model, see how it is fitted, and check it. We finish with overdispersion, because that is where the Poisson model most often breaks.

Count data and the Poisson distribution

Definition: Poisson distribution

A random variable Y that counts events follows a Poisson distribution with mean μ if

$$f(y) = \frac{\mu^y e^{-\mu}}{y!}, \qquad y = 0, 1, 2, \ldots$$

We write $Y \sim \mathrm{Po}(\mu)$. Its mean and its variance are the same number: $\mathbb{E}(Y) = \mathrm{Var}(Y) = \mu$ (Dobson and Barnett, 2018, p. 197).

The last sentence of the box is the key to everything that follows. If the average count is larger, the spread is larger too. This is exactly the behaviour the rabbit-birth data needed in the exploration phase. It also means the Poisson model has no spare parameter for the spread. Once we choose μ, the variance is decided. When the real data disagree, we have a problem, and we deal with it in the last section.
The mean μ is usually a *rate*. We might count crashes per 1 000 drivers, or births per colony per week. Counts are then measured per unit of *exposure*, such as person-years at risk or the number of cars insured (Schall, nd, pp. 89–90). Two situations come up often (Schall, nd, pp. 89–90):

- The exposure differs between observations. We then use Poisson regression, with the exposure as an offset (see below).
- The exposure is the same for every observation and the predictors are mostly categories. The data then fit in a table, and we use log-linear models.

From the multinomial back to the Poisson

The multinomial from the second video fixes the total number of trials n. The Poisson does not fix any total. The two are closely related. Suppose $Y_1, \dots, Y_k$ are independent with $Y_j \sim \mathrm{Po}(\mu_j)$. Their sum $T = \sum_j Y_j$ is $\mathrm{Po}(\sum_j \mu_j)$. If we now *condition* on the total being n, we get

$$\Pr(Y_1 = y_1, \dots, Y_k = y_k \mid T = n) = \frac{\prod_j e^{-\mu_j} \mu_j^{y_j} / y_j!}{e^{-\sum \mu_j} \left(\sum \mu_j\right)^n / n!} = \frac{n!}{\prod_j y_j!} \prod_{j=1}^{k} \left(\frac{\mu_j}{\sum_l \mu_l} \right)^{y_j}.$$

This is a multinomial with probabilities $p_j = \mu_j / \sum_l \mu_l$.
Think of the gym example. If people arrive at the gym at random, the number choosing each activity is Poisson. Once we know that 8 people came in, the split between strength training, cardio, yoga and sport is multinomial.
For a table of counts there are three standard probability models (Schall, nd, pp. 99–100):

- **Poisson:** the cell counts are independent, with no restrictions.
- **Multinomial:** only the grand total n is fixed.
- **Product multinomial:** more margins are fixed, for example the total in each row, which happens in a clinical trial with fixed group sizes.

All three are built on the Poisson distribution. This is why one modelling tool covers all of them.

The Poisson regression model

Let Y_i be the number of events for the i-th covariate pattern, observed over an exposure n_i. The expected count is $\mu_i = n_i \theta_i$, where θ_i is the rate per unit of exposure. We let the predictors act on the rate through an exponential function, $\theta_i = e^{x_i^\top \beta}$. In GLM language there are three parts, and the first matches the list in the exploration phase (Schall, nd, pp. 90–91):

1. **Distribution:** $Y_i \sim \mathrm{Po}(\mu_i)$, independent.
2. **Linear predictor:** $\eta_i = x_i^\top \beta$.
3. **Link:** the natural logarithm.

Putting them together gives

$$\log \mu_i = \underbrace{\log n_i}_{\text{offset}} + x_i^\top \beta \qquad \Longleftrightarrow \qquad \mu_i = n_i \, e^{x_i^\top \beta}.$$

The term $\log n_i$ is called the *offset*. It is a known number, so it is easy to put into the fitting. When every exposure is equal, as in the rabbit data where each colony is watched for one week at a time, the offset is the same for everyone and can be left out (Schall, nd, pp. 90–91).
Why the log link? It is the natural link for the Poisson distribution (Dobson and Barnett, 2018, p. 199), and it fixes the two problems from the first video. The mean $\mu_i = e^{\eta_i}$ can never be negative. And the variance μ_i grows with the mean, so higher counts are allowed to vary more.
Reading the coefficients. The log link makes effects multiply. For a 0/1 predictor the rate ratio of “present” against “absent” is

$$\mathrm{RR} = \frac{\mathbb{E}(Y_i \mid \text{present})}{\mathbb{E}(Y_i \mid \text{absent})} = e^{\beta_j},$$

as long as the other predictors stay the same. For a continuous predictor, one extra unit multiplies the rate by e^{β_k} (Schall, nd, pp. 91–92). In the rabbit data, a coefficient of -0.25 for `week` means the expected count is multiplied by $e^{-0.25} \approx 0.78$ each week, a drop of about 22%.

Parameter estimation

We use maximum likelihood. For independent Poisson counts the log-likelihood is

$$\ell(\mu; y) = \sum_{i=1}^{N} \big[y_i \log \mu_i - \mu_i - \log(y_i!) \big].$$

Take the slope with respect to β_j and use $\partial \mu_i / \partial \eta_i = \mu_i$ for the log link. The score collapses to something very simple (Schall, nd, pp. 43–44):

$$U_j = \frac{\partial \ell}{\partial \beta_j} = \sum_{i=1}^{N} (y_i - \mu_i)\, x_{ij}.$$

Look at what this equation contains. The data enter only through y_i, and the model enters only through the mean μ_i. The variance does not appear. This is why the coefficient estimates stay sensible even when the variance is wrong, which the first question in the Reflection phase asks about. If the model has an intercept, the score for it says $\sum y_i = \sum \hat{\mu}_i$: the fitted counts add up to the observed total.
We cannot solve $U_j = 0$ in closed form, so we use iteratively reweighted least squares (IRLS). Each round solves the weighted least-squares equations

$$X^\top W^{(m-1)} X\, \beta^{(m)} = X^\top W^{(m-1)} z^{(m-1)},$$

where, for the Poisson model with log link (Schall, nd, pp. 44–45),

$$W = \mathrm{diag}(\mu_i), \qquad z_i = \eta_i + \frac{y_i - \mu_i}{\mu_i}.$$

The weights and z depend on the current estimate, so we repeat until the estimates stop changing. This is what `glm()` does in R.
The same matrix gives the uncertainty. In large samples $\hat{\beta} \approx N\big(\beta, (X^\top W X)^{-1}\big)$. So the standard error of $\hat{\beta}_j$ comes from the j-th diagonal element of $(X^\top W X)^{-1}$, and

$$\frac{\hat{\beta}_j - \beta_j}{\mathrm{s.e.}(\hat{\beta}_j)} \sim N(0, 1)$$

approximately (Schall, nd, p. 92). This gives Wald tests and confidence intervals. We can also use the score or likelihood-ratio statistic, or compare nested models by their fit (Dobson and Barnett, 2018, p. 199). Here is the catch. Because $W = \mathrm{diag}(\mu_i)$ was built from $\mathrm{Var}(Y_i) = \mu_i$, these standard errors are only right if the variance really equals the mean.

Residuals and goodness of fit

Write $O_i = y_i$ for the observed counts and $E_i = \hat{\mu}_i = n_i e^{x_i^\top \hat{\beta}}$ for the fitted counts. Since $\mathrm{Var}(Y_i) = \mathbb{E}(Y_i)$, the standard error of Y_i is estimated by $\sqrt{E_i}$. So the *Pearson residual* is

$$r_i = \frac{O_i - E_i}{\sqrt{E_i}}, \qquad X^2 = \sum_{i=1}^{N} r_i^2 = \sum_{i=1}^{N} \frac{(O_i - E_i)^2}{E_i}.$$

The sum of squares is the usual Pearson chi-square statistic (Schall, nd, p. 92). The *deviance residual* and the deviance are

$$d_i = \mathrm{sign}(O_i - E_i)\sqrt{2\left[O_i \log \frac{O_i}{E_i} - (O_i - E_i)\right]}, \qquad D = \sum_{i=1}^{N} d_i^2.$$

When the model has an intercept, $\sum O_i = \sum E_i$, so the second part drops out and $D = 2\sum O_i \log(O_i/E_i)$ (Schall, nd, p. 93). Both residuals can be made more even by dividing by $\sqrt{1-h_i}$, where h_i is the leverage.

A Taylor expansion shows that D and X^2 are close to each other (Schall, nd, p. 93). There is good news for Poisson data: no extra parameter like σ^2 in the normal model is needed, so both can be used directly as measures of fit. If the model is right, each follows roughly a $\chi^2(N-p)$ distribution, where p is the number of fitted parameters. The approximation is usually better for X^2 than for D (Dobson and Barnett, 2018, pp. 93, 200–201). This gives a simple rule of thumb: both should be about as large as their degrees of freedom. *The catch is the same as before. The rule only works if there is no overdispersion.*

Two more summaries are common. The likelihood-ratio statistic $C = 2[\ell(b) - \ell(b_{\min})]$ compares the fitted model with a model that has only an intercept. It tests whether all slopes are zero, against $\chi^2(p-1)$. A rough measure of fit is the pseudo-$R^2 = [\ell(b_{\min}) - \ell(b)]/\ell(b_{\min})$ (Dobson and Barnett, 2018, p. 201).

Log-linear models for tables of counts

When the counts fill a table, the log link turns the model into a sum. Since $\mathbb{E}(Y_i)$ is a product of parameters and offsets, $\log \mathbb{E}(Y_i) = \text{constant} + x_i^\top \beta$. Any model of this form is a *log-linear model* (Schall, nd, p. 101).

The standard use is to test whether two classifying variables are independent. They are independent if each cell probability is a product of the margins, $\theta_{jk} = \theta_{j+}\theta_{+k}$. On the log scale we compare two models:

$$\begin{aligned} \log \mathbb{E}(Y_{jk}) &= \log n + \log \theta_{j+} + \log \theta_{+k} \quad \text{(independence)}, \\ \log \mathbb{E}(Y_{jk}) &= \log n + \log \theta_{jk} \quad \text{(saturated)}. \end{aligned}$$

This works like a two-way ANOVA: the first has main effects only, and the second adds the interaction (Schall, nd, p. 101). The interaction terms are the interesting ones, because they measure the association between the variables (Dobson and Barnett, 2018, p. 212).

One result makes this practical. Birch showed that for any log-linear model, the maximum likelihood estimates are the same under the Poisson, multinomial and product multinomial models, as long as the parameters for the fixed margins stay in the model. So for estimation we can always pretend the cells are Poisson, and every GLM tool applies (Dobson and Barnett,

2018, p. 212).

Overdispersion

Definition: Overdispersion

The counts are *overdispersed* when $\mathrm{Var}(Y_i) > \mathbb{E}(Y_i)$, although the Poisson model requires $\mathrm{Var}(Y_i) = \mathbb{E}(Y_i)$ (Dobson and Barnett, 2018, p. 216).

Real counts often vary more than a Poisson allows. One cause is a lack of independence, for example animals from the same colony. Another is a missing predictor. A third is a mean that changes from one subject to the next in ways the model does not capture. When independence is the problem, methods for correlated data can help (Dobson and Barnett, 2018, p. 216).

What goes wrong. The estimates of β are usually still fine, because the score depends only on the mean. But the information matrix $X^\top W X$ assumes the variance equals the mean. If the true variance is larger, the standard errors are too small. The Wald z-values are then too big, and the p-values too optimistic.

How to spot it. Compare X^2 or D with their degrees of freedom $N - p$. A $\chi^2(N-p)$ variable has mean $N-p$, so a ratio far above 1 is a warning (Schall, nd, p. 93). We can also test directly, as shown below.

The lecture notes suggest two ways to repair the model: use a negative binomial distribution, or estimate a dispersion parameter and scale by it (Schall, nd, p. 107).

Fix 1: quasi-Poisson. Keep the Poisson mean model and assume $\mathrm{Var}(Y_i) = \phi\,\mu_i$ for an unknown ϕ. The fit does not change, so the coefficients are identical to the Poisson ones. Only the standard errors change: they are multiplied by $\sqrt{\hat\phi}$, where

$$\hat\phi = \frac{X^2}{N-p}.$$

The Pearson version is preferred over $D/(N-p)$ (Schall, nd, pp. 81–82, 107). A quasi-Poisson model has no full likelihood, so it has no AIC.

Fix 2: the negative binomial. Here we give the counts a real distribution with extra spread. Imagine each observation has its own hidden Poisson mean λ_i. The λ_i vary from one observation to the next according to a gamma distribution with mean μ and variance $\alpha\mu^2$. Then the counts follow a negative binomial distribution, with

$$\mathbb{E}(Y) = \mu, \qquad \mathrm{Var}(Y) = \mu + \alpha\mu^2, \qquad \alpha = \frac{1}{\theta}.$$

The variance is the Poisson part μ plus an extra part $\alpha\mu^2$ that grows faster than the mean. The probability function is

$$f(y) = \frac{\Gamma(y+\theta)}{\Gamma(\theta)\,y!}\left(\frac{\theta}{\theta+\mu}\right)^{\theta}\left(\frac{\mu}{\theta+\mu}\right)^{y}.$$

When $\theta \to \infty$ (so $\alpha \to 0$) the extra term disappears and we are back at the Poisson. We still use the log link and the same linear predictor. The new parameter θ is estimated by maximum likelihood together with β. Dobson and Barnett also name the negative binomial as the alternative model for overdispersed counts (Dobson and Barnett, 2018, p. 216).

Testing for overdispersion. Because the Poisson is the special case $\alpha = 0$ of the negative

binomial, we can test

$$H_0 : \alpha = 0 \quad \text{against} \quad H_1 : \alpha > 0.$$

There is one subtlety. Under H_0 the parameter sits on the edge of its allowed range ($\alpha \geq 0$). So the usual χ^2_1 reference distribution for the likelihood-ratio statistic does not apply. A common fix is to use an equal mixture of χ^2_0 and χ^2_1, which simply halves the p-value of the χ^2_1 test. A Wald test also works, but it should be one-sided. Since it relies on the shape of the likelihood at $\hat{\alpha}$, it is best used together with the likelihood-ratio test and not on its own.

Choosing between models. The negative binomial has one more parameter than the Poisson, namely θ. So we compare them with $\text{AIC} = -2\ell(\hat{\beta}) + 2p$, which charges each model for its parameters. Lower is better.

	Poisson	Quasi-Poisson	Negative binomial
Variance	μ	$\phi\,\mu$	$\mu + \alpha\mu^2$
Coefficients	Standard	Same as Poisson	Slightly different
Standard errors	From $X^\top W X$	Poisson ones $\times \sqrt{\hat{\phi}}$	Re-estimated
Full likelihood, AIC	Yes	No	Yes
Extra parameter	None	ϕ	θ ($\alpha = 1/\theta$)

Table 7.2.: Three ways to handle count data, from least to most flexible in the variance.

In short

- Counts are modelled with a Poisson distribution and the log link, so effects are *rate ratios* e^{β}.
- The score equations use only the mean, but the standard errors also use the variance $\text{Var}(Y) = \mu$.
- If $X^2/(N-p)$ is far above 1, the standard errors are too small. Scale them (quasi-Poisson) or switch to the negative binomial.
- A table of counts is just a Poisson model on the log scale, and its independence test is a comparison of two log-linear models.

Reflection **Application**

Before running anything, the questions this phase has to answer are worth naming.

Question: Does overdispersion bias the Poisson coefficients?

No. The Poisson score equations only need the *mean* to be specified correctly, so the estimates of β stay (approximately) right even when $\text{Var}(Y) \neq \mu$. What breaks is the information matrix: $W = \text{diag}(\mu_i)$ in $I = X'WX$ assumes $\text{Var}(Y_i) = \mu_i$, so when the true variance is larger the standard errors come out too small, the Wald z values too large, and the p-values over-confident.

Question: What is the overdispersion parameter α, and why test $\alpha = 0$?

The negative binomial adds a quadratic term to the variance, $\mathrm{Var}(Y) = \mu + \alpha\mu^2$ with $\alpha = 1/\theta$. As $\alpha \to 0$ the extra term vanishes and we recover the Poisson, so testing $H_0 : \alpha = 0$ against $H_1 : \alpha > 0$ asks whether the extra variance is real, or whether $\mathrm{Var} = \mu$ is good enough.

Question: Can the Wald test for α be taken at face value?

Not entirely. Under H_0 the parameter sits on the boundary of its space ($\alpha \geq 0$), so the null distribution is not a plain standard normal and the test should be one-sided. A Wald test also rests on the curvature of the likelihood at $\hat{\alpha}$, which is a shaky guide for variance-type parameters. The likelihood-ratio test is therefore run as a cross-check, using the boundary-corrected null $\frac{1}{2}\chi^2_1$ for its p-value.

Question: Why is AIC a fair way to compare the two models?

Both models are fitted by maximum likelihood to the same response, and the Poisson is the limiting case of the negative binomial. $\mathrm{AIC} = 2p - 2\ell(\hat{\beta})$ charges each model for its parameters ($p = 3$ for the Poisson, $p = 4$ for the negative binomial, which adds θ), so a lower AIC means the better fit is not just bought with extra parameters.

To put this into practice, a dataset in the spirit of the rabbit-birth example was simulated in R: 8 colonies observed weekly for 12 weeks, with a binary feed `supplement`. The true mean is $\mu = \exp(4.4 - 0.25\,\text{week} + 0.30\,\text{supplement})$ and the counts are drawn from a negative binomial with $\theta = 5$, so overdispersion is built in. Since the data are simulated this way, the negative binomial winning below is expected; the point is to see *how* each diagnostic shows it.

The data

Listing 7.1: Simulating the rabbit-birth data

```
library(MASS)                          # glm.nb()
options(width = 78)
set.seed(2026)

# Rabbit births: 8 colonies observed weekly for 12 weeks (week 0 to 11)
rabbits <- expand.grid(week = 0:11, colony = 1:8)
rabbits$supplement <- rbinom(nrow(rabbits), 1, 0.5)     # 1 = extra feed

# True mean decays with the week; counts are over-dispersed (NB, size = 5)
mu_true <- exp(4.4 - 0.25 * rabbits$week + 0.30 * rabbits$supplement)
rabbits$births <- rnbinom(nrow(rabbits), mu = mu_true, size = 5)

head(rabbits)
summary(rabbits$births)
```

Listing 7.2: Output of head() and summary()

```
  week colony supplement births
1    0      1          1    144
2    1      1          1     31
3    2      1          0     49
4    3      1          0     51
5    4      1          1     32
6    5      1          0     10
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1.00   10.75   20.00   36.29   53.00  144.00
```

Poisson regression

The Poisson model with its canonical log link comes first.

Listing 7.3: Fitting the Poisson GLM

```
pois <- glm(births ~ week + supplement, family = poisson, data = rabbits)
summary(pois)
```

Listing 7.4: Output of summary(pois)

```
Call:
glm(formula = births ~ week + supplement, family = poisson, data = rabbits)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  4.440825   0.030856 143.920   <2e-16 ***
week        -0.247867   0.006006 -41.268   <2e-16 ***
supplement   0.332466   0.034116   9.745   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 2765.49  on 95  degrees of freedom
Residual deviance:  583.44  on 93  degrees of freedom
AIC: 1066

Number of Fisher Scoring iterations: 4
```

The residual deviance is 583.4 on 93 degrees of freedom, about six times larger than a well-fitting Poisson model would give (it should be close to its degrees of freedom), which is already a warning sign.

Negative binomial regression

The same linear predictor and log link are used, but the variance is now allowed to exceed the mean.

Listing 7.5: Fitting the negative binomial GLM

```
nb <- glm.nb(births ~ week + supplement, data = rabbits)
summary(nb)
```

Listing 7.6: Output of summary(nb)

```
Call:
glm.nb(formula = births ~ week + supplement, data = rabbits,
    init.theta = 6.366637583, link = log)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  4.41711    0.09376  47.110  < 2e-16 ***
week        -0.24670    0.01385 -17.807  < 2e-16 ***
supplement   0.36095    0.09246   3.904 9.47e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(6.3666) family taken to be 1)

    Null deviance: 454.96  on 95  degrees of freedom
Residual deviance: 101.40  on 93  degrees of freedom
AIC: 741.89

Number of Fisher Scoring iterations: 1


              Theta:  6.37
          Std. Err.:  1.20

 2 x log-likelihood:  -733.895
```

Effect of overdispersion on the estimates

The two models are placed side by side, with the ratio of standard errors showing how much the Poisson understates the uncertainty.

Listing 7.7: Comparing coefficients and standard errors

```
cmp <- data.frame(
  Pois_Est = coef(pois),  Pois_SE = sqrt(diag(vcov(pois))),
  NB_Est   = coef(nb),    NB_SE   = sqrt(diag(vcov(nb)))
)
cmp$SE_ratio <- cmp$NB_SE / cmp$Pois_SE      # how much Poisson understates SEs
round(cmp, 4)
```

Listing 7.8: Poisson versus negative binomial coefficients

```
            Pois_Est Pois_SE  NB_Est  NB_SE SE_ratio
(Intercept)   4.4408  0.0309  4.4171 0.0938   3.0387
week         -0.2479  0.0060 -0.2467 0.0139   2.3066
supplement    0.3325  0.0341  0.3609 0.0925   2.7101
```

The point estimates barely move (for example -0.2479 against -0.2467 for `week`), which agrees with the first reflection question. The standard errors do change: the Poisson ones are roughly 2.3 to 3.0 times too small. The effect of feed is still significant under the negative binomial, but its z value falls from 9.7 to 3.9.

Testing for overdispersion

Listing 7.9: Pearson statistic, Wald test for α and likelihood-ratio check

```
# (i) Pearson dispersion statistic of the Poisson fit (should be about 1)
pear <- sum(residuals(pois, type = "pearson")^2)
c(Pearson_X2 = pear, df = df.residual(pois), phi_hat = pear / df.residual(
    pois))

# (ii) Wald test of H0: alpha = 0 (Poisson) against alpha > 0 (NB),
#      where Var(Y) = mu + alpha * mu^2 and alpha = 1/theta
theta    <- nb$theta
alpha    <- 1 / theta
se_alpha <- nb$SE.theta / theta^2            # delta method
z_wald   <- alpha / se_alpha
c(alpha = alpha, SE = se_alpha, z = z_wald,
  p_one_sided = pnorm(z_wald, lower.tail = FALSE))

# (iii) Likelihood-ratio cross-check (null is on the boundary: halve the p-
    value)
lr <- as.numeric(2 * (logLik(nb) - logLik(pois)))
c(LR = lr, p = 0.5 * pchisq(lr, df = 1, lower.tail = FALSE))
```

Listing 7.10: Output of the overdispersion tests

```
Pearson_X2          df     phi_hat
572.817956   93.000000    6.159333
       alpha            SE             z  p_one_sided
1.570688e-01 2.958911e-02 5.308330e+00 5.531697e-08
          LR             p
3.260677e+02 3.453367e-73
```

Three results agree:

- The Pearson statistic divided by its degrees of freedom is $\hat{\phi} = 572.8/93 \approx 6.16$, far above the value of 1 expected for a Poisson model. Its square root, about 2.5, matches the typical size of the standard-error ratios above.
- The Wald test gives $\hat{\alpha} = 0.157$ (SE 0.030), so $z = 5.31$ and a one-sided $p \approx 5.5 \times 10^{-8}$: the data reject $\alpha = 0$.
- The likelihood-ratio statistic is 326.1, a far stronger signal than the Wald $z^2 \approx 28$. The two tests reach the same conclusion, but this gap is the reason the Wald test for a dispersion parameter should not be the only evidence.

AIC comparison

Listing 7.11: AIC and BIC for both models

```
AIC(pois, nb)
aic <- AIC(pois, nb)$AIC
c(delta_AIC = aic[1] - aic[2])                    # positive favours NB
BIC(pois, nb)
```

Listing 7.12: Output of AIC() and BIC()

```
      df       AIC
pois   3 1065.9624
nb     4  741.8947
delta_AIC
 324.0677
      df       BIC
pois   3 1073.6554
nb     4  752.1521
```

The negative binomial has the lower AIC by $\Delta\text{AIC} = 324.1$, even after paying for its extra parameter; the BIC agrees (752.2 against 1073.7). A difference of this size is overwhelming evidence for the negative binomial.

Residual diagnostics

Listing 7.13: Residual check and Figure 7.4

```
# Share of Pearson residuals beyond +/-2 (about 5% expected for a good fit)
c(Poisson = mean(abs(residuals(pois, type = "pearson")) > 2),
  NegBin  = mean(abs(residuals(nb,   type = "pearson")) > 2))

png("figures/ch07_overdispersion.png", width = 2000, height = 900, res = 220)
par(mfrow = c(1, 2), mar = c(4.2, 4.2, 2.6, 1))
teal <- "#14706F"; orange <- "#D97B2D"

# (a) data with both fitted mean curves (supplement = 0)
nd <- data.frame(week = seq(0, 11, 0.1), supplement = 0)
plot(rabbits$week, rabbits$births, pch = 16, col = adjustcolor("grey40", 0.5),
     xlab = "Week", ylab = "Number of births", main = "Fitted means")
lines(nd$week, predict(pois, nd, type = "response"), col = teal,   lwd = 2)
lines(nd$week, predict(nb,   nd, type = "response"), col = orange, lwd = 2, lty = 2)
legend("topright", c("Poisson", "Negative binomial"), col = c(teal, orange),
       lty = c(1, 2), lwd = 2, bty = "n", cex = 0.85)

# (b) Pearson residuals against fitted values
plot(fitted(pois), residuals(pois, type = "pearson"), pch = 16, log = "x",
     col = adjustcolor(teal, 0.6), xlab = "Fitted mean (log scale)",
     ylab = "Pearson residual", main = "Residual spread")
points(fitted(nb), residuals(nb, type = "pearson"), pch = 17,
       col = adjustcolor(orange, 0.6))
abline(h = c(-2, 2), lty = 3)
legend("topright", c("Poisson", "Negative binomial"), col = c(teal, orange),
       pch = c(16, 17), bty = "n", cex = 0.85)
invisible(dev.off())
```

Listing 7.14: Share of Pearson residuals beyond ± 2

```
   Poisson     NegBin
0.35416667 0.05208333
```

About 35% of the Poisson Pearson residuals fall outside ± 2, against roughly 5% for the negative binomial, which is what a well-specified model should give.

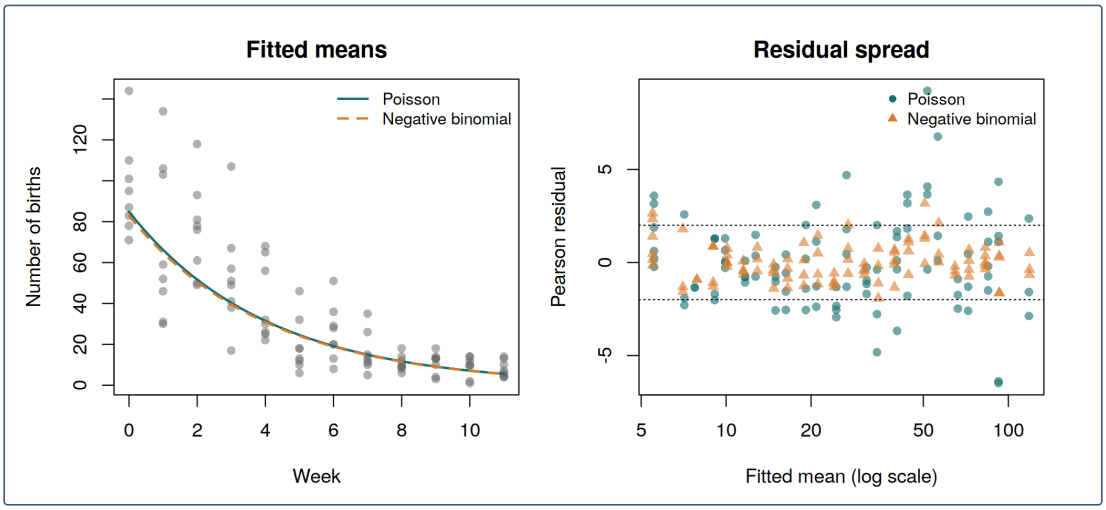


Figure 7.4.: Left: both models give essentially the same fitted mean. Right: the Poisson Pearson residuals fan out far beyond ± 2, the negative binomial ones do not.

	Poisson	Negative binomial
Residual deviance / df	$583.4/93$	$101.4/93$
SE of `week`	0.0060	0.0139
Share of $\lvert r_{\text{Pearson}} \rvert > 2$	35.4%	5.2%
AIC	1066.0	741.9
BIC	1073.7	752.2

Table 7.3.: Poisson against negative binomial on the simulated rabbit-birth data.

Conclusion. The Poisson model captures the mean but not the spread. Its dispersion statistic ($\hat{\phi} \approx 6.2$), the Wald and likelihood-ratio tests for α, the residuals and the AIC all point the same way, and the negative binomial is the model to report.

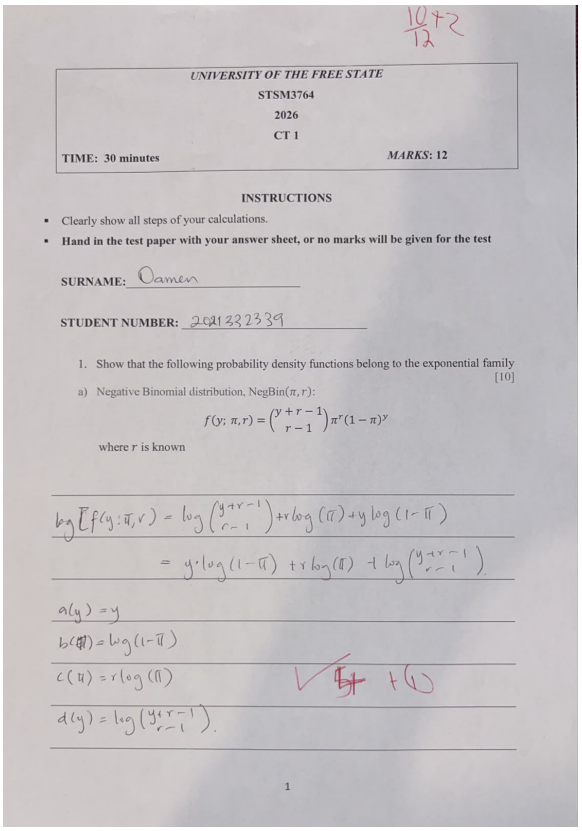
PART IV.

Appendices

Class Tests

Class Tests 1 to 4, written throughout the semester, are reproduced as marked.

Class Test 1



10+2
12

UNIVERSITY OF THE FREE STATE
STSM3764
2026
CT 1

TIME: 30 minutes MARKS: 12

INSTRUCTIONS

- Clearly show all steps of your calculations.
- **Hand in the test paper with your answer sheet, or no marks will be given for the test**

SURNAME: Damen

STUDENT NUMBER: 2021323339

1. Show that the following probability density functions belong to the exponential family [10]

a) Negative Binomial distribution, NegBin(π, r):

$$f(y;\pi,r) = \binom{y+r-1}{r-1}\pi^r(1-\pi)^y$$

where r is known

$\log[f(y;\pi,r)] = \log\binom{y+r-1}{r-1} + r\log(\pi) + y\log(1-\pi)$

$= y\cdot\log(1-\pi) + r\log(\pi) + \log\binom{y+r-1}{r-1}$

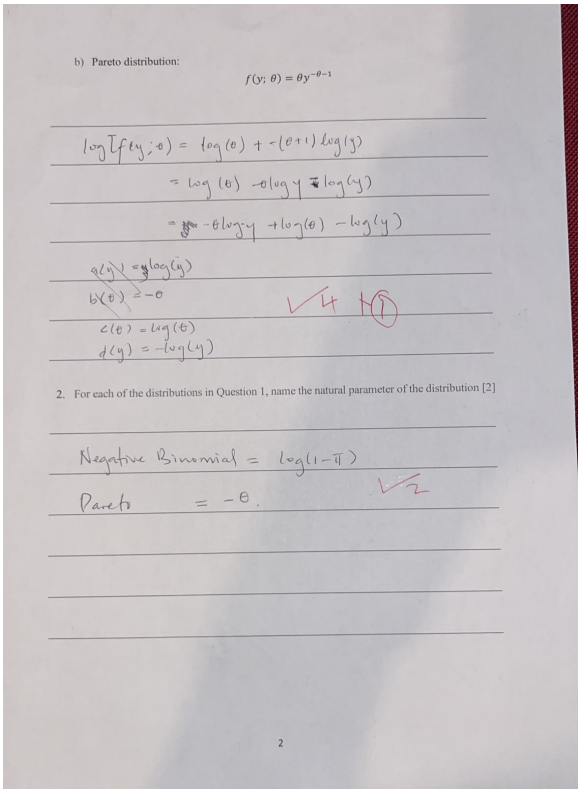
$a(y) = y$

$b(\pi) = \log(1-\pi)$

$c(\pi) = r\log(\pi)$ ✓4 +(1)

$d(y) = \log\binom{y+r-1}{r-1}$.

1



b) Pareto distribution:

$f(y;\theta) = \theta y^{-\theta-1}$

$\log[f(y;\theta)] = \log(\theta) + -(\theta+1)\log(y)$

$= \log(\theta) - \theta\log y - \log(y)$

$= -\theta\log y + \log(\theta) - \log(y)$

$a(y) = y\log(y)$

$b(\theta) = -\theta$ ✓4 +(1)

$c(\theta) = \log(\theta)$

$d(y) = -\log(y)$

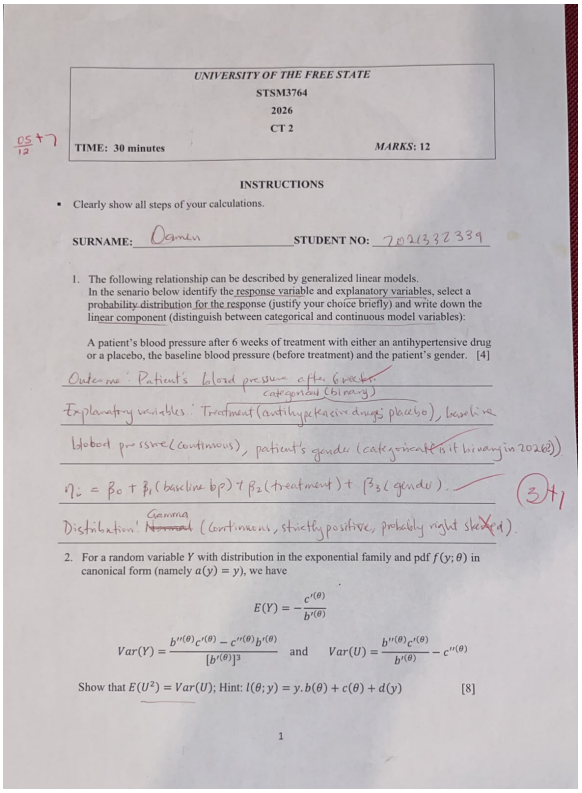
2. For each of the distributions in Question 1, name the natural parameter of the distribution [2]

Negative Binomial $= \log(1-\pi)$

Pareto $= -\theta$. ✓2

2

Class Test 2



UNIVERSITY OF THE FREE STATE
STSM3764
2026
CT 2

TIME: 30 minutes MARKS: 12

INSTRUCTIONS

- Clearly show all steps of your calculations.

SURNAME: Damen STUDENT NO: 2021322339

1. The following relationship can be described by generalized linear models. In the senario below identify the response variable and explanatory variables, select a probability distribution for the response (justify your choice briefly) and write down the linear component (distinguish between categorical and continuous model variables):

   A patient's blood pressure after 6 weeks of treatment with either an antihypertensive drug or a placebo, the baseline blood pressure (before treatment) and the patient's gender. [4]

Outcome: Patient's blood pressure after 6 weeks (categorical (binary))

Explanatory variables: Treatment (antihypertensive drugs placebo), baseline blood pressure (continuous), patient's gender (categorical is it binary in 2026?))

$\eta_i = \beta_0 + \beta_1$(baseline bp) $+ \beta_2$(treatment) $+ \beta_3$(gender).

Distribution: ~~Normal~~ Gamma (continuous, strictly positive, probably right skewed).

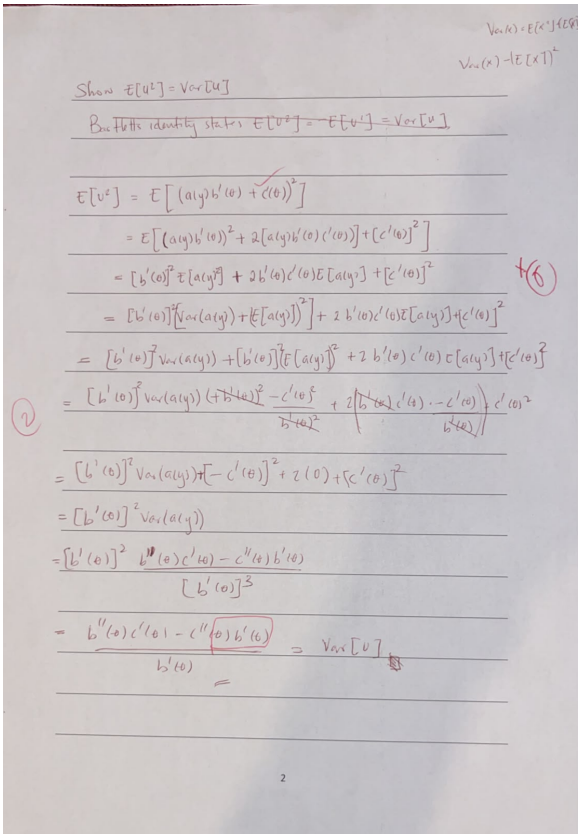
2. For a random variable Y with distribution in the exponential family and pdf $f(y;\theta)$ in canonical form (namely $a(y) = y$), we have

$$E(Y) = -\frac{c'(\theta)}{b'(\theta)}$$

$$Var(Y) = \frac{b''(\theta)c'(\theta) - c''(\theta)b'(\theta)}{[b'(\theta)]^3} \quad \text{and} \quad Var(U) = \frac{b''(\theta)c'(\theta)}{b'(\theta)} - c''(\theta)$$

Show that $E(U^2) = Var(U)$; Hint: $l(\theta;y) = y.b(\theta) + c(\theta) + d(y)$ [8]

1



Show $E[U^2] = Var[U]$

Bartlett's identity states $E[U^2] = E[U'] = Var[U]$

$E[U^2] = E\left[(a(y)b'(\theta) + c'(\theta))^2\right]$

$= E\left[(a(y)b'(\theta))^2 + 2[a(y)b'(\theta)c'(\theta)] + [c'(\theta)]^2\right]$

$= [b'(\theta)]^2 E[a(y)^2] + 2b'(\theta)c'(\theta)E[a(y)] + [c'(\theta)]^2$

$= [b'(\theta)]^2\left[Var(a(y)) + [E[a(y)]]^2\right] + 2b'(\theta)c'(\theta)E[a(y)] + [c'(\theta)]^2$

$= [b'(\theta)]^2 Var(a(y)) + [b'(\theta)]^2[E[a(y)]]^2 + 2b'(\theta)c'(\theta)E[a(y)] + [c'(\theta)]^2$

$= [b'(\theta)]^2 Var(a(y)) + [b'(\theta)]^2\frac{-c'(\theta)}{b'(\theta)} + 2\left[b'(\theta)c'(\theta)\cdot\frac{-c'(\theta)}{b'(\theta)}\right] + c'(\theta)^2$

$= [b'(\theta)]^2 Var(a(y)) + [-c'(\theta)]^2 + 2(0) + [c'(\theta)]^2$

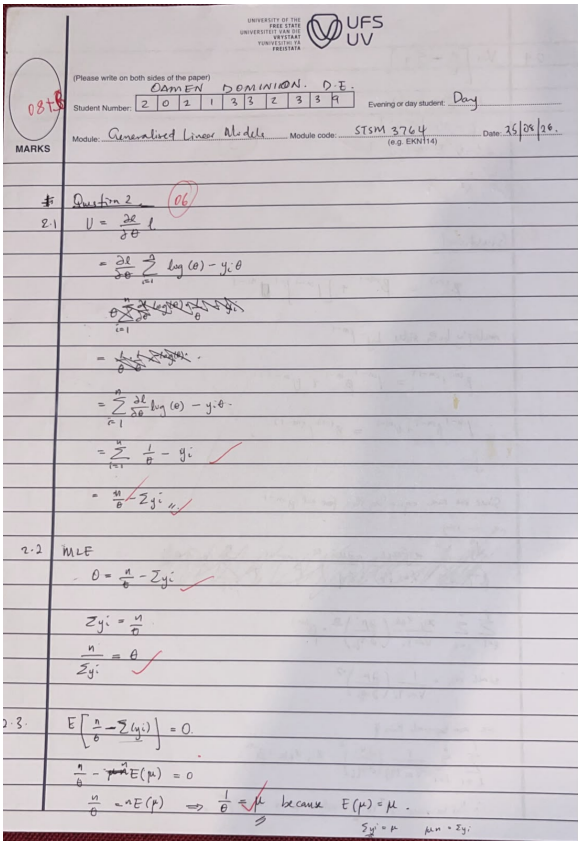
$= [b'(\theta)]^2 Var(a(y))$

$= [b'(\theta)]^2 \frac{b''(\theta)c'(\theta) - c''(\theta)b'(\theta)}{[b'(\theta)]^3}$

$= \frac{b''(\theta)c'(\theta) - c''(\theta)b'(\theta)}{b'(\theta)} = Var[U]$

2

Class Test 3



(Please write on both sides of the paper)
DAMEN DOMINIQUE. O.E.
Student Number: 2021331384 Evening or day student: Day
Module: Generalized Linear Models Module code: STSM 3764 (e.g. EKN114) Date: 25/08/26.
MARKS

Question 2 (06)

2.1 $U = \frac{\partial l}{\partial \theta}$

$= \frac{\partial}{\partial \theta} \sum_{i=1}^{n} \log(\theta) - y_i \theta$

$= \sum_{i=1}^{n} \frac{\partial l}{\partial \theta} \log(\theta) - y_i \theta$

$= \sum_{i=1}^{n} \frac{1}{\theta} - y_i$

$= \frac{n}{\theta} - \sum y_i$

2.2 MLE

$0 = \frac{n}{\theta} - \sum y_i$

$\sum y_i = \frac{n}{\theta}$

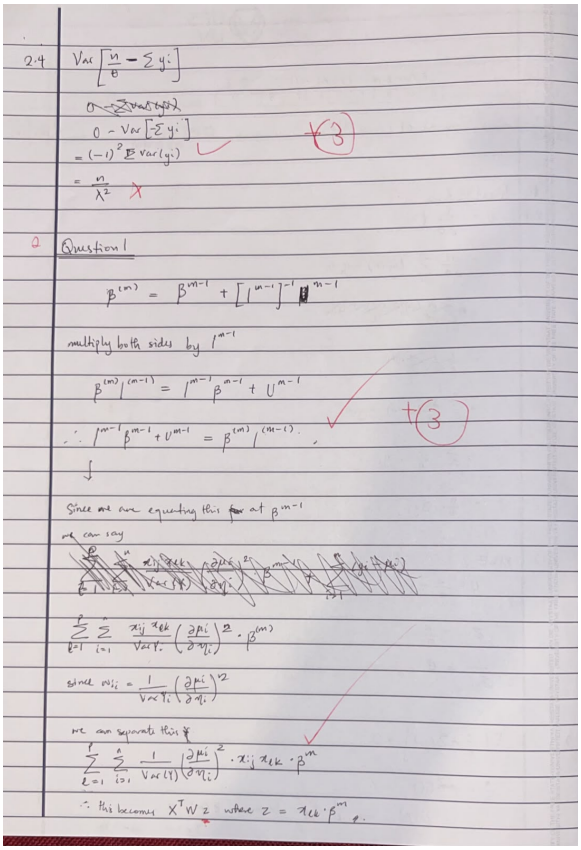
$\frac{n}{\sum y_i} = \theta$

2.3 $E\left[\frac{n}{\theta} - \sum(y_i)\right] = 0$

$\frac{n}{\theta} - nE(\mu) = 0$

$\frac{n}{\theta} = nE(\mu) \Rightarrow \frac{1}{\theta} = \mu$ because $E(\mu) = \mu$.

$\sum y_i = \mu$ $\mu n = \sum y_i$



2.4 $\mathrm{Var}\left[\frac{n}{\theta} - \sum y_i\right]$

$0 - \mathrm{Var}\left[\sum y_i\right]$ (3)

$= (-1)^2 \sum \mathrm{Var}(y_i)$

$= \frac{n}{\lambda^2}$

Question 1

$\beta^{(m)} = \beta^{m-1} + \left[I^{m-1}\right]^{-1} \beta^{m-1}$

multiply both sides by I^{m-1}

$\beta^{(m)} I^{(m-1)} = I^{m-1} \beta^{m-1} + U^{m-1}$

$\therefore I^{m-1} \beta^{m-1} + U^{m-1} = \beta^{(m)} I^{(m-1)}$ (3)

Since we are equating this at β^{m-1}

we can say

$\sum_{k=1}^{p} \sum_{i=1}^{n} \frac{x_{ij} x_{ik}}{\mathrm{Var} Y_i} \left(\frac{\partial \mu_i}{\partial \eta_i}\right)^2 \cdot \beta^{(m)}$

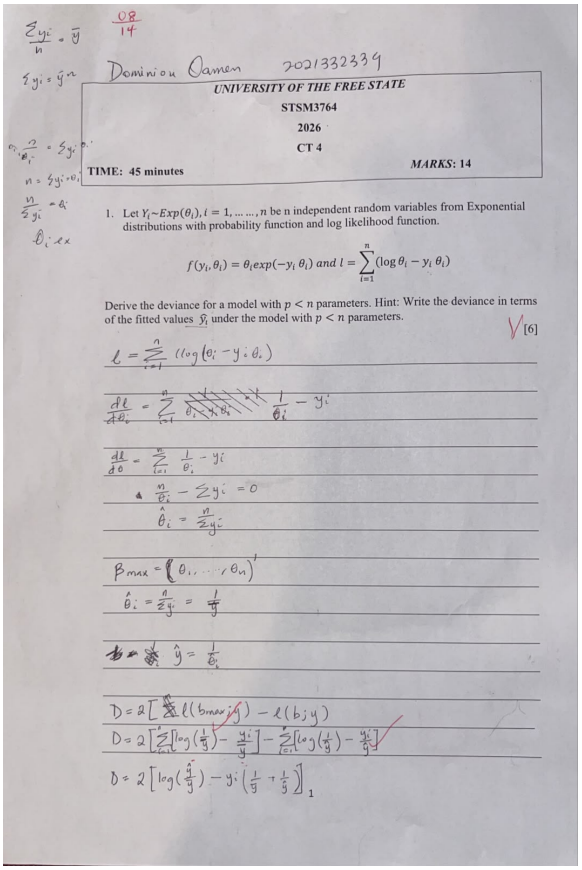
since $w_{ii} = \frac{1}{\mathrm{Var} Y_i} \left(\frac{\partial \mu_i}{\partial \eta_i}\right)^2$

we can separate this

$\sum_{k=1}^{p} \sum_{i=1}^{n} \frac{1}{\mathrm{Var}(Y)} \left(\frac{\partial \mu_i}{\partial \eta_i}\right)^2 \cdot x_{ij} x_{ik} \cdot \beta^{m}$

$\therefore$ this becomes $X^T W z$ where $z = x_{ik} \cdot \beta^{m}$

Class Test 4

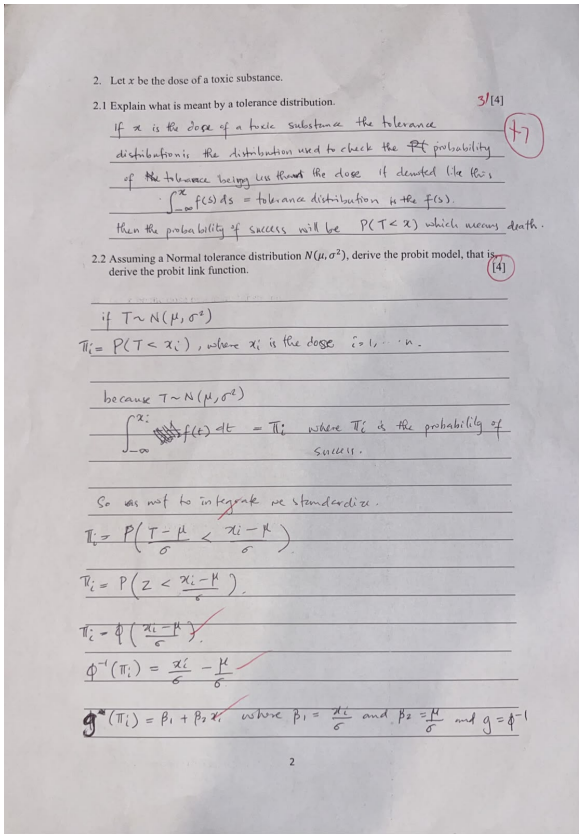


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2026
CT 4
TIME: 45 minutes MARKS: 14

1. Let $Y_i \sim Exp(\theta_i), i = 1, \ldots, n$ be n independent random variables from Exponential distributions with probability function and log likelihood function.

$$f(y_i, \theta_i) = \theta_i exp(-y_i\,\theta_i) \text{ and } l = \sum_{i=1}^{n} (\log \theta_i - y_i\,\theta_i)$$

Derive the deviance for a model with $p < n$ parameters. Hint: Write the deviance in terms of the fitted values $\hat{y}_i$ under the model with $p < n$ parameters. [6]



2. Let x be the dose of a toxic substance.

2.1 Explain what is meant by a tolerance distribution. [4]

2.2 Assuming a Normal tolerance distribution $N(\mu, \sigma^2)$, derive the probit model, that is, derive the probit link function. [4]

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Referencing is done using APA style.

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